

FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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ABSTRACT. Every quadratic pencil in dimension $n \geq 3$ is determined up to projective equivalence by the abstract $(n^2 + 2n - 4)$ -th neighbourhood of the socle Grassmannian in its multiplication-failure scheme. Every lower-order neighbourhood is independent of the pencil. We prove this by recovering the oriented tensor cone and its full Pluecker coefficient line intrinsically. The first relation spaces embed the entire pencil parameter scheme and give universal finite neighbourhoods compatible with arbitrary complex base change. For regular pencils they recover spectral sheaves, their Fitting incidence schemes, and the reciprocal critical correspondence. At fixed discriminant and reduced rank data, the finite invariant realizes and distinguishes every spectral dominance transition. For definite real pencils, spectral projectors give a congruence-covariant open family of data on which the critical correspondence is finite etale and all its points are real, simple, and globally labelled. The full web, contraction, polarized, and boundary theory is retained in the accompanying supplement.

1. INTRODUCTION

A degeneracy locus usually forgets the map that defines it. We ask whether its nonreduced scheme structure can recover that map, without an ambient embedding or a marking of the normal directions. For the multiplication-failure schemes of quadratic pencils the answer is exact: there is one uniform infinitesimal order at which the entire pencil first becomes visible. The inverse theorem applies to every pencil, including singular pencils. Its regular locus then carries the spectral and critical correspondences studied below.

1.1. The principal theorem. Let V be a complex vector space of dimension $n \geq 3$, and set

$$W = \mathrm{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \mathrm{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The algebra $B_R = \mathbb{C} \oplus V \oplus S_R$ has multiplication $\mathrm{Sym}^2 V \rightarrow S_R$, with $(V \oplus S_R)S_R = 0$. On $X_R = \mathrm{Gr}(n, V \oplus S_R)$ let $\mathcal{K}$ denote the tautological bundle. For any $m \geq 2$, define

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow B_R \otimes \mathcal{O}]). \quad (1.1)$$

This is the literal maximal-minor scheme of the multiplication map; it is independent of $m \geq 2$. Its reduction is the fixed Schubert divisor where $\mathcal{K} \rightarrow V$ drops rank. Let $B = \mathrm{Gr}(n, S_R)$, let $\mathfrak{b}_R$ be its ideal in $\widehat{D}_R$, and put

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

The neighbourhood is taken along the whole projective socle Grassmannian, not at one framed affine point.

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Theorem 1.1 (Sharp intrinsic finite reconstruction). *For fixed $n \geq 3$, the following assertions hold on all of $\mathrm{Gr}(2, \mathrm{Sym}^2 V)$.*

(i) *For every pair of pencils R, R' ,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

The isomorphism of neighbourhoods is abstract and unmarked. For each $0 \leq k < d$, the isomorphism class of $Z_R^{[k]}$ is independent of R . Thus d is the smallest uniform reconstruction order.

(ii) *In fixed tensor coordinates, the first relation spaces define a closed immersion*

$$\mathrm{Gr}(2, W) \longrightarrow \mathrm{Gr}\left(\binom{N}{2}, \mathrm{Sym}^d \mathrm{Hom}(U, V)^*\right), \quad \dim U = n.$$

This assertion includes infinitesimal parameter schemes and singular pencils.

(iii) *These relations glue to the actual universal order- d neighbourhood over the relative socle Grassmannian. It is finite flat there of rank*

$$\binom{n^2 + d}{d} - \binom{N}{2},$$

projective and flat over the pencil parameter scheme, and compatible with arbitrary complex base change with the relative socle retained.

Theorem 5.1 proves (i); Theorems 5.3 and 7.1 prove (ii)–(iii). The distinction between their inputs is essential: (i) uses an abstract global scheme, whereas (ii)–(iii) retain the relative socle and a tensor presentation. No equivalence of unmarked moduli stacks is asserted.

Here is the mechanism behind the theorem. In graph coordinates $T : \mathcal{U} \rightarrow V$, where $\mathcal{U}$ is tautological on B , the homogeneous ideal is

$$(\det T)I_p(\gamma_R \mathrm{Sym}^2 T), \quad \gamma_R : W \twoheadrightarrow S_R.$$

It is generated in degree d . The abstract finite scheme recovers B , its degree-one conormal, and the first relation module. The reduced determinant cone has two rank-one rulings. Only its $\mathbb{P}(V)$ -ruling is projectively trivial over B ; a Schubert line distinguishes the other one. Regular descent therefore recovers a single constant projective coordinate transformation on V , rather than unrelated changes for different coefficients. Cancelling the determinant line then recovers the residual coefficient module. For pencils its exterior representation is irreducible; exterior duality identifies its coefficient line with the full Pluecker line of R . The first visible degree alone is formal; these intrinsic identifications are the substantive inverse argument.

1.2. Spectral and critical correspondences. On the regular locus, the reconstructed pencil determines its torsion spectral sheaf, not merely its determinant. Theorems 6.3 and 7.4 recover both the elementary divisors and their scheme-theoretic Fitting incidence loci. Theorem 8.1 applies this to every fixed discriminant and reduced-rank stratum: the classical dominance specializations are realized by flat families of the finite neighbourhoods, and every strict change first becomes distinguishable at order d . The qualifier *reduced* matters; the full spectral Fitting schemes need not remain fixed.

There is a further functorial output. The reciprocal critical correspondence is the zero scheme of the two score equations on the invertible members of a pencil, with the data matrix allowed to vary. Theorem 10.1 recovers its isomorphism class from the finite failure invariant and constructs it in families. Thus the passage from finite failure to likelihood has a specified algebraic intermediate object, rather than a parallel numerical analogy. For a supplied definite real realization, Theorem 10.2 gives a congruence-covariant open family of data, a finite etale

critical algebra of rank $2r - 3$, and globally labelled real analytic critical sections. Its residue and separated-root argument builds on the complete proof of Theorem 9.1. That latter theorem proves the assertion formulated in [40, Conjecture 4.5], for unrestricted real data and arbitrary eigenvalue multiplicities. Neither a real form nor a definite member is inferred from an unmarked complex scheme, and positivity of the data or of all critical points is not asserted.

1.3. Proof dependencies and further results. The principal argument is independent of exterior recombination. The following table specifies the logical inputs; the implication from reconstructed matrices to the critical correspondence is formal, whereas its totally real open family requires an additional proof.

Step	Input	Output
Intrinsic coefficients	Literal Fitting ideal; oriented rank-one rulings	One projective frame and the full Pluecker line
Finite reconstruction	Coefficient inverse; single relation degree	Sharp uniform order; parameter immersion
Relative spectral read-out	Glued finite neighbourhood; regular pencil	Spectral sheaf and Fitting incidence schemes
Critical correspondence	Reconstructed pencil; universal score equations	Congruence-covariant algebraic correspondence
Totally real cover	Supplied real definite realization; residues and signs	Finite etale algebra and labelled real critical sections

The supplement retains every web, contraction, polarized, polar, and boundary theorem with its full proof. In particular, Theorem S1.1 recovers all basepoint-free four-dimensional webs with smooth Jacobian quartic; the all-ranks contraction theorem and exact Jacobian–Casimir factorization supply its recombination mechanism. They are substantial further results, not hidden hypotheses of the pencil theorem. The first-layer K3 construction, hyperelliptic interpretation, component-fibre classification, and primary boundary calculations remain available at their original level of strength.

1.4. Relation to classical results. The Weierstrass–Segre classification, its orbit closures, the harmonic decomposition, and exterior-support bounds are classical inputs, identified at the statements where they are used. In particular, [40, Theorem 1.1 and Corollary 2.1] classify regular pencils by their labelled elementary divisors; [44, equation (2)] gives the selfadjoint orthogonal orbit calculation underlying the fixed-spectrum discussion. The inverse problem here starts instead with an unmarked failure scheme. The exact differential dictionary for the auxiliary contraction is given in Section S12; it separates classical harmonic tools from the normalization of the specified map.

Ballico’s paper [2] is a directly relevant predecessor. Its complete text has not been obtained for the present comparison. Consequently we do not infer nonanticipation from its title, first page, or later citations. The six theorem-level comparisons requiring that text are recorded explicitly in the accompanying referee response. This documentary limitation is separate from the hypotheses and proofs of the theorems stated here.

2. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the supplement.

Lemma 2.1 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\mathrm{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is the zero scheme of*

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (2.1)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\bigwedge^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$. For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\mathrm{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\mathrm{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (2.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an

identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 2.2 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 2.1, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\mathrm{Spec} H^0(B, \mathcal{O}_B) = \mathrm{Spec} \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. For the degree-twelve construction below, both left Cauchy factors have the same degree twelve; the exterior-duality determinant multiplier is common as well. Thus neither line-bundle transition functions nor right changes of frame can introduce two different projective transformations on the left. $\square$

3. AN INTRINSIC COEFFICIENT PRINCIPLE

The web problem combines two operations: extracting coefficient data from an abstract scheme and reconstructing a relation space from those data. The first operation admits a formulation independent of webs or quadratic multiplication. We state the criterion and retain its polar-restriction application for linear systems in every dimension. Theorem 4.1 subsequently derives all of its geometric hypotheses directly for a full Grassmannian family of the original quadratic multiplication schemes.

3.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \text{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \text{rank } \mathcal{U} = n$.
- (ii) The reduction of $\text{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (3.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 3.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \twoheadrightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 2.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 2.2 then gives one constant projective map $[g]$. After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form $T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

4. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

4.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma : \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (1.1) of $\widehat{D}_R$, with A_R in place of B_R . Thus $\widehat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (4.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 4.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\widehat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (4.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 3.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \text{Sym}^2 M \end{pmatrix}. \quad (4.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1+n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \text{Sym}^2 M). \quad (4.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \text{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (4.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \text{Tot Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \text{Sym}^2 T). \quad (4.5)$$

There are no degree-one equations. The ambient bundle in (4.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 2.1. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\text{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 2.2 applies: an abstract isomorphism gives a morphism $B \rightarrow \text{PGL}(V)$, constant because B is connected projective and $\text{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (4.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \text{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (4.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (4.1); see [37, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (4.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the supplement. For pencils that issue disappears altogether.

4.2. Every pencil, without a regularity hypothesis.

Lemma 4.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 4.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N - 2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (4.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N - 2 > n$. By Theorem 4.1 and Lemma 4.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (4.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (4.4). $\square$

4.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [40, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 4.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A'h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the

size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (4.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 4.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (4.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 4.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 4.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [41, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic, so simultaneous diagonalization identifies the pencil with $(\sum x_i^2, \sum \lambda_i x_i^2)$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 4.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (4.4). Finally the two binary forms in (4.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

5. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \text{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (5.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \text{Gr}(n, S_R) \subset \hat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 4. Write $\mathfrak{b} \subset \mathcal{O}_{\hat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\hat{D}_R}/\mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (5.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 5.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \text{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \hat{D}_R \simeq \hat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \text{PGL}(V). \end{aligned} \quad (5.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

5.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (4.4) gives

$$\mathcal{I}_R = (\det T) I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (5.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 5.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\hat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (5.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\mathrm{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (5.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\mathrm{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \mathrm{Sym}_{\mathcal{O}_B} L/(K_R). \quad (5.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 5.1. For $k < d$, equations (5.4) and (5.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \mathrm{Sym} E^*/\mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 5.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 4.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (5.4) recovers the residual ideal. Finally Lemma 4.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (4.8) recovers the Pluecker line of R . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (5.3), also using Theorem 4.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

5.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \mathrm{Sym}^2 V, \quad F(U) = \bigwedge^p \mathrm{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \mathrm{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \mathrm{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \mathrm{Sym}^2 T)$ in these fixed coordinates.

Theorem 5.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \mathrm{Gr}(2, \mathrm{Sym}^2 V) \longrightarrow \mathrm{Gr}(M, \mathrm{Sym}^d H^*), \quad R \longmapsto K_R, \quad (5.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 5.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \mathrm{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 5.3. The coefficient map (4.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \mathrm{Sym}^{2p} H^*.$$

It is injective: by Lemma 4.2 its source is irreducible for $\mathrm{GL}(V) \times \mathrm{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (5.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\mathrm{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 5.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 5.5 (A flat family of finite normal algebras). *Over $\mathrm{Gr}(2, W)$, the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (5.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 5.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\mathrm{Sym}^j H^*$, and its degree- d piece is $\mathrm{Sym}^d H^* / K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

5.3. Classical moduli at a sharp order.

Corollary 5.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 5.1 with Corollary 4.6, and substitute $n = 2g + 2$ in (5.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 5.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

6. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 5.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

6.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (6.1)$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [40, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma 6.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma 6.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_\lambda (x - \lambda)^{a_\lambda}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_\lambda^2 = \lambda$, and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_\lambda}$, gives $h_\lambda(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem 6.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 5.1. A congruence of pencils induces an isomorphism of (6.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma 6.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma 6.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x} / \mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \quad (6.2)$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R,x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (6.3)$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

6.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2}N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (6.4)$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (6.5)$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem 6.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (6.6)$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*
- (3) *The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).*

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} \begin{pmatrix} P_t & I_2 \end{pmatrix}. \quad (6.7)$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (6.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (6.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem 6.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem 7.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\text{Gr}(2, \text{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \text{Sym}^2 V/R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (5.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \text{Sym}^j E^* \oplus (\text{Sym}^d E^*/K_{R_t}). \quad (6.8)$$

By Theorem 5.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (6.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

6.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [40, §3].

Corollary 6.5 (A conic specializes to a line). *For the four-dimensional family (6.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem 6.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

7. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem 7.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \mathrm{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \mathrm{Spec}_{\mathcal{B}} \left(\mathrm{Sym} \mathcal{E}^* / (\mathcal{K}, \mathrm{Sym}^{d+1} \mathcal{E}^*) \right). \quad (7.1)$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (7.2)$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\mathrm{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \mathrm{Sym}^j \mathcal{E}^$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.*

Proof. The locus in $\mathrm{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\mathrm{Tot} \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \mathrm{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (5.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T) I_p(\gamma \mathrm{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \mathrm{Sym}^2 \mathcal{U} \longrightarrow \mathrm{Sym}^d \mathcal{E}^*. \quad (7.3)$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \mathrm{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \mathrm{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 5.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \mathrm{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (7.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (7.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (7.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph

description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark 7.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

7.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma 7.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \text{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\text{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \text{coker}(X^a : D^n \rightarrow D^n). \quad (7.4)$$

Equivalently, the a -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem 7.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 5.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (7.5)$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\text{length}(\mathcal{T}_z/\mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 5.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma 7.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (7.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (6.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

7.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (7.6)$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem 7.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [40, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem 6.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (7.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

8. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section 7 and the example of Section 6 lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct $\alpha_1, \dots, \alpha_s \in \mathbb{C}$, positive m_i with $\sum m_i = n \geq 3$, and $1 \leq c_i \leq m_i$. Fix a nonsingular symmetric A , and let $\mathfrak{p}_A$ be its self-adjoint endomorphisms. Consider the reduced locally closed variety

$$X_{\mathbf{m},c}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (8.1)$$

Exclude $s = 1, c_1 = n$, for which C is scalar and $\langle A, AC \rangle$ is not a pencil. Every other datum is realized. Put $R_C = \langle A, AC \rangle$, $d = n^2 + 2n - 4$, and $Z_C = Z_{R_C}^{[d]}$. The full binary determinant is $\det(A) \prod_i (u + \alpha_i v)^{m_i}$. The coranks and labelled roots specify all reduced determinantal loci on the line.

Write $\mathcal{P}(m, c)$ for partitions into exactly c positive parts. For equal-size partitions, $\mu \preceq \lambda$ means $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$ for every a ; write λ' for the conjugate partition.

Theorem 8.1 (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The $O(A)$ -orbits in $X_{\mathbf{m},c}(A)$ are indexed by $\lambda \in \prod_i \mathcal{P}(m_i, c_i)$, with dimensions*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (8.2)$$

Unmarked isomorphism classes of Z_C are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.

- (2) *A specialization from λ to μ inside (8.1) exists if and only if*

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (8.3)$$

Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with A fixed and with constant full determinant and reduced rank loci.

- (3) *Every such pencil family induces a flat projective family of Z_C . After shrinking the curve, every truncation of order $k < d$ is constant. If at least one dominance inequality is strict, the generic and special Z_C are nonisomorphic; their first distinguishing order is d .*
- (4) *The recovered spectral module at α_i has length profile*

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (8.4)$$

Every $L_{i,a}$ can only increase in the stated specialization. All these inequalities together are equivalent to (8.3), giving an intrinsic numerical criterion for every admissible transition.

Proof. We specify the classical input. On a complex nondegenerate symmetric space of dimension m , full orthogonal orbits of self-adjoint nilpotents are indexed by all partitions λ of m . Their dimensions are $(m^2 - \sum_a (\lambda'_a)^2)/2$, and the closure order is dominance. These are the statements in [44, §1, p. 3, equation (2) and the following dominance criterion]. They concern self-adjoint matrices, not the skew-adjoint orthogonal Lie algebra. The Grassmannian Segre-stratum comparison is [40, Theorem 5.1].

The primary spaces $V_i = \ker(C - \alpha_i I)^{m_i}$ are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint C , hence self-adjoint projections. Each restricted form is nondegenerate, and $C|_{V_i} = \alpha_i I + N_i$, with N_i self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension $(n^2 - \sum_i m_i^2)/2$. Adding the nilpotent orbit dimensions proves (8.2). These are dimensions in the fixed- A matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem 6.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks m_i . The rank of N_i^a cannot increase at specialization. Since $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$, this gives the length inequalities. Summing the first a columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of (V, A) . The closure criterion puts a chosen representative of μ_i in the closure of the orbit for λ_i . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator $\bigoplus_i (\alpha_i I + N_i(t))$ is self-adjoint for the same A . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem 7.1 along the curve's morphism to $P = \text{Gr}(2, \text{Sym}^2 V)$. That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby R_C , identifying S_{R_C} , B , and E . There is no relation below degree d , so all lower

neighbourhoods become constant. For a strict dominance transition, $\sum_{i,a}(\lambda'_{i,a})^2$ strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- d neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes. $\square$

Corollary 8.2 (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is $\prod_i \#\mathcal{P}(m_i, c_i)$. It is one precisely when, for every i , either $c_i = 1$ or $m_i - c_i \leq 1$. The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at $(m, c) = (4, 2)$, with partitions $(3, 1)$ and $(2, 2)$.*

Proof. Subtracting one from every part identifies $\mathcal{P}(m, c)$ with partitions of $m - c$ into at most c parts. There is one if $c = 1$ or $m - c \leq 1$. Otherwise $(m - c)$ and $(m - c - 1, 1)$ are distinct admissible partitions. The theorem gives the count and the minimal example. $\square$

8.1. The first relation as a standard graded Betti space. The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

Proposition 8.3 (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 5.3, let $S = \text{Sym } H^*$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $Q_R = S/((K_R) + S_{\geq d+1})$. Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (8.5)$$

This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

is represented by $\text{Gr}(M, S_d)$. The pencil locus is a closed copy of $\text{Gr}(2, \text{Sym}^2 V)$. Its degree- d first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.

Proof. Put $J = (K_R) + S_{\geq d+1}$ and $S_+ = \bigoplus_{j>0} S_j$. The augmented sequence $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$ gives $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$, since its map to S/S_+ is zero. Its degree- d part is K_R . The same argument over a base algebra applies to the degree- d subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree d , a rank- M subbundle in degree d , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree d . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 5.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence. $\square$

9. A TOTALLY REAL RECIPROCAL LIKELIHOOD FIBRE

The spectral data also lead to a question independent of reconstruction: can every critical point of reciprocal likelihood on a definite real pencil be made real by a choice of real data? The assertion was formulated as Conjecture 4.5 in [40]. We give an explicit construction valid for every number of distinct eigenvalues and every multiplicity pattern. Its proof counts the critical points directly; it does not assume the complex likelihood-degree formula.

A real symmetric pencil containing a positive definite matrix can be written, after a real congruence and a choice of generators, as $\langle I, \text{diag}(a_1, \dots, a_n) \rangle$. Indeed, make the definite matrix the

identity and orthogonally diagonalize the other symmetric matrix. Group its distinct eigenvalues as $\alpha_1 < \dots < \alpha_r$, with multiplicities $m_1, \dots, m_r > 0$ and $n = \sum_i m_i$. A two-dimensional pencil has $r \geq 2$. If σ_i is the sum of the data entries in the i -th eigenspace, the reciprocal score equations are the critical equations of

$$\tilde{\ell}(x, y) = -\sum_{i=1}^r m_i \log(\alpha_i x + y) - \sum_{i=1}^r \frac{\sigma_i}{\alpha_i x + y} \quad (9.1)$$

on the complement of $\prod_i (\alpha_i x + y) = 0$. The derivatives are rational and independent of the local branches of the logarithms. On the real complement one can equivalently use $\log |\alpha_i x + y|$.

Theorem 9.1 (Explicit totally real likelihood data). *Choose any real numbers*

$$\alpha_r < \beta_1 < \dots < \beta_{r-1}, \quad D(t) = \prod_{i=1}^r (t - \alpha_i), \quad Q(t) = \prod_{j=1}^{r-1} (t - \beta_j),$$

and prescribe

$$\sigma_i = \frac{Q(\alpha_i)}{D'(\alpha_i)}. \quad (9.2)$$

Then (9.1) has exactly $2r - 3$ critical points on its complex domain. Every one is real and nondegenerate. There is one critical point with $t = -y/x$ in each interval (α_i, α_{i+1}) , and one in each interval (β_j, β_{j+1}) ; there are no others. The same conclusion holds on a nonempty Euclidean open subset of the real data space. Thus the assertion of Conjecture 4.5 in [40] holds for all definite real pencils.

The data here are unrestricted real data, exactly as in that assertion. Neither positivity of the data matrix nor the assertion that all the critical points lie in the positive definite cone is part of the theorem. For example, the residues in (9.2) alternate in sign.

Lemma 9.2 (Two separated strings of roots). *With the notation of the theorem, put*

$$H(t) = \sum_{i=1}^r (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (9.3)$$

The polynomial F has degree exactly $2r - 3$. It has a simple root in each of the intervals in the theorem, has no other root over $\mathbb{C}$, and is relatively prime to DQ .

Proof. Since $\sum_i (n - m_i) = n(r - 1)$, the terms of degree $2r - 2$ in F cancel. At a zero of D ,

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i).$$

The factors $n - m_i$ are positive, all the values $Q(\alpha_i)$ have the same sign, and consecutive values $D'(\alpha_i)$ have opposite signs. Thus F changes sign in each of the $r - 1$ intervals between the α_i . At a zero of Q ,

$$F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

Here $D(\beta_j) > 0$, and consecutive values of Q' have opposite signs. There are therefore a further $r - 2$ roots in the intervals between the β_j . These intervals are pairwise disjoint and are disjoint from all the former intervals. The upper bound $2r - 3$ on the degree forces exactly one root in each interval, all simple, and excludes any remaining real or complex root. The displayed endpoint values are nonzero, proving coprimality. This argument includes $r = 2$, when the second list of intervals is empty. $\square$

Proof of Theorem 9.1. Partial fractions give

$$\frac{Q(t)}{D(t)} = \sum_i \frac{\sigma_i}{t - \alpha_i}, \quad S(t) := \sum_i \frac{\sigma_i}{\alpha_i - t} = -\frac{Q(t)}{D(t)}. \quad (9.4)$$

On the chart $x \neq 0$, put $y = -xt$. Locally, up to a constant, the function becomes

$$-n \log x - \sum_i m_i \log(\alpha_i - t) - \frac{S(t)}{x}.$$

The first critical equation gives $x = S(t)/n$. If $S(t) = 0$ it cannot be satisfied on this chart. When $S(t) \neq 0$, the remaining critical equation is

$$\left. \frac{\partial \tilde{\ell}}{\partial t} \right|_{x=S(t)/n} = - \sum_i \frac{m_i}{t - \alpha_i} - n \frac{S'(t)}{S(t)} = - \frac{F(t)}{D(t)Q(t)} = 0. \quad (9.5)$$

The lemma now gives exactly $2r - 3$ solutions in this chart, with the real coordinates

$$x = -\frac{Q(t)}{nD(t)}, \quad y = -xt. \quad (9.6)$$

At each of these points $x \neq 0$ and $\alpha_i x + y \neq 0$, so none is an extraneous solution introduced by clearing denominators.

We must also exclude the omitted chart $x = 0$. Comparing the coefficients of t^{-1} and t^{-2} in (9.4) gives

$$\sum_i \sigma_i = 1, \quad \sum_i \alpha_i \sigma_i = \sum_i \alpha_i - \sum_j \beta_j.$$

At $x = 0$, the y -equation would imply $y = 1/n$. The x -equation would then require

$$\sum_i (n - m_i) \alpha_i = n \sum_j \beta_j. \quad (9.7)$$

But the left side is strictly smaller than $n(r - 1)\alpha_r$, whereas the right side is strictly larger. Thus there is no critical point on $x = 0$. This proves the assertion about the entire complex domain, not just the chosen coordinate chart.

For nondegeneracy, at a critical point in (x, t) -coordinates one has $\tilde{\ell}_{xx} = -n/x^2 \neq 0$. The derivative of the eliminated equation (9.5) is $-F'(t)/(D(t)Q(t)) \neq 0$. It is the Schur complement of $\tilde{\ell}_{xx}$ in the Hessian. Hence the Hessian is nonsingular. The coordinate change $(x, t) \mapsto (x, -xt)$ has nonzero Jacobian at these points, so nondegeneracy also holds in the original coordinates.

Finally allow Q to have any positive leading coefficient and $r - 1$ distinct real roots to the right of α_r . These polynomials form a nonempty Euclidean open set in their r -dimensional coefficient space. The linear map $Q \mapsto (Q(\alpha_i)/D'(\alpha_i))_{i=1}^r$ is an isomorphism, by polynomial interpolation. Scaling Q does not change its critical t -values and scales (x, y) by the same nonzero factor. Thus its image is the required open set of grouped data. The map from the original n data entries to their r group sums is surjective, so its inverse image is open and nonempty as well. $\square$

Example 9.3. For $(\alpha_1, \alpha_2, \alpha_3) = (1, 2, 3)$, $m_1 = m_2 = m_3 = 1$, and $(\beta_1, \beta_2) = (4, 5)$, the data are $(6, -6, 1)$ and

$$F(t) = 15t^3 - 130t^2 + 345t - 278.$$

Its three roots lie respectively in $(1, 2)$, $(2, 3)$, and $(4, 5)$. The theorem proves the same assertion for arbitrary r , not just for examples reached by root finding.

Remark 9.4 (Relation to the intrinsic invariant). The finite failure neighbourhood recovers the complex projective pencil and its spectral sheaf. To apply the real theorem one specifies a real structure and a definite member; these are not asserted to be recoverable from an unmarked complex scheme. After that specification, the construction uses only the ordered spectral points, their multiplicities, and the freely chosen β_j . The result is an independently stated consequence in reciprocal likelihood geometry, rather than a new name for Weierstrass–Segre classification. It proves the cited conjectural assertion without claiming that every real critical point is a statistically admissible maximum.

10. THE RECONSTRUCTED CRITICAL CORRESPONDENCE

This section identifies the algebraic object joining the finite inverse theorem to the real likelihood result. The reconstruction step is a functorial consequence of recovering the pencil. The additional content is a coordinate-free data construction and a relative critical algebra whose real points split into globally labelled sections.

10.1. The universal score scheme. For matrix formulas we regard quadrics as symmetric bilinear forms on a vector space; this amounts to dualizing the vector-space convention used earlier. Write $\mathcal{W} = \text{Sym}^2 V^*$. For a regular pencil $R \subset \mathcal{W}$, let $R^\times = \{K \in R : \det K \neq 0\}$. A data matrix S belongs to the same space $\mathcal{W}$. Define a section of the trivial bundle with fibre R^* on $R^\times \times \mathcal{W}$ by

$$\eta_{K,S}(H) = -\text{tr}(K^{-1}H) + \text{tr}(SK^{-1}HK^{-1}), \quad H \in R. \quad (10.1)$$

Its zero scheme $\mathfrak{C}_R$ is the reciprocal critical correspondence, with its projection to the data space $\mathcal{W}$. Over the reals these are the critical equations of $-\log |\det K| - \text{tr}(SK^{-1})$; the scheme itself uses no choice of logarithm and is defined over the complex field.

Theorem 10.1 (Reconstruction and base change of the critical correspondence). *Let $n \geq 3$ and $d = n^2 + 2n - 4$. For every regular pencil R , the abstract scheme $Z_R^{[d]}$ determines the isomorphism class of the correspondence*

$$\mathfrak{C}_R \longrightarrow \mathcal{W}$$

up to simultaneous congruence on the pencil and data space. On the regular pencil parameter scheme the score sections define a universal zero scheme. Its pullback along any complex base change is the corresponding relative critical scheme. The classifying map of a family in the framed first-relation locus therefore recovers this relative correspondence, including over nonreduced bases. No chosen data matrix or real form is part of the unmarked conclusion.

Proof. If g is invertible, put $K' = g^\dagger K g$, $H' = g^\dagger H g$, and $S' = g^\dagger S g$. Cyclic invariance of trace gives

$$\eta_{K',S'}(H') = \eta_{K,S}(H).$$

Thus simultaneous congruence identifies the score sections and their zero ideals, not just their closed points. By Theorem 5.1, an isomorphism of the finite neighbourhoods recovers the pencil up to projective congruence. A linear lift of that projective transformation supplies the displayed isomorphism of correspondences. Different lifts differ by a common scalar change of coordinates on the pencil and data space, so they leave its isomorphism class unchanged.

On $G^{\text{reg}} \subset \text{Gr}(2, \mathcal{W})$, let $\mathcal{R}$ be the tautological pencil bundle. On the determinant open in $\text{Tot } \mathcal{R}$, inversion of the universal form is a regular operation. Formula (10.1) is a section of the pullback of $\mathcal{R}^*$ after taking the product with $\mathcal{W}$. In a frame it has two entries. Their generated ideal is independent of the frame and its quotient commutes with arbitrary base change, as does localization at the determinant. This proves the relative assertion directly, without an assertion that these score schemes are flat everywhere. Finally the closed immersion of Theorem 5.3 identifies a framed first-relation family with its classifying morphism to G . Pulling back the universal score construction proves the last assertion. It does not identify unmarked moduli stacks. $\square$

10.2. Spectral projectors and a congruence-covariant data section. Suppose now that A is real symmetric positive definite, B is real symmetric, and $R = \langle A, B \rangle$ is a pencil. Set $C = A^{-1}B$. Its distinct eigenvalues and multiplicities are

$$\alpha_1 < \cdots < \alpha_r, \quad m_1, \dots, m_r, \quad \sum_i m_i = n,$$

where $r \geq 2$. Its spectral idempotents are

$$P_i = \prod_{j \neq i} \frac{C - \alpha_j I}{\alpha_i - \alpha_j}, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_i P_i = I. \quad (10.2)$$

They are selfadjoint for A . Define the grouped data map

$$\tau_C : \mathcal{W}_{\mathbb{R}} \longrightarrow \mathbb{R}^r, \quad \tau_C(S)_i = \text{tr}(P_i A^{-1} S). \quad (10.3)$$

For $K = xB + yA$, matrix functional calculus and cyclic trace give

$$-\log |\det K| - \text{tr}(SK^{-1}) = -\log \det A - \sum_i m_i \log |\alpha_i x + y| - \sum_i \frac{\tau_C(S)_i}{\alpha_i x + y}. \quad (10.4)$$

Thus the entire score scheme, not only its critical count, depends on S through the surjective linear map τ_C .

Choose $c > 0$ and $\alpha_r < \beta_1 < \dots < \beta_{r-1}$, and set

$$\begin{aligned} D(t) &= \prod_i (t - \alpha_i), & Q(t) &= c \prod_j (t - \beta_j), \\ \sigma_i &= \frac{Q(\alpha_i)}{D'(\alpha_i)}, & S_0 &= A \sum_i \frac{\sigma_i}{m_i} P_i. \end{aligned} \quad (10.5)$$

The form S_0 is symmetric and $\tau_C(S_0) = \sigma$, since $\text{tr} P_i = m_i$. More generally the assignment $\sigma \mapsto A \sum_i (\sigma_i / m_i) P_i$ is a right inverse of τ_C ; all data with these grouped values are $S_0 + \ker \tau_C$. Under simultaneous congruence, $C \mapsto g^{-1} C g$, $P_i \mapsto g^{-1} P_i g$, and $S_0 \mapsto g^t S_0 g$. No eigenbasis has been chosen.

Theorem 10.2 (A totally real finite etale critical cover). *Fix the multiplicities $m_1, \dots, m_r$. On the locus of supplied real definite pairs (A, B) with these ordered multiplicities, there is a nonempty fibrewise open semialgebraic set of data defined by the following condition: the polynomial*

$$Q_S(t) = \sum_i \tau_C(S)_i \frac{D(t)}{t - \alpha_i} \quad (10.6)$$

has positive leading coefficient and $r - 1$ distinct real roots, all greater than α_r . On this set the reciprocal critical scheme has exactly $2r - 3$ points, all real and nondegenerate. They form globally labelled real analytic semialgebraic sections: $r - 1$ have their spectral coordinate in (α_i, α_{i+1}) , and $r - 2$ in the gaps between successive roots of Q_S .

In algebraic spectral coordinates the same family is the restriction of a finite etale algebra of rank $2r - 3$ over a specified Zariski open base. Its construction commutes with arbitrary base change on that base. The formula in (10.5) provides a congruence-covariant section of the data map, and its entire affine fibre has the same critical scheme.

Proof. Write $Q = Q_S$, and put

$$H(t) = \sum_i (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (10.7)$$

Both apparent leading terms have degree $2r - 2$ and cancel. Writing $Q = c \prod_j (t - \beta_j)$, the coefficient of t^{2r-3} is

$$a = c \left(n \sum_j \beta_j - \sum_i (n - m_i) \alpha_i \right) > 0. \quad (10.8)$$

The strict inequality follows from $\beta_j > \alpha_r$, $n - m_i > 0$, and $\sum_i (n - m_i) = n(r - 1)$. Hence F has degree exactly $2r - 3$.

At α_i and β_j one has

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i), \quad F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

The signs alternate along each string. The first string gives a root in every (α_i, α_{i+1}) , and the second gives one in every (β_j, β_{j+1}) . These $2r - 3$ disjoint intervals exhaust the degree. All roots are simple and F is coprime to DQ . This also treats $r = 2$, when the second string of intervals is empty.

We next prove the scheme statement and exclude the missing coordinate chart uniformly. Take the polynomial ring over $\mathbb{Q}$ in $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{r-1}, c$, and localize at c , the products of distinct spectral and β -differences, a , $\text{disc}_t F$, and $\text{Res}_t(F, DQ)$. Call this ring $\mathcal{B}$. Every point of the separated real chamber lies in $\text{Spec } \mathcal{B}(\mathbb{R})$, by the preceding sign argument. For a linear F , its discriminant is understood as 1.

The partial fraction identity $Q/D = \sum_i \sigma_i/(t - \alpha_i)$ shows that

$$\sum_i \sigma_i = c, \quad \sum_i \alpha_i \sigma_i = c \left(\sum_i \alpha_i - \sum_j \beta_j \right).$$

On the chart $x = 0$ of the invertible-pencil locus the score in y forces $y = c/n$. Substitution in the score in x then forces $a = 0$. Since a is inverted, the quotient of the critical coordinate ring by (x) is zero. Thus x is a unit on the whole critical scheme, including nilpotent test rings; excluding this chart is not merely a pointwise argument.

Set $t = -y/x$ and $U(t) = \sum_i \sigma_i/(\alpha_i - t) = -Q(t)/D(t)$. In these coordinates the grouped function is $-n \log x - \sum_i m_i \log(\alpha_i - t) - U(t)/x$, with the logarithms used only to write its rational differential. The score in x gives $x = U/n$; the remaining score becomes $-F/(DQ)$. Consequently the critical coordinate algebra is

$$\mathcal{B}[t]/(F/a), \quad x = -\frac{Q(t)}{nD(t)}, \quad y = \frac{tQ(t)}{nD(t)}. \quad (10.9)$$

The resultant makes D and Q units in this algebra, so these formulas and the coordinate change are genuine inverse homomorphisms. The polynomial F/a is monic of degree $2r - 3$; its discriminant is a unit. The algebra is therefore finite free of that rank, and its derivative is a unit modulo the polynomial, which proves that it is étale. These algebraic identities persist under every base change of $\mathcal{B}$. The statement is local in spectral coordinates; no global flatness of the unrestricted critical correspondence is inferred.

At a critical point the second derivative in the eliminated x direction is $-n/x^2$. The complementary Schur factor is $-F'(t)/(D(t)Q(t))$. Both are nonzero, so the Hessian is nondegenerate. The change from (x, t) to (x, y) has Jacobian determinant $-x \neq 0$, preserving this conclusion. Each specified real interval contains a unique root. The implicit function theorem therefore supplies a real analytic root function locally, and uniqueness glues these functions globally with the stated labels. Their graphs are semialgebraic, since they are specified by a polynomial equation and the interval inequalities. The ordered roots β_j of Q_S have the same local analytic and global semialgebraic property.

For fixed α , the map from the r coefficients of Q to σ is a linear isomorphism, with inverse (10.6). Polynomials with the specified simple separated roots and positive leading coefficient form a nonempty open semialgebraic coefficient set. Surjectivity and the explicit right inverse of τ_C make its preimage in the full data space nonempty and open. Spectral projectors give the same formulas along any real analytic family in the fixed multiplicity locus. Their congruence covariance was verified above; the kernel directions of τ_C do not occur in the score by (10.4). This completes all the assertions. $\square$

Corollary 10.3 (The finite-failure-to-critical construction). *An order- d finite failure invariant of a regular complex pencil, together with a supplied real definite realization and the spectral choices in (10.5), determines the above finite etale critical family up to simultaneous congruence. The all-real critical count and labelled real sections are independent of an eigenbasis. The real choices are additional inputs, not outputs of complex unmarked reconstruction.*

Proof. Apply Theorem 10.1, use the spectral sheaf to read the eigenvalues and their multiplicities in the supplied definite realization, and apply Theorem 10.2. Equations (10.2)–(10.5) verify invariance under all changes of eigenbasis and simultaneous congruence. The proof of total reality is an additional spectral argument, not a premise of the inverse theorem. $\square$

APPENDIX S1. WEB RECONSTRUCTION AND THE POLARIZED FIRST LAYER

The pencil inverse theorem uses a single irreducible coefficient module. For webs, the intrinsic normal cone instead supplies two projective components that must be recombined. This supplement keeps the full web, operator, polarized, and boundary theory. The statements below have their original hypotheses and conclusions.

S1.1. The multiplication failure scheme. Let V be a complex vector space of dimension $e \in \{3, 4\}$, put $p = e(e - 1)/2$, and let

$$R \subset \mathrm{Sym}^2 V$$

be an e -dimensional relation space. Set

$$S_R = \mathrm{Sym}^2 V / R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

with multiplication induced by the quotient $\mathrm{Sym}^2 V \twoheadrightarrow S_R$ and with $(V \oplus S_R)S_R = 0$. On

$$X_R = \mathrm{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle. For every $m \geq 2$ the multiplication map defines the same zeroth-Fitting scheme $\widehat{D}_R$ of (1.1). Its reduction is the Schubert divisor $\widehat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the invariant open G_e° used throughout, the web is basepoint-free and its Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \tag{S1.1}$$

is smooth. For $e = 4$ it is a quartic K3 surface.

S1.2. Recovering the original web. The web inverse theorem uses the full nonreduced scheme, including information at the deepest projection-corank stratum. The following conclusion holds without an additional genericity condition and is proved in Section S6.

Theorem S1.1 (Intrinsic web reconstruction). *For every pair $R, R' \in G_4^\circ$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \implies R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

Consequently the abstract failure scheme determines B_R throughout the smooth Jacobian locus and separates every finite web-to-K3 packet contained in this locus.

The proof first recovers $\mathrm{Gr}(4, S_R)$ by iterated reduced singular loci. Its nonreduced normal cone identifies the two components of

$$\bigwedge^4 \mathrm{Sym}^2 V = \mathbb{S}_{(4,3,1,0)} V \oplus \mathbb{S}_{(5,1,1,1)} V.$$

The second component is the Jacobian covariant; the first is not retained by the quartic alone. The two projective components define a line in the Pluecker space. The contraction-kernel theorem

(Lemma S6.10) computes the exact exterior support of the Jacobian component. For a smooth quartic that support has dimension ten. A secant or tangent four-vector on the Grassmannian has support at most eight. This excludes every second point and every infinitesimal ambiguity, so the line meets the Grassmannian in one reduced point. In fact three essential variables already suffice. The global tensor rulings of the normal cone exclude the transposition ambiguity of an isolated square-matrix cone. No extension of an abstract scheme isomorphism to the ambient Grassmannian is assumed.

The exceptional fibres of the two-component invariant are described in Theorem S16.1. A proper intersection of the component line with the Grassmannian has length at most two. Its second point can coincide with the first, lie at a discarded pure endpoint, or give a distinct candidate exchanged by a regular residual involution. A positive-dimensional fibre is exactly a Grassmann line specified by a three-plane inside a five-plane. The tangent condition $Qw_R \in \bigwedge^3 R \wedge W$ distinguishes the double-point case from a reduced secant pair. All these alternatives are excluded on G_4° by Theorem S6.13. The ambient obstruction is confined to the binary-Jacobian boundary, defined invariantly by the first catalecticant. Thus the smooth family has no residual secant involution, flag-line ambiguity, or ramification divisor to resolve. Outside that family the component-fibre classification is not promoted to a classification of abstract failure-scheme isomorphisms.

S1.3. The mechanism in arbitrary dimension. There is a second inverse theorem, for a family that is not confined to the two-summand plethysm in dimension four. If $\dim V = n$, polarization and the Jacobian determinant give natural maps

$$j_n : \det V \otimes \operatorname{Sym}^n V \longrightarrow \bigwedge^n \operatorname{Sym}^2 V, \quad C_n : \bigwedge^n \operatorname{Sym}^2 V \longrightarrow \det V \otimes \operatorname{Sym}^n V, \quad C_n j_n = 2 \operatorname{id}.$$

Put $Q_n = \frac{1}{2} j_n C_n$ and $P_n = 1 - Q_n$. Theorems S2.1–S2.6 prove the exact all-ranks contraction kernel, the resulting support formula, and the following consequence. For every $n \geq 4$, on the nonempty open set of relation spaces whose Jacobians have at least three essential variables, the pair $([P_n w_R], [Q_n w_R])$ determines R . The corresponding pencil intersection is a single section even after nonreduced base change. The complement $\ker C_n$ need not be irreducible.

The numerical mechanism is uniform: for $d < n$ essential variables, the exterior support is $\binom{n+1}{2} - \binom{n-d+1}{2}$, and for $d = n$ it is full except for a one-dimensional annihilator of a nondegenerate quadratic power when n is even. The resulting gap above $2n$ excludes secants and tangents. The skew-flattening support criterion itself is classical [35, 36]; the calculation here is its restriction to the specified Jacobian component, including all singular and nondegenerate contraction kernels. In dimension four the intrinsic failure scheme supplies the required component pair, thereby turning the uniform inverse mechanism into Theorem S1.1. The smooth application has four essential variables; the algebraic exclusion is stronger and already applies with three.

S1.4. Intrinsic reconstruction. On the projection-corank-at-most-one locus let D_R be the restriction of $\widehat{D}_R$, let $\mathcal{N}_R$ be its nilradical, and let E_R be the annihilator scheme of the first nilpotent layer. The first-layer reconstruction is the following result proved in Section S8.

Theorem S1.2 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

Then

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en))$$

as graded algebras. In particular, for $e = 4$ the abstract failure scheme determines the polarized quartic $K3$ $(Y_R, \mathcal{O}_{Y_R}(1))$.

APPENDIX S2. A UNIFORM CONTRACTION THEOREM AND ITS INVERSE CONSEQUENCE

The support mechanism is not confined to quartics. In this section let $\dim V = n \geq 2$, $W = \text{Sym}^2 V$, $N = \dim W = n(n+1)/2$, and $D = \det V$. Polarization defines a natural injection

$$j_n : D \otimes \text{Sym}^n V \longrightarrow \bigwedge^n W, \quad j_n(\epsilon \otimes \ell^n) = (\ell x_1) \wedge \cdots \wedge (\ell x_n), \quad \epsilon = x_1 \wedge \cdots \wedge x_n. \quad (\text{S2.1})$$

Changing the basis multiplies both sides by the same determinant. The map is nonzero and its source is irreducible, so it is injective. For $q \in W^* = \text{Sym}^2 V^*$ put $\kappa_{n,q} = \iota_q \circ j_n$. We regard q as the symmetric map $V \rightarrow V^*$; when it is invertible, $q^{-1} : V^* \rightarrow V$ defines an element of $\text{Sym}^2 V$. The pairing convention is $\langle q, xy \rangle = q(x, y)$.

Theorem S2.1 (The contraction kernel in every dimension). *For every nonzero $q \in \text{Sym}^2 V^*$, the following equalities hold:*

$$\ker \kappa_{n,q} = \begin{cases} D \otimes \text{Sym}^n(\text{rad } q), & q \text{ singular,} \\ 0, & q \text{ nondegenerate and } n \text{ odd,} \\ D \otimes \mathbb{C}(q^{-1})^{n/2}, & q \text{ nondegenerate and } n \text{ even.} \end{cases} \quad (\text{S2.2})$$

Thus for $\text{rank } q = r < n$ the kernel dimension is $\binom{2n-r-1}{n}$. For full rank it is zero or one according to the parity of n .

S2.1. Shears and the singular kernel.

Lemma S2.2 (Shear descent in a fixed weight). *Let $V = A \oplus B$, with $B \neq 0$, and let $m \geq 1$. Suppose that $N \subset D \otimes \text{Sym}^m V$ is invariant under $\text{id}_A \oplus t \text{id}_B$, $t \in \mathbb{C}^*$, and under all shears*

$$u_L(a+b) = a+b+L(a), \quad L \in \text{Hom}(A, B).$$

If $N \not\subset D \otimes \text{Sym}^m B$, then

$$N \cap (D \otimes A \otimes \text{Sym}^{m-1} B) \neq 0.$$

More precisely, a nonzero vector in the A -degree- k weight space, $k > 0$, has a nonzero image in the degree-one space under a product of $k-1$ infinitesimal rank-one shears. The same fixed nonzero vector of B can be used in every shear.

Proof. Put $s = \dim B$. The displayed torus acts on $D \otimes \text{Sym}^k A \otimes \text{Sym}^{m-k} B$ with weight $s+m-k$. These weights are distinct. Since a complex torus acts semisimply, its invariant subspace N is the direct sum of its intersections with these weight spaces. This justifies projecting a vector of N to one degree before differentiating; no cancellation between degrees is possible.

Choose a basis $a_1, \dots, a_r$ of A , a nonzero $b \in B$, and a basis vector ϵ of D . Write a nonzero degree- k vector as $\epsilon \otimes F$, where

$$F = \sum_{|\beta|=k} a^\beta f_\beta, \quad f_\beta \in \text{Sym}^{m-k} B.$$

Choose β with $f_\beta \neq 0$, and i with $\beta_i > 0$. Set $\alpha = \beta - e_i$. In $\partial_A^\alpha F$, the coefficient of a_i is exactly $\beta! f_\beta$. Indeed a monomial that contributes to this coefficient must have exponent $\alpha + e_i = \beta$. This coefficient is nonzero in characteristic zero, so the derivative is a nonzero element of $A \otimes \text{Sym}^{m-k} B$.

The shear sending a_j to $a_j + tb$ and fixing the other basis vectors has determinant one. Its infinitesimal action on $D \otimes \text{Sym}^m V$ is consequently $1_D \otimes b \partial_{a_j}$, with no extra action on D . Invariance under the shear implies invariance under its infinitesimal action: the orbit is a polynomial in t , and all its coefficients belong to the invariant linear subspace. These differential operators commute, because every ∂_{a_j} kills b . Their product with multiplicities α_j sends our vector to

$$\epsilon \otimes b^{k-1} \partial_A^\alpha F.$$

This vector belongs to N . It is nonzero, since multiplication by b^{k-1} is injective in the polynomial algebra $\text{Sym } B$. Its exact bidegree is $(1, m-1)$; thus its image in the corresponding weight-graded piece is also nonzero. The proof uses neither a special choice of b nor a restriction on r or s . When $k = 1$ the product of shears is the identity. This proves all assertions, including the low-dimensional cases. $\square$

Proof of Theorem S2.1 for singular q . Choose $V = A \oplus B$, where $B = \text{rad } q$ and $q|_A$ is nondegenerate. Pure powers from B are killed by contraction, so $D \otimes \text{Sym}^n B \subset \ker \kappa_{n,q}$. The stabilizer of q contains the B -scaling torus, all shears $a + b \mapsto a + b + L(a)$, and $O(A, q|_A) \times GL(B)$. Both the shears and the torus preserve q : adding a radical vector to either argument does not change its value. On $D \otimes \text{Sym}^n V$ the torus has the ordinary polynomial weights shifted by the constant $\dim B$, its determinant weight; no weight spaces are identified by that shift. Consequently any additional kernel vector would, by Lemma S2.2, give a nonzero kernel vector in $D \otimes A \otimes \text{Sym}^{n-1} B$. This module is irreducible for the last product group. Here the standard module of the full O_2 , not just SO_2 , is used: a reflection exchanges its two weight lines. The rank-one case is one-dimensional; in higher rank the standard orthogonal module is irreducible. The determinant character only twists this assertion.

Choose $a = x_1 \in A$ with $q(a, a) = 1$, a nonzero $b \in B$, and complete to a basis with $q(a, x_i) = 0$ for $i > 1$. Differentiate (S2.1) at $\ell = b + ta$ and contract. Only its first factor contributes in degree one in t , giving

$$n\kappa_{n,q}(\epsilon \otimes ab^{n-1}) = (bx_2) \wedge \cdots \wedge (bx_n) \neq 0. \quad (\text{S2.3})$$

Multiplication by b is injective on V , which proves the nonvanishing. The restriction to the irreducible degree-one module is therefore injective, a contradiction. This proves the singular case, including all radical dimensions. $\square$

S2.2. A cofactor identity for the scalar kernel. The exceptional even-dimensional line admits a proof that does not require a decomposition of the exterior-power target. The cofactor divergence cancellation is the exterior-valued version of the classical Piola identity, rather than a new divergence identity; see [39, equation (1.1), §3.1]. Its use below combines that cancellation with the algebraic Gaussian moment recurrence.

Lemma S2.3 (Algebraic cofactor cancellation). *For $q(x_i, x_j) = \delta_{ij}$ and $\rho = \sum_i x_i^2$, $\kappa_{n,q}(\epsilon \otimes \rho^{n/2}) = 0$ when n is even.*

Proof. Introduce indeterminates $t_1, \dots, t_n$, write $\ell(t) = \sum_i t_i x_i$, and set

$$a_i(t) = \ell(t)x_i, \quad A_i(t) = (-1)^{i-1} a_1 \wedge \cdots \wedge \widehat{a_i} \wedge \cdots \wedge a_n.$$

The symmetry $\partial_{t_i} a_j = x_i x_j = \partial_{t_j} a_i$ implies the polynomial identity

$$\sum_i \partial_{t_i} A_i = 0. \quad (\text{S2.4})$$

Indeed the term differentiating a_j in A_i cancels with the term differentiating a_i in A_j : for $i < j$, moving their common factor to the same position changes the sign by $(-1)^{j-i-1}$, opposite to their cofactor signs.

Define a linear functional $G : \mathbb{C}[t_1, \dots, t_n] \rightarrow \mathbb{C}$ by $G(t^\alpha) = 0$ if some α_i is odd, and otherwise $G(t^\alpha) = \prod_i (\alpha_i - 1)!!$, with $(-1)!! = 1$. The monomial recurrence gives $G(t_i F) = G(\partial_{t_i} F)$; this is an algebraic identity, not an appeal to convergence of a Gaussian integral. If $n = 2r$, the multinomial formula gives $G(\ell(t)^n) = (n-1)!! \rho^r$. On the other hand contraction of (S2.1) gives

$$\kappa_{n,q}(\epsilon \otimes \ell(t)^n) = \sum_i t_i A_i(t).$$

Apply G coefficientwise and use (S2.4). The result is $(n-1)!!\kappa_{n,q}(\epsilon \otimes \rho^r) = 0$, as required in characteristic zero. $\square$

Lemma S2.4 (Harmonics for the full orthogonal group). *Let $n \geq 2$, $\rho = x_1^2 + \cdots + x_n^2$, and $\mathcal{H}_a = \ker(\Delta : \text{Sym}^a \mathbb{C}^n \rightarrow \text{Sym}^{a-2} \mathbb{C}^n)$. For every $k \geq 0$,*

$$\text{Sym}^k \mathbb{C}^n = \bigoplus_{t=0}^{\lfloor k/2 \rfloor} \rho^t \mathcal{H}_{k-2t}. \quad (\text{S2.5})$$

Each $\mathcal{H}_a$ is irreducible for the full complex group O_n , and distinct nonnegative a give inequivalent modules. These statements remain true after tensoring every summand by the same determinant character. In particular they include $n = 2$.

Proof. For a harmonic homogeneous polynomial h of degree a ,

$$\Delta(\rho^t h) = 2t(2a + 2t + n - 2)\rho^{t-1}h.$$

Induction on k proves (S2.5): the formula makes Δ an isomorphism from $\rho \text{Sym}^{k-2} \mathbb{C}^n$ to $\text{Sym}^{k-2} \mathbb{C}^n$ after decomposing the latter by the induction hypothesis. Its complementary kernel is $\mathcal{H}_k$.

Here is a proof of the group assertion that also treats the disconnected case. Restriction to the real unit sphere gives $\mathcal{H}_a$ a positive definite $O(n, \mathbb{R})$ -invariant Hermitian inner product. It is positive definite because a homogeneous polynomial vanishing on that sphere vanishes on $\mathbb{R}^n$, hence identically. The stabilizer of e_1 is the full $O(n-1, \mathbb{R})$. A homogeneous polynomial of degree a fixed by this stabilizer has the form

$$\sum_{j=0}^{\lfloor a/2 \rfloor} c_j x_1^{a-2j} (x_2^2 + \cdots + x_n^2)^j.$$

Indeed, expand in powers of x_1 ; invariance makes each homogeneous coefficient constant on the real spheres in the other variables, and invariance under their simultaneous negation kills its odd degrees. The harmonic equation is the recurrence

$$2(j+1)(2j+n-1)c_{j+1} = -(a-2j)(a-2j-1)c_j.$$

All denominators are nonzero, so the fixed subspace in $\mathcal{H}_a$ is one-dimensional. This argument includes $n = 2$, when the stabilizer is $O(1) = \{1, -1\}$.

Let K_v represent evaluation at a point v of the sphere under this inner product. The vector K_{e_1} is nonzero, as witnessed by the harmonic polynomial $\frac{1}{2}((x_1 + \sqrt{-1}x_2)^a + (x_1 - \sqrt{-1}x_2)^a)$. The K_v span $\mathcal{H}_a$ and form one orthogonal-group orbit. If $M \subset \mathcal{H}_a$ is invariant, its orthogonal projection commutes with the group and sends K_{e_1} into its one-dimensional stabilizer-fixed space. Idempotence makes that scalar zero or one. The orbit spans, so the projection is zero or the identity. This proves irreducibility for the real compact group, and therefore for its containing complex orthogonal group.

Finally, writing $E = \sum_i x_i \partial_i$, the invariant operator

$$\Omega = \sum_{i < j} (x_i \partial_j - x_j \partial_i)^2 = \rho \Delta - E(E + n - 2)$$

acts on $\mathcal{H}_a$ by $-a(a+n-2)$. These eigenvalues are distinct for $a \geq 0$ and $n \geq 2$, which proves inequivalence. Tensoring by a common one-dimensional character is an invertible operation on representations. In dimension two the two nonzero SO_2 -weight lines are exchanged by a reflection; we have not asserted their irreducibility for SO_2 . For clarity, a complex subspace invariant under $O_n(\mathbb{C})$ is invariant under its compact real subgroup $O(n, \mathbb{R})$. The irreducibility just proved for that subgroup therefore gives irreducibility for the full complex group as asserted. $\square$

The Fischer decomposition and harmonic irreducibility are classical; compare [38, §1, equations (1)–(3)]. The proof above supplies the full-group and low-dimensional conventions needed here, rather than importing them from a stable-range statement.

Proof of Theorem S2.1 for nondegenerate q . Work in an orthonormal basis, and put $\mathcal{H}_m = \ker(\sum_i \partial_{x_i}^2 : \text{Sym}^m V \rightarrow \text{Sym}^{m-2} V)$. The classical harmonic decomposition is

$$D \otimes \text{Sym}^n V = \bigoplus_{t=0}^{\lfloor n/2 \rfloor} D \otimes \rho^t \mathcal{H}_{n-2t}. \quad (\text{S2.6})$$

The summands are irreducible and pairwise inequivalent for the full orthogonal group by Lemma S2.4. One can obtain the direct sum by applying the Laplacian to powers of ρ ; the nonzero coefficient on $\rho^t h_m$ is $2t(2m + 2t + n - 2)$. The harmonic summand has highest weight $(m, 0, \dots, 0)$ for $n \geq 3$. For $n = 2$ and $m > 0$ its two SO_2 -weight lines are exchanged by the other component of O_2 , so it is again irreducible. For $m = 0$ it is the scalar module. These facts also specify the determinant twists in (S2.6).

It suffices to show that every summand with $m = n - 2t > 0$ has nonzero image. Put

$$h_m = \frac{1}{2}((x_1 + \sqrt{-1}x_2)^m + (x_1 - \sqrt{-1}x_2)^m), \quad \tau = (x_1 x_2) \wedge \cdots \wedge (x_1 x_n).$$

Then $h_m \in \mathcal{H}_m$. Polarization in (S2.1) gives, for a degree- n polynomial f ,

$$[\tau] \kappa_{n,q}(\epsilon \otimes f) = [x_1^n] f - \binom{n}{2}^{-1} \sum_{i=2}^n [x_1^{n-2} x_i^2] f. \quad (\text{S2.7})$$

To check the coefficient, a contracted factor must be a square. Contracting the first column leaves τ only for x_1^n . Contracting column $i > 1$ leaves it only when that column is x_i^2 and the first is $x_1 x_i$; its wedge sign is minus. The source monomial has multinomial coefficient $\binom{n}{2}$. No other placement can leave all the factors of τ .

For $f = \rho^t h_m$, the coefficient of x_1^n is one and the sum in (S2.7) is $(n-1)t - \binom{m}{2}$. Hence

$$[\tau] \kappa_{n,q}(\epsilon \otimes \rho^t h_m) = \frac{m(n+m-2)}{n(n-1)} \neq 0 \quad (m > 0). \quad (\text{S2.8})$$

Equivariance and irreducibility imply injectivity on each such summand. Pairwise inequivalence and complete reducibility prevent cancellation between them. For odd n this exhausts the source. For even n , the remaining line is killed by Lemma S2.3. A change of basis treats every nondegenerate q , and proves the theorem. $\square$

S2.3. Exact support and reconstruction in all dimensions at least four. For $0 \neq z \in \wedge^n W$ its exterior support is the smallest subspace $S \subset W$ with $z \in \wedge^n S$. Its dimension is the rank of the first skew-flattening $c_z : W^* \rightarrow \wedge^{n-1} W$, $q \mapsto \iota_q z$. Indeed coordinates along $\ker c_z$ show that $S = (\ker c_z)^\perp$. This is the classical enclosing-space construction, not a new notion of rank [35, §10.1][36, §2.4]. The essential-variable space of a form f , in contrast, is a subspace of V , the annihilator of $\{\eta \in V^* : \partial_\eta f = 0\}$ [31].

Theorem S2.5 (Uniform support table). *Let $0 \neq f \in \text{Sym}^n V$ have d essential variables. For $0 \neq \epsilon \in D$, the exterior support of $j_n(\epsilon \otimes f)$ has dimension*

$$s_n(f) = \begin{cases} N - \binom{n-d+1}{2}, & d < n, \\ N - 1, & d = n, \text{ } n \text{ even, and } f \in \mathbb{C}^* h^{n/2} \text{ for a nondegenerate quadratic } h, \\ N, & \text{otherwise.} \end{cases} \quad (\text{S2.9})$$

For every $n \geq 4$, every form with at least three essential variables satisfies $s_n(f) > 2n$. Consequently its image is outside the closed second secant variety of $\text{Gr}(n, \text{Sym}^2 V)$.

Proof. If the essential-variable space is $H \neq V$, Theorem S2.1 says that the annihilating forms q are exactly $\text{Sym}^2(V/H)^*$. A nondegenerate annihilator would make f a power of a nondegenerate quadratic, which has essential-variable space V : differentiating that power and cancelling a nonzero polynomial proves the assertion. The first support formula follows by taking the codimension of $\text{Sym}^2(V/H)^*$ in W^* .

If $H = V$, there is no nonzero singular annihilator. The nondegenerate case of the same theorem leaves precisely the stated even-dimensional exception. The projective quadratic root of $h^{n/2}$ is unique, by factorization in a polynomial ring. Thus all nonzero annihilators in the exceptional case are proportional, and the annihilator space has dimension one.

For $3 \leq d < n$, the first formula is at least $3n - 3 > 2n$ when $n \geq 4$. For $d = n$, even its lower bound $N - 1$ is greater than $2n$. A sum of two decomposable n -vectors has support at most $2n$. The locus defined by this support bound is closed, being a minors locus, so it contains the closed second secant variety. This last containment is classical [34, §6.1][35, §10.1]. $\square$

The determinant of the gradients defines a natural Jacobian map

$$C_n : \bigwedge^n \text{Sym}^2 V \longrightarrow D \otimes \text{Sym}^n V.$$

For $q_1 \wedge \cdots \wedge q_n$, in the basis used above, it is $\epsilon \otimes \det(\partial q_i / \partial x_j)$. Evaluating on $(x_1^2) \wedge (x_1 x_2) \wedge \cdots \wedge (x_1 x_n)$ and then using pure powers proves $C_n j_n = 2 \text{id}$. Hence $Q_n = \frac{1}{2} j_n C_n$ and $P_n = 1 - Q_n$ are canonical complementary projections. There is no assumption that the complement $\ker C_n$ is irreducible.

Theorem S2.6 (A family of inverse problems beyond dimension four). *Let $n \geq 4$. On $X = \text{Gr}(n, \text{Sym}^2 V)$, let U_n be the open locus where the Jacobian has at least three essential variables. This locus is nonempty. Both $P_n w$ and $Q_n w$ are nonzero there, and*

$$\mathbb{P}(\mathbb{C}P_n w \oplus \mathbb{C}Q_n w) \cap X = \{[w]\} \quad (\text{S2.10})$$

as schemes. The analogous relative intersection over U_n is the tautological section and remains so after every base change, including nonreduced base change. In particular the pair $([P_n w], [Q_n w])$ determines $[w]$, and an element of $\text{PGL}(V)$ transporting both members of the pair transports the underlying relation space.

Proof. The first-catalecticant rank condition shows that U_n is open. For $R = \langle x_1^2, \dots, x_n^2 \rangle$, the Jacobian is $2^n x_1 \cdots x_n$, with n essential variables, so this open is nonempty in every dimension under consideration. By Theorem S2.5, $Q_n w$ has support greater than $2n$, whereas w has support n . Neither projection can therefore vanish.

Every quadratic Pluecker relation restricts on the component line as $F(aP_n w + bQ_n w) = (b - a)\ell_F(a, b)$. If the linear forms ℓ_F span a two-dimensional space, their ideal sheaf on $\mathbb{P}^1$ is the unit ideal, and the intersection is the reduced point $[1 : 1]$. If their span has dimension one, the intersection is a degree-two divisor: either it contains a second decomposable point or it is doubled at $[w]$. A second point would express $Q_n w$ as a linear combination of two decomposable vectors. A double point would put $Q_n w$ in $\widehat{T}_w X = \bigwedge^{n-1} R \wedge W$. Every vector in that affine tangent space is a sum of at most n terms, one for each omitted basis vector of R ; it is supported on R and at most n further directions. Both cases thus contradict support greater than $2n$. If all ℓ_F vanish, the entire line is contained in X , giving the same tangent contradiction. This proves (S2.10).

For the relative assertion, the tautological Pluecker line and its two projections identify the component-line bundle with $\mathbb{P}^1 \times U_n$, with tautological section $[1 : 1]$. In local frames the coefficients of the ℓ_F form a matrix with two columns. Its rank is two on every geometric fibre, so its two-by-two minors generate the unit ideal on U_n . Locally two rows are invertible; hence the

restricted ideal sheaf is exactly $(b - a)$. That calculation is preserved under every base change, with no reducedness assumption on the new base. The final equivariance statement follows from the naturality of j_n and C_n . $\square$

Example S2.7 (A nondiagonal family in every dimension). For $n \geq 4$ and $t \neq 0$, put

$$R_t = \langle x_1^2 + tx_2x_3, x_2^2, \dots, x_n^2 \rangle.$$

Its Jacobian is $2^n x_1 \cdots x_n$, by expansion along the first column of the gradient matrix. The n first partials have distinct monomial supports, so the essential-variable dimension is n . The product is squarefree, hence not a positive power of a quadratic. Theorem S2.5 gives full exterior support $n(n+1)/2 > 2n$, and Theorem S2.6 applies.

This web is not a projective change of the diagonal web. A quadric $a_1(x_1^2 + tx_2x_3) + \sum_{i>1} a_i x_i^2$ with $a_1 \neq 0$ has rank at least two: its isolated nonzero x_1^2 block and its nonzero x_2x_3 block contribute separately. For $a_1 = 0$ it has rank one exactly at the $n-1$ square points $[x_2^2], \dots, [x_n^2]$. The diagonal web has n such points, an invariant distinction. This example belongs to the uniform component-reconstruction open, not to the smooth-Jacobian locus: its Jacobian is a normal-crossing hypersurface. For $n=5, t=1$ it gives an explicit nondiagonal example with exterior support fifteen.

Remark S2.8 (What is, and is not, dimension-specific). The kernel, support, and component-pair inverse theorems hold in all specified dimensions. For $n=4$, the intrinsic normal cone of the multiplication-failure scheme supplies precisely that pair, which is the additional geometric input proved below. We do not assert that the same failure-scheme readout has been proved for $n > 4$. Thus the uniform result is an actual nonempty family of inverse problems, not a claim that every step of the four-dimensional geometric theorem automatically generalizes. The classical skew-flattening containment is used only after the restricted kernel has been calculated. A separate intrinsic extension, for a different family of actual Fitting failure schemes in every dimension, is proved in Section 3.

APPENDIX S3. A JACOBIAN IDENTITY FOR THE CONTRACTION OPERATOR

The contraction theorem admits an operator-level interpretation that also determines its singular values. This interpretation distinguishes the exterior-valued map from classical differential operators while identifying an exact composite with them. It gives a second proof of the singular-rank formula without a descent argument. The proofs in Section S2 remain valid independently.

Let $D = \det V$, $\dim V = n \geq 2$, and $W = \text{Sym}^2 V$. Use the normalization of j_n in (S2.1), and let

$$C_n : \bigwedge^n W \longrightarrow D \otimes \text{Sym}^n V, \quad C_n(w_1 \wedge \cdots \wedge w_n) = \epsilon \otimes \det(\partial_{x_j} w_i).$$

For $h \in \text{Sym}^2 V$ and $q \in \text{Sym}^2 V^*$, write $\varepsilon_h(z) = h \wedge z$, $\Delta_q = q(\partial)$, and let $\mathcal{D}_{hq}$ be the derivation of $\text{Sym} V$ induced by $hq : V \rightarrow V$. For symmetric matrices H, Q ,

$$h = x^t H x, \quad \Delta_q = \sum_{i,j} Q_{ij} \partial_{x_i} \partial_{x_j}, \quad \mathcal{D}_{hq} = \sum_{i,j} (HQ)_{ij} x_i \partial_{x_j}.$$

Off-diagonal entries are counted twice in both quadratic expressions.

Theorem S3.1 (Mixed Jacobian–contraction identity). *For arbitrary $h \in \text{Sym}^2 V$, $q \in \text{Sym}^2 V^*$, and $f \in \text{Sym}^n V$,*

$$C_n \varepsilon_h \kappa_{n,q}(\epsilon \otimes f) = \epsilon \otimes \left(\frac{4}{n} \mathcal{D}_{hq} f - \frac{2}{n(n-1)} h \Delta_q f \right). \quad (\text{S3.1})$$

No nondegeneracy hypothesis on either form is required.

Proof. It suffices to take $f = \ell^n$, since powers span $\text{Sym}^n V$. Write $\ell = a^t x$. The Jacobian matrix of the ordered quadrics $\ell x_1, \dots, \ell x_n$ is $J = \ell I + x a^t$. The contraction coefficients are the entries of $b = Qa$. Inserting h after contraction gives the sum of determinants obtained by replacing row i of J by $2x^t H$, with coefficient b_i . The contraction sign is cancelled by moving the inserted first factor back to position i . On the open set $\ell \neq 0$,

$$\det J = 2\ell^n, \quad J^{-1} = \ell^{-1}I - \frac{x a^t}{2\ell^2}.$$

The row-replacement formula evaluates that sum as

$$2\ell^n(2x^t H)J^{-1}Qa = 4\ell^{n-1}x^t H Qa - 2\ell^{n-2}(x^t H x)(a^t Qa).$$

But $\mathcal{D}_{hq}\ell^n = n\ell^{n-1}x^t H Qa$ and $\Delta_q\ell^n = n(n-1)(a^t Qa)\ell^{n-2}$. Both sides are polynomials, so the equality holds everywhere, including isotropic powers and singular matrices. $\square$

Corollary S3.2 (The singular kernel from differential identities). *If $0 < \text{rank } q = r < n$, then*

$$\ker \kappa_{n,q} = D \otimes \text{Sym}^n(\text{rad } q), \quad \dim \ker \kappa_{n,q} = \binom{2n-r-1}{n}.$$

Proof. Suppose $\kappa_{n,q}(\epsilon \otimes f) = 0$. Choose $0 \neq b \in \text{rad } q$ and take $h = b^2$ in (S3.1). Since $hq = 0$, we obtain $b^2\Delta_q f = 0$, hence $\Delta_q f = 0$. Thus $\mathcal{D}_{hq}f = 0$ for every h . Choose coordinates with $q = \text{diag}(I_r, 0)$, and take $h = x_i^2$ for $i \leq r$. Then $x_i \partial_{x_i} f = 0$, so f involves only radical variables. Conversely, $j_n(\epsilon \otimes b^n)$ is killed by contraction for $b \in \text{rad } q$. Polarization proves the reverse inclusion, and counting monomials gives the dimension. The differential argument uses characteristic zero; the diagonalization is over $\mathbb{C}$. $\square$

S3.1. The angular Casimir and exact singular values. For nondegenerate q , set $\rho = q^{-1}$, $E = \sum x_i \partial_{x_i}$, and

$$\mathcal{C}_q = E(E + n - 2) - \rho \Delta_q.$$

This is the angular Casimir with nonnegative harmonic eigenvalues. The harmonic decomposition is classical; see [42, Theorem 9 and §4(a)]. The following identity identifies the particular exterior operator:

$$C_n \varepsilon_\rho \kappa_{n,q} = \frac{2}{n(n-1)}(1_D \otimes \mathcal{C}_q) \quad \text{on } D \otimes \text{Sym}^n V. \quad (\text{S3.2})$$

Indeed $\mathcal{D}_{\rho q} = E$, and $E = n$ on this degree.

For the metric statement, take $q = I$ in a specified real orthonormal frame and complexify. Equip the tensor, symmetric and exterior powers with their induced positive Hermitian products, and D with unit volume. In particular,

$$\|x^\alpha\|^2 = \frac{\alpha!}{k!} \quad (|\alpha| = k), \quad \|x_i x_j\|^2 = \frac{1}{2} \quad (i \neq j). \quad (\text{S3.3})$$

This is the degree-normalized Fischer product. Compare the unnormalized convention in [38, Definition 6.1 and Lemma 6.2].

Lemma S3.3 (Adjoint normalization). *For these products, $j_n^* = 2^{-n}C_n$ and $\iota_q^* = \varepsilon_\rho$. Consequently,*

$$\kappa_{n,q}^* \kappa_{n,q} = \frac{2^{1-n}}{n(n-1)}(1_D \otimes \mathcal{C}_q). \quad (\text{S3.4})$$

Proof. The Riesz representative of q in W is ρ , so the exterior adjoint formula gives the second assertion. For the first, test against $\epsilon \otimes \ell^n$ and a wedge $w_1 \wedge \dots \wedge w_n$. The quadratic product pairs ℓx_i with w_j as one half of the corresponding first derivative evaluated at ℓ , with the usual Hermitian conjugation. The determinant contributes 2^{-n} . The normalized symmetric

product pairs ℓ^n with a polynomial by evaluation. Thus the determinant of derivatives is the required adjoint. Powers and decomposable wedges span. Composition with (S3.2) proves the last formula. $\square$

Theorem S3.4 (Singular-value decomposition and a sharp inverse bound). *Under the preceding normalization, the squared singular value on $D \otimes \rho^{(n-m)/2} \mathcal{H}_m$, for $m = n, n-2, \dots$, is*

$$\sigma_m^2 = \frac{2^{1-n} m(m+n-2)}{n(n-1)}. \quad (\text{S3.5})$$

Its multiplicity is $\binom{n+m-1}{m} - \binom{n+m-3}{m-2}$, with the second term zero for $m < 2$. A zero singular value occurs precisely when n is even, with eigenspace $D \otimes \mathbb{C}\rho^{n/2}$. On the orthogonal complement of the kernel,

$$\text{cond}(\kappa_{n,q}) = \begin{cases} \sqrt{2n}, & n \text{ odd}, \\ \sqrt{n-1}, & n \text{ even}. \end{cases} \quad (\text{S3.6})$$

Let $\mathcal{C}_q^\dagger$ be its inverse on the positive harmonic summands and zero on the scalar summand. Then

$$\kappa_{n,q}^\dagger = \frac{n(n-1)}{2} (1_D \otimes \mathcal{C}_q^\dagger) C_n \varepsilon_\rho. \quad (\text{S3.7})$$

Proof. For $h_m \in \mathcal{H}_m$, differentiation gives $\Delta_q(\rho^t h_m) = 2t(2m+2t+n-2)\rho^{t-1} h_m$. With $m+2t = n$, the Casimir eigenvalue is $m(m+n-2)$. Equation (S3.4) proves the formula and kills precisely the scalar line. The dimension follows from the surjectivity of the Laplacian in the harmonic decomposition. The largest Casimir eigenvalue is $2n(n-1)$; the smallest positive one is $n-1$ in odd degree and $2n$ in even degree. Their ratios give the condition numbers, attained on the nonzero endpoint harmonic summands. Finally use $A^\dagger = (A^* A)^\dagger A^*$ and $\kappa_{n,q}^* = 2^{-n} C_n \varepsilon_\rho$. $\square$

The kernel statements, unlike the metric assertions, are invariant under arbitrary complex changes of basis. The two arguments above therefore independently prove Theorem S2.1. The condition number refers to the specified compact real form and induced norms; it is not invariant under $O(n, \mathbb{C})$, and no bound for arbitrary ill-conditioned coordinate changes is asserted. The mixed identity explains why comparison of codomains alone was insufficient: the exterior operator is not the apolar Laplacian, but its specified Jacobian composite is exactly a classical angular operator.

APPENDIX S4. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma S4.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \text{Sym}^2 \mathcal{H} \longrightarrow \text{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\text{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\text{Sym}^2 V = \text{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \text{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition S4.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (S1.1). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (\text{S4.1})$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e - 1$ and is the whole target. Intersecting $e - 1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma S4.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = \mathbb{C}\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition S4.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension $p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$

when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma S4.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

APPENDIX S5. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition S5.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\widehat{D}} = (\det M)I_p(\gamma \mathrm{Sym}^2 M). \quad (\text{S5.1})$$

The reduction of $\widehat{D}$ is the integral divisor $\widehat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\widehat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \mathrm{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (S5.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\mathrm{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \mathrm{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \mathrm{Tot}_{P \times \mathbb{P}(S)} \mathrm{Hom}(\mathcal{H}, S/\Lambda), \quad (\text{S5.2})$$

$$E \simeq \mathrm{Tot}_{Y \times \mathbb{P}(S)} \mathrm{Hom}(\mathcal{H}, S/\Lambda). \quad (\text{S5.3})$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem S5.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (\text{S5.4})$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \text{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\text{Sym}^2 H}$. Equation (S4.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (S5.1) is (t) . On Y , the restriction has rank $p - 1$ by Proposition S4.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (\text{S5.5})$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (S5.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma S5.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\text{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \longrightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \text{Spec } \mathcal{O}_{Z, \eta} \longrightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

APPENDIX S6. THE DEEPEST NORMAL CONE AND RECONSTRUCTION OF THE WEB

The polarized Jacobian retains only one of two irreducible components of the Pluecker vector of a web. The full failure scheme retains both. In this section we recover these two projective components intrinsically and prove that they determine every web in the smooth Jacobian open. Exterior support excludes every exceptional pencil in this geometric family. Thus the deeper boundary separates the finite ambiguity of the polarized-K3 invariant. The normal cone used here is along the deepest Schubert stratum, not along the reduction of the failure scheme.

Throughout this section $\dim V = 4$, $W = \text{Sym}^2 V$, and $E = \bigwedge^4 W$. There is a multiplicity-free decomposition

$$E = E_{175} \oplus E_{35}, \quad E_{175} = \mathbb{S}_{(4,3,1,0)} V, \quad E_{35} = \mathbb{S}_{(5,1,1,1)} V \simeq \det V \otimes \text{Sym}^4 V. \quad (\text{S6.1})$$

Write P and Q for its two equivariant projections, in that order. For a web R , write $[w_R] \in \mathbb{P}(E)$ for its Pluecker point. On the locus where both projections are nonzero, put

$$\mathcal{L}_R = \mathbb{P}(\mathbb{C}Pw_R \oplus \mathbb{C}Qw_R).$$

The quadratic ideal of $\text{Gr}(4, W)$ is denoted by $\mathcal{I}_{\text{Pl}, 2}$. Each of its quadrics satisfies

$$F(aPw_R + bQw_R) = (b - a)\ell_{F, R}(a, b), \quad F \in \mathcal{I}_{\text{Pl}, 2}, \quad (\text{S6.2})$$

for a linear form $\ell_{F,R}$, since w_R is decomposable. Put $X = \text{Gr}(4, W)$, and let $X^\times$ be the open where both projections are nonzero. Set $H_w = \langle \ell_{F,w} : F \in \mathcal{I}_{\text{Pl},2} \rangle$, and let $\pi : X^\times \rightarrow \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35})$ send $[w]$ to its pair of component lines. Define $G_4^{\text{rec}} \subset G_4^\circ$ by the conditions

$$Pw_R \neq 0, \quad Qw_R \neq 0, \quad \dim \langle \ell_{F,R} : F \in \mathcal{I}_{\text{Pl},2} \rangle = 2. \quad (\text{S6.3})$$

These conditions are independent of a representative of $[w_R]$ and invariant under $\text{PGL}(V)$.

Theorem S6.1 (Intrinsic reconstruction on the smooth Jacobian locus). *The reconstruction open is the whole smooth geometric locus: $G_4^{\text{rec}} = G_4^\circ$. Let $R \in G_4^\circ$ and $R' \in G_4^\circ$. Every isomorphism of complex schemes*

$$\hat{D}_R \simeq \hat{D}_{R'}$$

implies $R' = gR$ for some $g \in \text{PGL}(V)$. In particular, the abstract nonreduced failure scheme determines the cube-zero algebra B_R , up to isomorphism, throughout G_4° .

More precisely, the normal cone of $\hat{D}_R$ along its intrinsically determined deepest Schubert stratum recovers $([Pw_R], [Qw_R])$, up to the same projective change of coordinates in V . The webs with this pair of projective components are exactly

$$(\mathcal{L}_R \cap \text{Gr}(4, W)) \setminus \{[Pw_R], [Qw_R]\}. \quad (\text{S6.4})$$

On G_4^{rec} , this intersection is the single reduced point $[w_R]$, before taking projective-coordinate orbits.

S6.1. The intrinsic deepest stratum and its oriented tensor cone. For clarity, set

$$\Sigma_r(R) = \{K \in \text{Gr}(4, V \oplus S_R) : \text{rank}(K \rightarrow V) \leq r\}, \quad 0 \leq r \leq 3,$$

with reduced structure. Thus $\Sigma_3(R) = \hat{\Delta}_R$ and $\Sigma_0(R) = \text{Gr}(4, S_R)$.

Lemma S6.2 (The deepest normal cone). *Starting with the reduction of $\hat{D}_R$, three iterations of taking the reduced singular locus give $\Sigma_0(R)$. Consequently $\Sigma_0(R)$ and $\mathcal{C}_R = C_{\Sigma_0(R)/\hat{D}_R}$ are intrinsic. Let $\mathcal{K}_0$ denote the tautological rank-four bundle on $\Sigma_0(R)$. The ambient vector bundle determined by the degree-one part is $\text{Hom}(\mathcal{K}_0, V)$. At a point $K_0 \subset S_R$, choose bases and write $T : K_0 \rightarrow V$. The fibre cone is*

$$\text{Spec } \mathbb{C}[t_{ij}] / (d(T)J_R(T)), \quad d(T) = \det T, \quad J_R(T) = I_6(\gamma_R \text{Sym}^2 T). \quad (\text{S6.5})$$

Its ideal is generated in degree sixteen and its reduction is the determinant hypersurface. The two rulings of its rank-one locus are intrinsically distinguishable over $\Sigma_0(R)$: the V -ruling is the projectively trivial one.

Proof. At a point of projection rank r , invert an r -minor and take the Schur complement. The remaining $(4-r)^2$ entries are independent coordinates and the rank conditions become the usual matrix rank conditions, times smooth auxiliary coordinates. The Jacobian criterion for minors gives

$$(\text{Sing } \Sigma_r(R))_{\text{red}} = \Sigma_{r-1}(R), \quad 1 \leq r \leq 3.$$

This also follows directly by differentiating the minors: the rank- r stratum is smooth, whereas at rank at most $r-1$ the tangent dimension exceeds the local dimension. Since reduction and reduced singular locus are preserved by abstract isomorphisms, $\Sigma_0(R)$ is intrinsic.

Choose a splitting $S_R = K_0 \oplus S'$. A Grassmann chart near K_0 consists of graphs

$$(T, B) : K_0 \longrightarrow V \oplus S'.$$

The deepest stratum is $T = 0$; B provides its eight smooth coordinates. The Fitting factorization gives precisely $d(T)J_R(T)$, independently of B . All generators of J_R have degree twelve in T , so passing to the (t_{ij}) -adic associated graded gives the same homogeneous ideal. This proves (S6.5). There are no degree-one relations. The dual of the conormal bundle is therefore $\text{Hom}(\mathcal{K}_0, V)$, as

also follows by quotienting the Grassmannian tangent bundle along $\text{Gr}(4, S_R)$. If T is invertible, $\gamma_R \text{Sym}^2 T$ is onto, so its maximal minors do not vanish simultaneously. Thus $V(dJ_R) = V(d)$; the determinant is irreducible, and the Nullstellensatz gives $\sqrt{dJ_R} = (d)$. This argument does not require any higher-colon or primary-boundary calculation.

The rank-one subcone is recovered from the reduced determinant cone by iterated reduced singular loci in each fibre. Its projectivization is the relative Segre variety

$$\mathbb{P}(V) \times \mathbb{P}(\mathcal{K}_0^*) \longrightarrow \Sigma_0(R).$$

The two components of the relative parameter space of its maximal linear $\mathbb{P}^3$'s are $\mathbb{P}(V) \times \Sigma_0(R)$ and $\mathbb{P}(\mathcal{K}_0^*)$. Their identification as schemes is Lemma 2.1 below. On points, note that a vector space of rank-one tensors cannot contain both $v_1 \otimes u_1$ and $v_2 \otimes u_2$ with both pairs independent: their sum has rank two. Thus a maximal such space fixes one tensor factor.

The first parameter bundle is projectively trivial. The second is not. Restrict to a Schubert line in $\text{Gr}(4, 6)$, obtained by varying K_0 between a fixed three-plane and five-plane. On this line

$$\mathcal{K}_0^*|_{\mathbb{P}^1} \simeq \mathcal{O}_{\mathbb{P}^1}^{\oplus 3} \oplus \mathcal{O}_{\mathbb{P}^1}(1).$$

If its projectivization were a trivial $\mathbb{P}^3$ -bundle, the vector bundle would be a direct sum of four copies of one line bundle. This follows by comparing the relative $\mathcal{O}(1)$ on a projective-bundle isomorphism and pushing forward. Its determinant degree would then be divisible by four, contrary to the degree one displayed above. Triviality is invariant under an isomorphism of the base. Since the base is connected, an isomorphism of the cone cannot interchange these two parameter bundles on some fibres but not others. It must preserve the V -ruling. $\square$

Remark S6.3 (Why the ruling argument is needed). A single determinant cone of square matrices permits transposition. The preceding relative argument removes that ambiguity using only intrinsic data on $\Sigma_0(R)$; it does not assume that an isomorphism of failure schemes extends to the ambient Grassmannian. A linear isomorphism of tensor spaces preserving the two rank-one rulings has the form $g \otimes h$. Indeed, its restrictions to the two Segre factors are projective linear, and after lifting those maps the remaining linear map fixes every decomposable line. Applying it to sums with one common factor shows that its scalar is constant. Thus the coordinate change in the recovered V factor is an ordinary projective linear change, as required below.

The lemma applies after pulling the target bundles back along the intrinsic base isomorphism $\sigma : \Sigma_0(R) \rightarrow \Sigma_0(R')$. Here the orientation is supplied by Lemma S6.2; it is not an extra marking of the failure scheme. Nor is a canonical basis of the trivial ruling required: different choices alter its representative only by constant projective coordinate changes.

S6.2. The intrinsic residual ideal.

Lemma S6.4 (Residual graded data without a determinant equation). *On $B = \Sigma_0(R)$, put $\mathcal{A} = \text{Sym}(V^* \otimes \mathcal{K}_0)$, and let $\mathcal{I} \subset \mathcal{A}$ be the homogeneous ideal of $\mathcal{C}_R$ in its intrinsic degree-one ambient bundle. Let $\mathcal{D} \subset \mathcal{A}$ be the ideal of the reduced cone. Its degree-four part is the line bundle*

$$L = \mathcal{D}_4 = (\det V)^* \otimes \det \mathcal{K}_0 \subset \mathcal{A}_4, \quad \mathcal{D} = L \otimes \mathcal{A}(-4). \quad (\text{S6.6})$$

The graded ideal

$$\mathcal{J} = (\mathcal{I} : \mathcal{D}) = \{a \in \mathcal{A} : a\mathcal{D} \subset \mathcal{I}\} \quad (\text{S6.7})$$

is intrinsic, and its degree-twelve part is the maximal-minor coefficient space used in the Schur readout. Multiplication gives canonical isomorphisms

$$L \otimes \mathcal{J}_{12} \xrightarrow{\sim} \mathcal{I}_{16}, \quad \mathcal{J}_{12} \simeq L^* \otimes \mathcal{I}_{16}. \quad (\text{S6.8})$$

The construction is preserved by graded cone isomorphisms. Its local residual-ideal calculation also commutes with arbitrary base change when $\mathcal{D}$ is transported as the determinant ideal.

Proof. The graded cone algebra determines $\mathcal{A}$ and its homogeneous kernel $\mathcal{I}$. Its reduction determines $\mathcal{D}$. In frames for V and $\mathcal{K}_0$, Lemma S6.2 says $\mathcal{D} = (d)$ and $\mathcal{I} = dJ$, where $d = \det T$ and $J = I_6(\gamma_R \text{Sym}^2 T)$. The determinant coefficient spans $\Lambda^4 V^* \otimes \Lambda^4 \mathcal{K}_0$ inside $\text{Sym}^4(V^* \otimes \mathcal{K}_0)$, proving (S6.6) with its precise twist.

Multiplication by the generic determinant is injective over any coefficient ring. For example, choose a diagonal term order; its leading coefficient is a unit, so the leading term of df cannot vanish for any nonzero polynomial f . Therefore $(dJ : d) = J$. The ideal J is generated in degree twelve, which gives (S6.8). If a frame changes d to ud , with u a unit, then $(dJ : d) = (dJ : ud)$. The local residual ideals hence glue in $\mathcal{A}$; the inverse line-bundle twist in (S6.8) is exactly what cancels the transition functions of the determinant generators.

Every graded isomorphism transports $\mathcal{I}$, $\mathcal{D}$, and their colon, so it preserves this operation before any scalar equation has been chosen. After arbitrary base change the same identity $I_T = d_T J_T$ holds for the image ideals in the new polynomial algebra. The determinant remains a nonzerodivisor there, so $(I_T : d_T) = J_T$ again. No general claim that colons commute with base change is being used: it is the factorization and this universal nonzerodivisor that prove it here. On a nonreduced new base one transports the determinant ideal; one does not recompute the absolute reduction of the total cone, which need not commute with base change. $\square$

S6.3. The universal Schur coefficient map. The distinction between an isotypic component and a specified line in its multiplicity space is essential here. We establish the coefficient identity before fixing the web. All vector spaces in this subsection are complex and four-dimensional. Put

$$\mathcal{F}(X) = \bigwedge^6 \text{Sym}^2 X, \quad \Lambda = \{(5, 4, 2, 1), (4, 4, 4, 0)\}.$$

For a linear map T write $\mathcal{F}(T) = \bigwedge^6 \text{Sym}^2 T$.

Lemma S6.5 (Universal coefficient identity). *The natural degree-twelve map*

$$\begin{aligned} \Phi_{V,U} : \mathcal{F}(V)^* \otimes \mathcal{F}(U) &\longrightarrow \text{Sym}^{12}(V^* \otimes U), \\ \Phi_{V,U}(\beta \otimes \xi)(T) &= \beta(\mathcal{F}(T)\xi) \end{aligned} \tag{S6.9}$$

is $GL(V) \times GL(U)$ -equivariant. Under the functorial multiplicity-free decompositions

$$\mathcal{F}(X) = \mathbb{S}_{(5,4,2,1)} X \oplus \mathbb{S}_{(4,4,4,0)} X \tag{S6.10}$$

and the Cauchy decomposition

$$\text{Sym}^{12}(V^* \otimes U) = \bigoplus_{\substack{\lambda \vdash 12 \\ \ell(\lambda) \leq 4}} \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U,$$

the restriction of $\Phi_{V,U}$ to $\mathbb{S}_\lambda V^ \otimes \mathbb{S}_\mu U$ is zero if $\lambda \neq \mu$. For $\lambda = \mu \in \Lambda$, it is a nonzero scalar multiple of the canonical Cauchy inclusion. With compatible functorial normalizations this scalar is one. Consequently, for every fixed $\beta = \sum_\lambda \beta_\lambda$,*

$$\Phi_{V,U}(\mathbb{C}\beta \otimes \mathcal{F}(U)) = \bigoplus_{\lambda \in \Lambda} \mathbb{C}\beta_\lambda \otimes \mathbb{S}_\lambda U, \tag{S6.11}$$

where each tensor product on the right is placed in its Cauchy summand. A zero component contributes the zero subspace.

Proof. The action on matrices is $T \mapsto gTh^{-1}$; the action on functions is $f(T) \mapsto f(g^{-1}Th)$. Functoriality of $\mathcal{F}$ shows directly that (S6.9) intertwines this action with $\beta \otimes \xi \mapsto (\mathcal{F}(g^{-1})^* \beta) \otimes \mathcal{F}(h)\xi$. Thus equivariance is a statement about the universal map, not about a fixed- β slice.

For completeness, the exterior character identity used here is

$$[z^6] \prod_{1 \leq i \leq j \leq 4} (1 + zx_i x_j) = s_{(5,4,2,1)}(x) + s_{(4,4,4,0)}(x).$$

Equivalently, it follows by exterior duality from (S6.1) and $\det(\mathrm{Sym}^2 X) = (\det X)^5$. The two dimensions are 175 and 35, whose sum is $\binom{10}{6}$. The characteristic-zero Schur decomposition is functorial; the Cauchy formula is multiplicity-free as a representation of the product group [18, 8]. The source of Φ is the direct sum of the four irreducible product representations indexed by $\Lambda \times \Lambda$. Only the two diagonal ones occur in the target. Schur's lemma kills the off-diagonal blocks and makes each diagonal block a scalar multiple of its Cauchy inclusion. This also proves that no other left multiplicity line can enter the formula.

Here is an explicit nonvanishing check for both scalars. For an ordered basis $x_1, \dots, x_4$, order the monomials of $\mathrm{Sym}^2 X$ as

$$(b_1, \dots, b_{10}) = (x_1^2, x_1 x_2, x_1 x_3, x_1 x_4, x_2^2, x_2 x_3, x_2 x_4, x_3^2, x_3 x_4, x_4^2).$$

Highest-weight vectors for the two summands are

$$\xi_{5421} = b_1 \wedge b_2 \wedge b_3 \wedge b_4 \wedge b_5 \wedge b_6, \quad \xi_{4440} = b_1 \wedge b_2 \wedge b_3 \wedge b_5 \wedge b_6 \wedge b_8.$$

Their weights are respectively $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Every raising operator $x_i \partial / \partial x_j$, $i < j$, either gives zero or repeats a wedge factor, so both are highest vectors. Complete reducibility and the dimension sum above show that these two generated irreducible submodules exhaust $\mathcal{F}(X)$; this also proves (S6.10) directly. Choose β_λ in the dual summand with $\beta_\lambda(\xi_\lambda) = 1$, and extend it by zero on the other summand. After identifying the ordered bases of U and V , evaluation at $T = \mathrm{diag}(t_1, t_2, t_3, t_4)$ gives

$$\begin{aligned} \Phi(\beta_{5421} \otimes \xi_{5421})(T) &= t_1^5 t_2^4 t_3^2 t_4, \\ \Phi(\beta_{4440} \otimes \xi_{4440})(T) &= t_1^4 t_2^4 t_3^4. \end{aligned} \tag{S6.12}$$

In particular both evaluations are one at the identity. This proves nonvanishing without a computed rank or a choice of numerical web. The canonical coefficient realization of the Cauchy inclusion is $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$. Compatible functorial splittings identify the restriction of $\mathcal{F}(T)$ with $\mathbb{S}_\lambda(T)$, giving scalar one; other scalar normalizations have the same image. Finally let ξ vary independently in its two summands. The off-diagonal vanishing and the nonzero diagonal scalars give (S6.11). $\square$

Lemma S6.6 (Exterior duality and the common determinant twist). *Set $W = \mathrm{Sym}^2 V$. The canonical isomorphism*

$$\begin{aligned} \bigwedge^4 W \otimes (\det W)^* &\longrightarrow \bigwedge^6 W^*, \\ w \otimes \omega &\longmapsto [\eta \mapsto \omega(w \wedge \eta)] \end{aligned} \tag{S6.13}$$

sends the Pluecker line of $R = \ker \gamma_R$ to the line of $\beta_R = \gamma_R^(\mathrm{vol}_{S_R})$. It identifies $E_{175} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(5,4,2,1)} V^*$, and $E_{35} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(4,4,4,0)} V^*$.*

Proof. Extend a basis of R to a basis of W . Both sides of the first assertion are nonzero alternating six-forms annihilating a wedge containing a vector of R , and on the six complementary basis vectors they agree after one scalar normalization. This proves the line identity. The map in (S6.13) is natural for all isomorphisms of W . The highest weights of the two twisted summands are $(-1, -2, -4, -5)$ and $(0, -4, -4, -4)$, respectively: these are the negatives of the reversed partitions $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Thus both labels and the common twist are as stated. Changing the volume multiplies both components by the same scalar. $\square$

S6.4. Reading both Pluecker components from the residual ideal.

Lemma S6.7 (The two-component readout). *At any fibre of the oriented cone in Lemma S6.2, its degree-sixteen ideal recovers the projective pair $([Pw_R], [Qw_R])$ whenever both components are nonzero. It detects the vanishing of either component as well.*

Proof. Lemma S6.4 recovers the residual ideal as $\mathcal{J} = (\mathcal{I} : \mathcal{D})$, and its canonical degree-twelve coefficient module as $\mathcal{J}_{12} \simeq L^* \otimes \mathcal{I}_{16}$. In a fibre this is exactly $(J_R)_{12}$. This uses the nonreduced ideal, not just its support, and is independent of a local generator of the determinant line.

Let $U = K_0$ and choose a volume on S_R . Its pullback $\beta_R \in \wedge^6(\text{Sym}^2 V)^*$ pairs with $\mathcal{F}(T)$ to give exactly the maximal minors of $\gamma_R \text{Sym}^2 T$. Lemma S6.5 now gives, as an equality of subspaces in the Cauchy decomposition,

$$(J_R)_{12} = \bigoplus_{\lambda \in \Lambda} \mathbb{C} \beta_{R,\lambda} \otimes S_\lambda U. \quad (\text{S6.14})$$

This is the fixed-slice consequence of the universal block identity. It does not treat the fixed slice as a $GL(V)$ -representation. Intersecting with a Cauchy summand and contracting its right factor with all elements of $(S_\lambda U)^*$ recovers exactly $\mathbb{C} \beta_{R,\lambda}$, including its possible vanishing. Lemma S6.6 identifies these two lines with $[Pw_R]$ and $[Qw_R]$, with their labels and one common determinant twist. Right coordinate changes preserve the left lines; left coordinate changes act on both by the same representation of the coordinate transformation. Naturality under an abstract isomorphism is isolated in Lemma S6.8 below. $\square$

S6.5. Naturality under an abstract isomorphism.

Lemma S6.8 (One projective transformation for both coefficient lines). *Let $\varphi : \widehat{D}_R \xrightarrow{\sim} \widehat{D}_{R'}$ be an abstract isomorphism, with $R, R' \in G_4^\circ$. It induces an isomorphism $\sigma : \Sigma_0(R) \rightarrow \Sigma_0(R')$ and a grading-preserving isomorphism of their normal cones. There is one $g \in \text{PGL}(V)$, independent of the point of $\Sigma_0(R)$, which transports both recovered nonzero Pluecker component lines:*

$$[Pw_{R'}] = [P((\bigwedge^4 \text{Sym}^2 g)w_R)], \quad [Qw_{R'}] = [Q((\bigwedge^4 \text{Sym}^2 g)w_R)].$$

The corresponding vanishing statements hold for zero components. No extension of φ to either ambient Grassmannian is required.

Proof. The deepest stratum is intrinsic by Lemma S6.2. If $\mathfrak{b}$ and $\mathfrak{b}'$ are its ideals in the two failure schemes, pullback by φ identifies every power of these ideals. Hence it induces the graded algebra isomorphism

$$\sigma^* \left(\bigoplus_{m \geq 0} \mathfrak{b}'^m / \mathfrak{b}'^{m+1} \right) \simeq \bigoplus_{m \geq 0} \mathfrak{b}^m / \mathfrak{b}^{m+1}. \quad (\text{S6.15})$$

The symmetric algebra of degree one surjects canonically onto either algebra. Thus (S6.15) also identifies its homogeneous kernel, not merely the cone's reduction; this is the functorial normal-cone construction [19]. Put $\mathcal{G} = \text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$, where $\mathfrak{b}$ is the deepest-stratum ideal. Its degree-one part is the conormal bundle $\mathcal{G}_1 = V^* \otimes \mathcal{K}_0$; its dual, not $\mathcal{G}_1$ itself, is $\text{Hom}(\mathcal{K}_0, V)$. Thus the vector bundle is recovered before choosing either tensor factor.

Here is the whole construction in one diagram. The arrows $\rightsquigarrow$ mean assignments functorial under abstract isomorphisms, not morphisms or chosen retractions between the displayed schemes. Write $\mathcal{A} = \text{Sym } \mathcal{G}_1$, with homogeneous kernel $\mathcal{I}$, and $\mathcal{D} = \sqrt{\mathcal{I}}$.

$$\begin{aligned} \widehat{D}_R &\rightsquigarrow B = \Sigma_0(R) \rightsquigarrow \mathcal{G} \rightsquigarrow (\mathcal{G}_1, \mathcal{A}, \mathcal{I}, \mathcal{D}), \\ \text{Spec}_B(\mathcal{A}/\mathcal{D}) &\rightsquigarrow \{\text{rank} \leq 1\} \rightsquigarrow \text{Fano}_4 \rightsquigarrow \mathbb{P}(V) \times B, \\ (\mathcal{I}, \mathcal{D}) &\rightsquigarrow L = \mathcal{D}_4 \rightsquigarrow (\mathcal{I} : \mathcal{D})_{12} = L^* \otimes \mathcal{I}_{16} \rightsquigarrow \{\mathbb{C} \beta_{R,\lambda}\}_{\lambda \in \Lambda} \rightsquigarrow ([Pw_R], [Qw_R]). \end{aligned} \quad (\text{S6.16})$$

The first row uses the reduced-singular-locus construction and powers of its ideal. The second uses reduced rank strata, the closed relative Fano scheme of Lemma 2.1, and its uniquely projectively trivial component. The third uses the intrinsic colon, Cauchy projection and contraction of the entire right factor, followed by exterior duality. It involves no selected generator of L . Every row is transported by φ ; comparison of the two trivial ruling bundles, after pullback along σ , is what produces $[g]$. There is no coordinate map $[g]$ attached to an individual scheme before comparing its projective factors.

The relative rank-one Segre variety and its two ruling parameter bundles are preserved. Their distinction by projective triviality was proved in Lemma S6.2; in particular the V -ruling cannot be exchanged with the other ruling. Lemma 2.2, applied after pullback by σ , gives a regular isomorphism of the trivial $\mathbb{P}(V)$ -bundles. It explicitly accounts for the line-bundle ambiguity of its local linear lifts. Since $\Sigma_0(R)$ has only constant global regular functions, the resulting $\mathrm{PGL}(V)$ -valued morphism is constant. Denote its value by g .

Fix a point $U = K_0$ and its image $U' = K'_0$. A linear isomorphism preserving both rank-one rulings has, after a lift of g , the form $\alpha(T) = gTh^{-1}$ for some isomorphism $h : U \rightarrow U'$. Indeed the two Segre factors give projective linear maps. After lifting them, the remaining linear map fixes each decomposable line; sums of tensors with a common factor force its scalar to be constant. This scalar can be absorbed in h .

The only frame-dependent display is the following naturality square; all its objects were already recovered in (S6.16). A change of a local lift g_i to $u_{ij}g_i$ is a line-bundle transition, not another projective coordinate map. A homogeneous degree-twelve coefficient functor acquires the common factor u_{ij}^{12} (or its inverse on a dual). It therefore preserves the recovered projective lines. The right-frame action is shown on the right tensor factor in the square:

$$\begin{array}{ccc} \mathcal{F}(V)^* \otimes \mathcal{F}(U') & \xrightarrow{\Phi_{V,U'}} & \mathrm{Sym}^{12}(V^* \otimes U') \\ \mathcal{F}(g)^* \otimes \mathcal{F}(h^{-1}) \downarrow & & \downarrow \alpha^* \\ \mathcal{F}(V)^* \otimes \mathcal{F}(U) & \xrightarrow{\Phi_{V,U}} & \mathrm{Sym}^{12}(V^* \otimes U). \end{array}$$

In fact both routes evaluate at T as $\beta'(\mathcal{F}(g)\mathcal{F}(T)\mathcal{F}(h^{-1})\xi')$. The induced graded map transports the determinant ideal line and the degree-sixteen cone ideal. By Lemma S6.4 it therefore transports $(\mathcal{I}' : \mathcal{D}')_{12}$ to $(\mathcal{I} : \mathcal{D})_{12}$, that is, $(J_{R'})_{12}$ to $(J_R)_{12}$ in the chosen fibres. This is the intrinsic residual operation, not a choice of scalar determinant equations. The universal diagonal-block identity gives

$$\mathbb{C}\beta_{R,\lambda} = \mathbb{C}\mathcal{F}(g)^*\beta_{R',\lambda} \quad (\lambda \in \Lambda)$$

in the respective left factors; h acts only on the right factor and cannot change either left line. Exterior duality in Lemma S6.6 now gives the displayed Pluecker identities. More explicitly the last arrow is the natural map

$$\bigwedge^4 \mathrm{Sym}^2 V \otimes (\det \mathrm{Sym}^2 V)^* \longrightarrow \bigwedge^6 (\mathrm{Sym}^2 V)^*, \quad (\det \mathrm{Sym}^2 V)^* = (\det V)^{-5}.$$

The factor $(\det V)^{-5}$ is common to both summands, so it disappears projectively; it cannot introduce two independent left transformations. Every step arose from the isomorphism of pairs, establishing the last assertion. $\square$

S6.6. Exterior support of the Jacobian component. The obstruction to recombining two projective Schur components has a simple linear-algebraic description. A secant or tangent vector of $\mathrm{Gr}(4, W)$ uses at most eight directions in W . The Jacobian component of a smooth quartic uses all ten. We prove a sharper statement which also applies to singular quartics.

Fix a volume $\epsilon = x_1 \wedge x_2 \wedge x_3 \wedge x_4$ and suppress it when writing quartics. Define the equivariant map

$$j : \det V \otimes \text{Sym}^4 V \longrightarrow \bigwedge^4 \text{Sym}^2 V$$

by polarization of

$$j(\epsilon \otimes \ell^4) = (\ell x_1) \wedge (\ell x_2) \wedge (\ell x_3) \wedge (\ell x_4). \quad (\text{S6.17})$$

The right side changes by the determinant of a change of the x_i , so this definition does not depend on the chosen volume. More explicitly,

$$j(\epsilon \otimes v_1 v_2 v_3 v_4) = \frac{1}{24} \sum_{\sigma \in \mathfrak{S}_4} (v_{\sigma(1)} x_1) \wedge \cdots \wedge (v_{\sigma(4)} x_4). \quad (\text{S6.18})$$

This is nonzero. Its irreducible source and (S6.1) identify its image with E_{35} . For the Jacobian map C , evaluation at $\ell = x_1$ gives $Cj(x_1^4) = 2x_1^4$; hence

$$Cj = 2 \text{ id}, \quad Q = \frac{1}{2} jC. \quad (\text{S6.19})$$

These identities also give $j = C^*/24$ in the Fischer normalization used below.

For $q \in W^* = \text{Sym}^2 V^*$, regard q as the symmetric bilinear functional $q(uv)$, and let ι_q denote exterior contraction. A quartic has *essential-variable space* $H \subset V$ if H is the smallest subspace for which it lies in $\text{Sym}^4 H$. Equivalently, $H^\perp$ is the kernel of its first differentiation map $V^* \rightarrow \text{Sym}^3 V$. This is the first-catalecticant characterization of essential variables [31, Proposition 1]; it also follows by choosing coordinates along the kernel and using characteristic zero. This subspace of V is distinct from the exterior-support subspace of $W = \text{Sym}^2 V$ defined below.

S6.6.1. An independent four-dimensional character calculation. The general shear lemma is Lemma S2.2. The following character calculation gives a second proof of the even-dimensional scalar-kernel step when $n = 4$. Write $D = \det V$.

For the next lemma write $[a, b] = \text{Sym}^a U \otimes \text{Sym}^b U'$, where U, U' are the two standard two-dimensional representations of $SL_2 \times SL_2$. Only even weights occur below, so all representations descend through its double cover of SO_4 .

Lemma S6.9 (The exterior-cube character). *Let q be nondegenerate on the four-dimensional space V , and put $W = \text{Sym}^2 V$. Under $SL_2 \times SL_2 \rightarrow SO(V, q)$, choose $V = [1, 1]$. Then*

$$\begin{aligned} \bigwedge^3 W &\simeq [6, 0] \oplus [0, 6] \oplus [4, 4] \oplus 2[4, 2] \oplus 2[2, 4] \\ &\quad \oplus [2, 2] \oplus 2[2, 0] \oplus 2[0, 2]. \end{aligned} \quad (\text{S6.20})$$

In particular $(\bigwedge^3 W)^{SO(V, q)} = 0$, and $\text{Hom}_{O(V, q)}(D, \bigwedge^3 W) = 0$.

Proof. Put $A_2 = \text{Sym}^2 U$ and $B_2 = \text{Sym}^2 U'$. The symmetric Cauchy formula gives

$$W = (A_2 \otimes B_2) \oplus \mathbb{C}\rho, \quad \rho = q^{-1},$$

where ρ is invariant. Write $T = A_2 \otimes B_2$. For any two three-dimensional vector spaces, antisymmetrization in $(A_2 \otimes B_2)^{\otimes r}$, for $r = 2, 3$, gives

$$\begin{aligned} \bigwedge^2 T &= (\text{Sym}^2 A_2 \otimes \bigwedge^2 B_2) \oplus (\bigwedge^2 A_2 \otimes \text{Sym}^2 B_2), \\ \bigwedge^3 T &= (\text{Sym}^3 A_2 \otimes \bigwedge^3 B_2) \oplus (\mathbb{S}_{(2,1)} A_2 \otimes \mathbb{S}_{(2,1)} B_2) \oplus (\bigwedge^3 A_2 \otimes \text{Sym}^3 B_2). \end{aligned}$$

For clarity, the reason for the conjugate partitions in these formulas is that tensoring a symmetric-group representation with the sign representation conjugates its partition. Schur–Weyl decomposition of the two tensor powers therefore gives exactly the displayed summands. This is the usual skew Cauchy formula, not an additional hypothesis about this particular q .

Here are all the required one-factor calculations:

functor on $\text{Sym}^2 U$	restriction to SL_2
Sym^2	$\text{Sym}^4 U \oplus \mathbb{C}$
$\bigwedge^2$	$\text{Sym}^2 U$
$\bigwedge^3$	$\mathbb{C}$
Sym^3	$\text{Sym}^6 U \oplus \text{Sym}^2 U$
$\mathbb{S}_{(2,1)}$	$\text{Sym}^4 U \oplus \text{Sym}^2 U$

(S6.21)

The weights of $\text{Sym}^2 U$ are $2, 0, -2$. Taking unordered pairs or triples of these weights verifies the symmetric-power rows; taking distinct weights verifies the exterior rows. Complete reducibility identifies the resulting characters with the stated representations. Alternatively the final row follows from $A_2 \otimes \bigwedge^2 A_2 = \mathbb{S}_{(2,1)} A_2 \oplus \bigwedge^3 A_2$: the tensor product $\text{Sym}^2 U \otimes \text{Sym}^2 U$ is $\text{Sym}^4 U \oplus \text{Sym}^2 U \oplus \mathbb{C}$, and the exterior cube removes the scalar summand.

Substitution gives

$$\bigwedge^2 T = [4, 2] \oplus [0, 2] \oplus [2, 4] \oplus [2, 0],$$

$$\bigwedge^3 T = [6, 0] \oplus [2, 0] \oplus [4, 4] \oplus [4, 2] \oplus [2, 4] \oplus [2, 2] \oplus [0, 6] \oplus [0, 2].$$

Since $\bigwedge^3 W = \bigwedge^3 T \oplus (\bigwedge^2 T) \wedge \rho$, these equalities prove (S6.20). The dimensions are $84 + 36 = 120 = \binom{10}{3}$. There is no summand $[0, 0]$. Finally the determinant character of $O(V, q)$, namely its representation on D , restricts to the trivial character of $SO(V, q)$. Any equivariant image of D would therefore be a nonzero $SO(V, q)$ -invariant vector, which the decomposition excludes. This tracks the determinant twist without choosing an extension of each nontrivial SO_4 summand to O_4 . $\square$

Lemma S6.10 (The contraction kernel). *For $0 \neq q \in \text{Sym}^2 V^*$, let $\kappa_q : D \otimes \text{Sym}^4 V \rightarrow \bigwedge^3 \text{Sym}^2 V$ be the map $\epsilon \otimes f \mapsto \iota_q j(\epsilon \otimes f)$. If q is singular, then*

$$\ker \kappa_q = \det V \otimes \text{Sym}^4(\text{rad } q). \quad (\text{S6.22})$$

If q is nondegenerate, its inverse is a quadratic tensor $q^{-1} \in \text{Sym}^2 V$, and

$$\ker \kappa_q = \det V \otimes \mathbb{C}(q^{-1})^2. \quad (\text{S6.23})$$

Proof. For each fixed q , the map in the statement is explicitly

$$\kappa_q : D \otimes \text{Sym}^4 V \longrightarrow \bigwedge^3 \text{Sym}^2 V, \quad \epsilon \otimes f \longmapsto \iota_q j(\epsilon \otimes f).$$

Naturality of polarization and contraction makes this map equivariant for the stabilizer of q in $GL(V)$. The source carries the determinant factor D ; the target has its ordinary exterior representation. We use that equivariance in both cases.

Suppose first that q has rank $r \in \{1, 2, 3\}$. Choose $V = A \oplus B$, where $B = \text{rad } q$ and $q|_A$ is nondegenerate. Every pure power b^4 , $b \in B$, is killed by contraction in (S6.17), since $q(bx) = 0$ for all $x \in V$. Pure powers span $\text{Sym}^4 B$, so $D \otimes \text{Sym}^4 B \subset \ker \kappa_q$.

The stabilizer contains $O(A, q|_A) \times GL(B)$, the B -scaling torus, and all shears in Lemma S2.2. On the degree-one weight space

$$D \otimes A \otimes \text{Sym}^3 B$$

the product-group representation is irreducible. Indeed its untwisted factors are the standard representation of the full orthogonal group and the irreducible symmetric cube for $GL(B)$. The determinant factor restricts to the product character $\det_A \det_B$, which does not change irreducibility. For $r = 1$ the standard factor is one-dimensional; for $r = 2$ its two isotropic weight lines for SO_2 are exchanged by a reflection in O_2 , so neither is invariant under the full group. The usual standard representation is irreducible also for $r = 3$.

The restriction of κ_q to this space is not zero. Choose an orthogonal basis with $a = x_1 \in A$, $q(a, a) = 1$, and $b = x_4 \in B$, and set $\epsilon = x_1 \wedge \cdots \wedge x_4$. In the coefficient of st^3 in $\iota_q j(\epsilon \otimes (sa + tb)^4)$, only the contraction of the first factor contributes. Thus

$$4\kappa_q(\epsilon \otimes ab^3) = (bx_2) \wedge (bx_3) \wedge b^2 \neq 0.$$

The three factors are independent because multiplication by b is injective on V . Irreducibility now gives injectivity on the degree-one space. If the kernel contained any other positive A -degree, Lemma S2.2, with $m = 4$, would give a nonzero kernel vector in that degree-one space, a contradiction. This proves (S6.22) for every singular rank, without a genericity condition on q .

Now let q be nondegenerate. After a change of coordinates, $q(x_i x_j) = \delta_{ij}$ and $\rho = q^{-1} = x_1^2 + x_2^2 + x_3^2 + x_4^2$. The harmonic decomposition is

$$D \otimes \text{Sym}^4 V = (D \otimes \mathcal{H}_4) \oplus (D \otimes \rho \mathcal{H}_2) \oplus (D \otimes \mathbb{C} \rho^2). \quad (\text{S6.24})$$

Here $\mathcal{H}_m$ is the kernel of the Laplacian in degree m . To check the decomposition directly, the symmetric Cauchy formula for $V = U \otimes U'$ in degree four has partitions $(4), (3, 1), (2, 2)$, giving SO_4 -types $[4, 4], [2, 2], [0, 0]$ of dimensions 25, 9, 1. Multiplication by the invariant ρ identifies the degree-two and scalar pieces; differentiation verifies the harmonic description. The three summands are irreducible and pairwise nonisomorphic for SO_4 . The determinant factor restricts trivially to this group.

The last summand is killed: as an $O(V, q)$ -representation, $D \otimes \mathbb{C} \rho^2 \simeq D$, and Lemma S6.9 proves that the target of κ_q has no such character. It remains to show that the other two restrictions are nonzero. Put

$$h_4 = x_1^4 - 6x_1^2 x_2^2 + x_2^4 \in \mathcal{H}_4, \quad h_2 = x_1^2 - x_2^2 \in \mathcal{H}_2.$$

In the basis triple $\tau = (x_1 x_2) \wedge (x_1 x_3) \wedge (x_1 x_4)$, formula (S6.18) gives

$$[\tau] \kappa_q(\epsilon \otimes h_4) = 2, \quad [\tau] \kappa_q(\epsilon \otimes \rho h_2) = \frac{2}{3}.$$

For the first equality the terms x_1^4 and $-6x_1^2 x_2^2$ contribute 1, 1. For the second the terms $x_1^4, x_1^2 x_3^2, x_1^2 x_4^2$ contribute 1, $-1/6, -1/6$; all other terms contribute zero. These are coefficient identities, not numerical rank assumptions. Each nonzero restriction from an irreducible summand is injective, and the images of the two nonisomorphic summands cannot intersect. The kernel is therefore precisely $D \otimes \mathbb{C} \rho^2$. Equivariance transports this conclusion back to the original form q , proving (S6.23). $\square$

For $z \in \wedge^4 W$, write $s(z)$ for the dimension of its smallest exterior support: the smallest subspace S with $z \in \wedge^4 S$. In any basis,

$$s(z) = 10 - \dim\{q \in W^* : \iota_q z = 0\}. \quad (\text{S6.25})$$

Indeed the vanishing of contraction by a dual basis vector is exactly the absence of all wedge monomials containing its corresponding basis vector. Applying this after a change of basis proves both the formula and the existence of the smallest support.

Proposition S6.11 (Exact support of the Schur embedding). *Let $0 \neq f \in \text{Sym}^4 V$ have essential-variable dimension d . With the determinant factor understood,*

d	condition on f	$s(j(f))$
1	arbitrary	4
2	arbitrary	7
3	arbitrary	9
4	f not the square of a nondegenerate quadratic	10
4	f the square of a nondegenerate quadratic	9

(S6.26)

In particular, $s(j(f)) \geq 9$ if $d \geq 3$, and it is 10 if the projective quartic $V(f)$ is smooth.

Proof. If $d < 4$ and H is the essential-variable space, the contraction kernel lemma says that the annihilating bilinear forms are exactly those with $H \subset \text{rad } q$. They form $\text{Sym}^2(V/H)^*$, of dimension $(4-d)(5-d)/2$. No nondegenerate form can annihilate $j(f)$, since the square of a nondegenerate quadratic has four essential variables. This gives the first three entries of (S6.26).

If $d = 4$, no singular form annihilates $j(f)$. A nondegenerate form does so exactly when f is proportional to its inverse square. In this case its projective class is unique: proportional squares of nonzero quadratics over $\mathbb{C}$ have proportional square roots, and inversion is projectively unique. The annihilator therefore has dimension one in the square case and zero otherwise. Formula (S6.25) gives the last two entries.

A quartic with fewer than four essential variables is a cone and is singular at its vertex. A square of a quadratic is nonreduced and has vanishing gradient along that quadratic. Neither defines a smooth quartic surface, proving the last assertion. $\square$

Lemma S6.12 (The eight-dimensional secant and tangent bound). *A sum of two decomposable four-vectors in $\Lambda^4 W$ has exterior support at most eight. Every vector in the affine tangent space $\Lambda^3 R \wedge W$ to the Pluecker cone at $w_R \neq 0$ also has exterior support at most eight.*

Proof. For a sum, take the sum of the two supporting four-planes. For a tangent vector, choose a basis $r_1, \dots, r_4$ of R . Every element of $\Lambda^3 R \wedge W$ has the form

$$\sum_{i=1}^4 r_1 \wedge \dots \wedge r_{i-1} \wedge u_i \wedge r_{i+1} \wedge \dots \wedge r_4$$

for four vectors $u_i \in W$, and is supported in $R + \langle u_1, \dots, u_4 \rangle$. $\square$

Theorem S6.13 (No exceptional recombination above three essential variables). *Let $R \in \text{Gr}(4, \text{Sym}^2 V)$ have nonzero Jacobian quartic $f_R = Cw_R$ with at least three essential variables. Then both Pw_R, Qw_R are nonzero and*

$$\dim H_{w_R} = 2, \quad \mathcal{L}_R \cap \text{Gr}(4, W) = \{[w_R]\}$$

as schemes. In particular,

$$G_4^{\text{rec}} = G_4^{\circ}. \quad (\text{S6.27})$$

The secant, double-point, pure-endpoint and flag-line strata of the component pencil are all disjoint from G_4° .

Proof. By (S6.19) and Proposition S6.11, the vector Qw_R has support at least nine. It is nonzero and cannot be proportional to the decomposable vector w_R ; hence $Pw_R \neq 0$. On the component line, the quadratic Pluecker ideal restricts to $(b-a)H_{w_R}$, where $H_{w_R} = \langle \ell_{F,R} \rangle \subset H^0(\mathbb{P}^1, \mathcal{O}(1))$. If this space has dimension two, its ideal sheaf is $(b-a)$. If it has dimension one, the intersection is the degree-two divisor of $(b-a)\ell$, including multiplicities. If it is zero, the entire line is contained in the Grassmannian. These assertions follow directly from the ideal generated by the Pluecker quadrics; no classification of an ambient boundary is used.

If the closed pencil has a distinct second decomposable point $z = aPw_R + bQw_R$ with $a \neq b$, then $z = aw_R + (b - a)Qw_R$. Thus Qw_R is a linear combination of two decomposable vectors, contradicting Lemma S6.12. This argument includes a pure endpoint. If the residual point coincides with $[w_R]$, then the line is tangent at $[w_R]$: its restricted equations have zero first derivative there. Its direction is Qw_R modulo $\mathbb{C}w_R$, so $Qw_R \in \wedge^3 R \wedge W$, giving the same contradiction. A whole Grassmann line also has that tangent property. All alternatives to rank two are excluded.

Smoothness of Y_R implies four essential variables, so every $R \in G_4^\circ$ satisfies this conclusion. The equality in (S6.27) holds as an equality of open subschemes. Here G_4° has the reduced open-subscheme structure inherited from $\text{Gr}(4, 10)$. The complement is defined by the two-by-two minors of the two-column coefficient matrix of the residual linear forms. It is a closed subscheme of finite type over $\mathbb{C}$ with no geometric points, hence is empty, regardless of possible nilpotents in that determinantal presentation. The statement for all the named exceptional strata follows at the same time. $\square$

Corollary S6.14 (An invariant location of the ambient obstruction). *On $X^\times$, the support of the rank-defect locus $\dim H_w \leq 1$ is contained in the inverse image under the Jacobian covariant of the subspace variety of binary quartics. That variety is defined set-theoretically by rank at most two of the first catalecticant $V^* \rightarrow \text{Sym}^3 V$. In particular, it produces no exceptional divisor or ramification locus in the smooth geometric family, and no pair of non-projectively-equivalent smooth webs with isomorphic full failure schemes.*

Proof. The containment is the contrapositive of Theorem S6.13; the catalecticant characterization is the essential-variable characterization above. The inverse-scheme assertion follows from the intrinsic readout and Theorem S6.1. No equality with the whole binary-Jacobian inverse image, or classification of its ambient irreducible components, is asserted. $\square$

Proof of Theorem S6.1. Theorem S6.13 proves that the rank-two condition holds on all of G_4° , rather than merely on a nonempty subopen. The coefficient-rank condition in (S6.3) defines this open algebraically. Nonemptiness of the geometric locus is independently witnessed by the smooth integral web C_* of Lemma S9.1; Appendix S26 retains its exact separating quadrics and the square-web normalization.

Indeed the restricted quadrics are $(b - a)H_{w_R}$. When H_{w_R} is the full space of linear forms, those forms generate the unit ideal sheaf on the line, and the intersection has ideal sheaf $(b - a)$. It is the single reduced point $[w_R]$. Without that rank condition, the points with the same two nonzero projective components are precisely the decomposable points $[aPw_R + bQw_R]$ with $ab \neq 0$, which proves (S6.4). This is an exact characterization of the ambiguity of the *two-component invariant*; it need not describe every additional invariant of the full failure scheme on an exceptional locus. If the line is not contained in the Grassmannian, its scheme-theoretic intersection has length at most two, because the Grassmannian ideal is generated by quadrics.

An abstract isomorphism $\widehat{D}_R \simeq \widehat{D}_{R'}$ induces the graded normal-cone map of Lemma S6.8. Its commutative coefficient square and Lemma S6.5 give one $g \in \text{PGL}(V)$ for which the pairs $([Pw_R], [Qw_R])$ and $([P(gw_R)], [Q(gw_R)])$ agree. The two components are transported by the same g ; the source-factor change acts only on the right Cauchy factors. Replace R' by $g^{-1}R'$. Its Pluecker point belongs to the pencil $\mathcal{L}_R$, and both components are nonzero. The reduced one-point intersection gives $[w_{R'}] = [w_R]$, hence $R' = R$. The algebra-orbit equivalence of Lemma S4.4 gives the assertion for B_R . $\square$

Corollary S6.15 (Separation of the Torelli packet). *On the finite etale locus of the web-to- $K3$ map in Theorem S15.3, every fibre is separated by the isomorphism class of the full failure scheme. No further restriction to a reconstruction subopen is necessary. Distinct projective web classes in*

one such fibre have nonisomorphic failure schemes. On the regular Reye locus, the full failure scheme therefore also determines the web's Reye congruence and its Enriques datum.

Proof. Every web in the indicated finite étale cover belongs to $G_4^\circ = G_4^{\text{rec}}$. Theorem S6.1 therefore separates all its projective web classes directly. The Reye and Enriques constructions on the regular web locus are functorial under projective equivalence, which proves the last assertion. $\square$

The result concerns the actual degree-sixteen ideal of the deepest transverse failure cone. It uses information that is lost on passing to the reduced incidence image or to the Jacobian quartic alone. It neither identifies an incidence-resolution decomposition with a primary decomposition of W_3 or W_4 , nor asserts that all higher embedded primary components have been classified. Its geometric conclusion is instead an intrinsic inverse theorem: for every smooth Jacobian web, the deeper nonreduced boundary recovers the original relation space.

APPENDIX S7. THE WEB INSTANCE AND POLAR RESTRICTIONS

We apply the intrinsic coefficient principle of Theorem 3.1 to the web normal cone and to a separate natural class of determinant-evaluation maps. Neither application is an additional hypothesis of the pencil inverse theorem.

Corollary S7.1 (The web instance of the criterion). *For $R \in G_4^\circ$, the scheme $\widehat{D}_R$ satisfies (i)–(iii) with*

$$B = \text{Gr}(4, S_R), \quad n = 4, \quad m = 12, \quad \Lambda = \{(5, 4, 2, 1), (4, 4, 4, 0)\}.$$

The left subspaces are the two lines $\mathbb{C}\beta_{R,\lambda}$. Their intrinsic recovery, exterior duality, and the support gap imply Theorem S1.1.

Proof. The intrinsic base and determinant normal cone are Lemma S6.2. The nontriviality of the other ruling is its Schubert-line test. The residual factorization is Lemma S6.4, and (3.1) is exactly Lemma S6.5. Exterior duality is Lemma S6.6. The final uniqueness is Theorem S6.13; the smooth Jacobian has four essential variables. Thus every hypothesis is verified on the full stated smooth locus, not on a newly restricted generic subset. $\square$

S7.1. Determinantal failure for polar restrictions. Fix $n \geq 2$, $d \geq 1$, a vector space H of dimension $s > n$, and an n -dimensional vector space V . Put $B = \text{Gr}(n, H)$, with tautological bundle $\mathcal{U}$, and $M = \text{Tot Hom}(\mathcal{U}, V)$. Let $0 \neq L \subset \text{Sym}^d V^*$ be a linear system, not furnished with a preferred basis. The tautological map $T : \mathcal{U} \rightarrow V$ on M gives the polar-restriction evaluation

$$\text{ev}_{L,T} : L \otimes \text{Sym}^d \mathcal{U} \longrightarrow \mathcal{O}_M, \quad f \otimes \xi \longmapsto f(\text{Sym}^d(T)\xi).$$

Here and below bundles from B are pulled back to M . Define a vector-bundle map and its zeroth-Fitting failure scheme by

$$\Psi_L = \begin{pmatrix} \det T & 0 \\ 0 & \text{ev}_{L,T} \end{pmatrix} : \det \mathcal{U} \oplus (L \otimes \text{Sym}^d \mathcal{U}) \longrightarrow \det V \oplus \mathcal{O}_M, \quad (\text{S7.1})$$

$$Z_L = V(\text{Fitt}_0 \text{coker } \Psi_L) \subset M. \quad (\text{S7.2})$$

In local frames, if $f_1(T), \dots, f_a(T)$ are the entries of the evaluation row and $\delta = \det T$, this is literally

$$\Psi_L = \begin{pmatrix} \delta & 0 & \cdots & 0 \\ 0 & f_1(T) & \cdots & f_a(T) \end{pmatrix}.$$

Its only nonzero maximal minors are $\delta f_i(T)$. The two blocks ask simultaneously whether T is invertible and whether the restricted system is nonzero. This is a different failure map from cube-zero multiplication; neither a relation web nor its Jacobian covariant is used in its definition.

Theorem S7.2 (Reconstruction of linear systems from polar failure). *For fixed n, d, s as above and any nonzero systems $L \subset \text{Sym}^d V^*$, $L' \subset \text{Sym}^d V'^*$,*

$$Z_L \simeq Z_{L'} \iff g^* L' = L \text{ for a linear isomorphism } g : V \rightarrow V'. \quad (\text{S7.3})$$

The isomorphism on the left is abstract: it need not preserve the projection to B , the zero section, the grading, or an ambient embedding. All reductions $(Z_L)_{\text{red}}$ are the same determinant hypersurface in M . Thus the nonreduced scheme, not its reduction, separates the projective-equivalence classes of these linear systems. No smoothness hypothesis on the hypersurfaces or their base locus is required.

Proof. In a frame, write J_L for the ideal generated by all coefficients of $f \circ T$, for $f \in L$. The maximal minors of the block map (S7.1) give

$$I(Z_L) = (\det T) J_L. \quad (\text{S7.4})$$

No zero of J_L has invertible T : an invertible substitution cannot annihilate a nonzero polynomial. Therefore $V(J_L)$ is contained in the determinant hypersurface, and the reduction in (S7.4) is precisely that hypersurface with its reduced structure. This also follows locally from the primality of the generic determinant and the Nullstellensatz, since B is smooth.

The reduced singular locus of the rank-at-most- r matrix variety, for $0 < r < n$, is the rank-at-most- $r - 1$ variety. For clarity, invert an r -minor and use block elimination: the remaining lower-right block is zero, giving a smooth chart. At smaller rank a similar block reduction leaves minors of size at least two, with zero linear terms and tangent dimension exceeding the local dimension. These charts prove the assertion, also after product with a smooth base. Applying it repeatedly to $(Z_L)_{\text{red}}$ recovers its zero section B intrinsically. Its normal-cone algebra is the original graded algebra $\text{Sym}(V^* \otimes \mathcal{U})/((\det T) J_L)$: the ideal is homogeneous and generated in degree $n + d$. In particular its degree-one conormal is $V^* \otimes \mathcal{U}$, before any factorization is chosen by an isomorphism.

The two ruling parameter bundles are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Only the first is projectively trivial. To check the second, restrict to a Schubert line in $\text{Gr}(n, H)$. There $\mathcal{U}^* = \mathcal{O}^{n-1} \oplus \mathcal{O}(1)$. If its projectivization were trivial, the vector bundle would be a direct sum of n copies of one line bundle: pull back the relative $\mathcal{O}(1)$ and push it down to see this assertion. Its determinant would have degree divisible by n , whereas the displayed splitting has determinant degree one. This is impossible for $n \geq 2$.

Finally the homogeneous coefficient map

$$\text{Sym}^d V^* \otimes \text{Sym}^d \mathcal{U} \longrightarrow \text{Sym}^d(V^* \otimes \mathcal{U}) \quad (\text{S7.5})$$

is the (d) -Cauchy inclusion. It is injective even before the Cauchy decomposition is invoked. Evaluate on a rank-one map $T = v \otimes a$, with $a \in \mathcal{U}^*$: a tensor $f \otimes \xi$ evaluates to $f(v)a^d(\xi)$. These products identify the tensor product with the polynomials bihomogeneous of degree (d, d) in (v, a) , so a zero coefficient polynomial has zero tensor. Consequently

$$(J_L)_d = L \otimes \text{Sym}^d \mathcal{U} \quad (\text{S7.6})$$

as an actual subspace of the specified Cauchy component. The ideal has no lower-degree generators. All hypotheses of Theorem 3.1 are now verified, with $m = d$ and $\Lambda = \{(d)\}$, proving the forward implication. For the reverse implication the bundle map $T \mapsto gT$ transports both block maps, up to invertible changes in their domain and target. It therefore transports their Fitting schemes. Scalar choices of g do not alter the system. $\square$

For a single equation $L = \mathbb{C}f$, this yields an intrinsic scheme-valued invariant of a degree- d hypersurface in $\mathbb{P}(V)$, up to projective equivalence. For example $n = 3, d = 4$ gives all plane quartics, smooth or singular, using a degree-seven Fitting ideal over $\text{Gr}(3, H)$. Higher-dimensional

systems give Grassmannian families of equations rather than isolated examples. The support is fixed throughout each such family, and the varying coefficient system is recovered solely from its nonreduced structure. This construction concerns arbitrary homogeneous systems. The original multiplication construction has the different natural extension proved in Section 4. Nor does (S7.3) assert an equivalence of moduli stacks or a classification of all automorphisms of Z_L ; it is the stated separation of abstract isomorphism classes.

Example S7.3 (An explicit polar nilpotent layer). For $n = d = 2$, $f = x_1^2 + x_2^2$, and $T = \begin{pmatrix} a & b \\ c & e \end{pmatrix}$, put $\delta = ae - bc$, $A = a^2 + c^2$, $B_0 = ab + ce$, and $C = b^2 + e^2$. The polar ideal is $J = (A, 2B_0, C)$, and $I(Z_{\mathbb{C}f}) = \delta J$. The identity $AC - B_0^2 = \delta^2$ gives $\delta^3 \in I$. But $\delta \notin J_2 = \langle A, B_0, C \rangle$, since the monomials ae, bc do not occur in that span. Cancellation in the polynomial domain gives $\delta^2 \notin I$. Since $\sqrt{I} = (\delta)$, the nilradical therefore has nilpotency index exactly three on this chart. The example displays scheme structure lost by the common reduced determinant support.

APPENDIX S8. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (S5.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \text{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma S8.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \text{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \text{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (\text{S8.1})$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\text{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\text{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\text{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\text{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\text{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every translation of $\text{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (S8.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition S8.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (\text{S8.2})$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (\text{S8.3})$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_{\Delta}|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem S5.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (S8.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem S1.2. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition S8.2, Lemma S8.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition S5.1 recovers D from $\widehat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark S8.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\text{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark S8.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

APPENDIX S9. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem S17.3. Work on the standard chart of $\text{Gr}(4, \text{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0x_1, x_0x_2, x_0x_3, x_1x_2, x_1x_3, x_2x_3).$$

For $C \in \text{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (\text{S9.1})$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma S9.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (\text{S9.2})$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\text{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\text{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \text{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (\text{S9.3})$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (\text{S9.4})$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix S28 specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C_*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$. Every monomial of those degrees belongs to the respective homogeneous ideal over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition S9.2. *The Jacobian map on $G_4^{\mathcal{O}}$ has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma S9.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition S4.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem S17.3. Combine Proposition S9.2 with the recovery of the polarized quartic in Theorem S1.2 and the algebra-orbit equivalence of Lemma S4.4. $\square$

Corollary S9.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.*

Proof. Flatness is Corollary S13.3. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem S1.2. The graph description of Δ in (S5.2) is independent of R . $\square$

APPENDIX S10. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by $q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (\text{S10.1})$$

Proposition S10.1. *For $R \in G_4^2$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem S1.2 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth. Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (\text{S10.2})$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition S10.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

APPENDIX S11. SUBSPACE VARIETIES AND THE SUPPORT CALCULATION

The support argument has two distinct ingredients. The description of tensors by contraction flattenings, the tangent formula for a Grassmannian, and the Schur decomposition are classical. The additional calculation here is the kernel of

$$\kappa_q : \det V \otimes \mathrm{Sym}^4 V \longrightarrow \bigwedge^3 \mathrm{Sym}^2 V$$

for every nonzero symmetric bilinear form q , now extended to all dimensions in Theorem S2.1, followed by its application to the two component lines recovered from the intrinsic normal cone. We give a geometric formulation that distinguishes this calculation from general rank or orbit questions.

For a vector space W define the exterior subspace variety

$$\mathrm{Sub}_r^\wedge(W) = \{[z] \in \mathbb{P}(\bigwedge^4 W) : z \in \bigwedge^4 S \text{ for some } S \subset W, \dim S \leq r\}.$$

It is the reduced closed set defined by rank at most r of $W^* \rightarrow \bigwedge^3 W$, $q \mapsto \iota_q z$. Indeed the support-annihilator identity (S6.25) identifies that rank with the smallest support dimension. Thus the minors give its support; we make no assertion here that their ideal is radical. Similarly, $\mathrm{Sub}_d^{\mathrm{sym}}(V) \subset \mathbb{P}(\mathrm{Sym}^4 V)$ is the reduced closed locus of quartics in at most d essential variables. It is the first-catalecticant rank locus by [31, Proposition 1]. The spaces being measured are different: V for essential variables and $W = \mathrm{Sym}^2 V$ for exterior support.

Proposition S11.1 (Exact reduced pullbacks of support varieties). *Let $j : \mathbb{P}(\mathrm{Sym}^4 V) \hookrightarrow \mathbb{P}(\bigwedge^4 \mathrm{Sym}^2 V)$ be the projectivization of j ; the one-dimensional determinant twist does not affect this map. Let $\mathcal{Q} = \{[h^2] : 0 \neq h \in \mathrm{Sym}^2 V\}$, the closed quadratic-square locus. As reduced closed subvarieties,*

$$(j^{-1} \mathrm{Sub}_4^\wedge(W))_{\mathrm{red}} = \mathrm{Sub}_1^{\mathrm{sym}}(V), \quad (\mathrm{S11.1})$$

$$(j^{-1} \mathrm{Sub}_7^\wedge(W))_{\mathrm{red}} = (j^{-1} \mathrm{Sub}_8^\wedge(W))_{\mathrm{red}} = \mathrm{Sub}_2^{\mathrm{sym}}(V), \quad (\mathrm{S11.2})$$

$$(j^{-1} \mathrm{Sub}_9^\wedge(W))_{\mathrm{red}} = \mathrm{Sub}_3^{\mathrm{sym}}(V) \cup \mathcal{Q}. \quad (\mathrm{S11.3})$$

In particular a quartic with at least three essential variables has image outside the second secant variety $\sigma_2(\mathrm{Gr}(4, W))$, including its limiting secants.

Proof. The support values in Proposition S6.11 give the first two equalities on complex points. They also give the third: its four-essential-variable part consists exactly of squares of nondegenerate quadratics. A degenerate quadratic and its square have the same essential-variable space, since $\partial(h^2) = 2h\partial h$ in a polynomial domain. Their squares therefore already belong to $\mathrm{Sub}_3^{\mathrm{sym}}(V)$. The locus $\mathcal{Q}$ is closed as the image of the projective morphism $[h] \mapsto [h^2]$, which has no base points. All the displayed loci are closed and have the specified reduced structures, so equality on complex points proves the assertions.

A sum of two decomposable four-vectors lies in $\mathrm{Sub}_8^\wedge(W)$. That rank locus is closed, so it contains the closure of all such secant lines. Equation (S11.2) then proves the last assertion. This is an exclusion statement; it does not identify the entire intersection of the binary locus with $\sigma_2(\mathrm{Gr}(4, W))$. $\square$

S11.1. The classical inputs and the additional assertion. The tangent-space identity in [30, Lemma 2.1] is precisely $\widehat{T}_{w_R} \text{Gr}(4, W) = \bigwedge^3 R \wedge W$. Its eight-dimensional support bound is the immediate linear-algebra consequence used in Lemma S6.12. We do not claim either formula, or the general study of Grassmannian secant dimensions, as new. The issue here is the position of one specified Schur subspace relative to those secants, not the dimension of a general secant variety.

The directly relevant exterior formulation is Landsberg–Ottaviani’s first skew-flattening discussion [34, §6.1], incorporated into the merged paper [35, §10.1]. It defines the exterior subspace varieties by support, gives their set-theoretic minors description, and records $\sigma_r(\text{Gr}(k, W)) \subset \text{Sub}_{rk}(\bigwedge^k W)$. Sheridan’s enclosing-space formulation [36, §2.4, Definitions 2.10–2.11 in the cited preprint] identifies the same minimal subspace by contraction. These are antecedents of precisely the support notion used here, not merely analogies for symmetric tensors. We do not claim the containment, the enclosing space, or its interpretation as a rank locus as new.

Those cited statements concern arbitrary exterior tensors. They do not state the rank-dependent kernel formula for the restriction $\iota_q j_n$, the even-dimensional quadratic-power exception, or the displayed reduced pullbacks along j_4 . Theorems S2.1 and S2.5 supply that restricted calculation; the secant exclusion is then its immediate consequence using the classical containment. This identifies the additional assertion relative to the statements actually compared, without purporting to certify that no other historical source contains it.

Landsberg–Weyman’s subspace varieties are defined for products of tensor factors; their Theorem 3.1 describes their equations and singularities [32]. That framework explains the use of flattening ranks as support conditions. It is not a theorem about the contraction of our determinant-twisted quartic submodule, nor does it identify the reduced pullbacks (S11.1)–(S11.3). The exterior rank identity needed here was proved directly, so no scheme-theoretic conclusion for alternating tensors is imported from the tensor-product statement.

The occurrence of $\mathbb{S}_{(5,1,1,1)} V$ is a plethysm calculation, not a novelty claim. Its multiplicity is one in $\bigwedge^4 \text{Sym}^2 V$, whose two highest weights are $(4, 3, 1, 0)$ and $(5, 1, 1, 1)$. For example its character identity is

$$[t^4] \prod_{i \leq j} (1 + tx_i x_j) = s_{(4,3,1,0)}(x) + s_{(5,1,1,1)}(x).$$

The highest-weight and exterior-duality verification is given in Lemma S6.5; polarization fixes the normalization of its unique quartic embedding. The skew Cauchy and SL_2 character calculations used for the scalar kernel are likewise standard representation operations, expanded in Lemma S6.9 rather than cited as a new decomposition method.

The precise additional assertions proved here are the dimension-uniform all-ranks contraction kernels of Theorem S2.1 (and their four-dimensional proof in Lemma S6.10), the resulting exact support table, and the reconstruction consequence of combining that table with the intrinsic common-coordinate readout. Exterior support is neither the least number of decomposable summands nor a classification of $GL(W)$ -orbits in $\bigwedge^4 W$. The reduced support pullbacks above, and the containment of the ambient reconstruction obstruction over binary quartics, must not be mistaken for either such classification. The primary comparisons just made establish which cited tools are classical; they are not an exhaustive historical priority certificate. In particular the six-way theorem comparison with the unread complete Ballico source [2] remains separately unresolved; the publisher first-page definitions do not settle its later theorem statements.

S11.2. Exterior symmetric powers, harmonics, and cofactor identities. There are closer representation-theoretic antecedents than general support varieties. Cheng–Wang [37, §2, Theorem 3.5] record the classical exterior-symmetric-square decomposition and its superalgebra extension.

Specializing the odd dimension to zero gives the multiplicity-free summands indexed by nested hooks, including $\mathbb{S}_{(n+1, 1^{n-1})} V = \det V \otimes \text{Sym}^n V$. Thus its occurrence, its multiplicity, and the two four-dimensional character identities used here are classical plethysm. The specified pure-power formula fixes a scalar normalization; it does not create a new representation or a new branching rule.

The harmonic decomposition is likewise classical. De Bie–Eelbode–Roels [38, §1, equations (1)–(3)] describe its one-variable form; their Theorem 2.1 develops the two-variable Howe picture under their dimension convention. We do not import that convention into the present low-dimensional cases. Lemma S2.4 proves exactly the full O_n assertion, including O_2 , used in the kernel calculation. The reduction of a kernel to inequivalent harmonic summands is a standard representation argument. What must still be computed for this particular map is which restrictions are zero.

The identity $\sum_i \partial_i A_i = 0$ in Lemma S2.3 is the exterior-valued cofactor cancellation underlying the Piola identity; compare [39, equation (1.1), §3.1]. Commutation of mixed partials and the algebraic Gaussian integration-by-parts recurrence are classical tools. Their use here establishes the scalar exceptional kernel. The nonzero coefficient $m(n+m-2)/(n(n-1))$ establishes injectivity on every other harmonic summand. Neither cofactor cancellation nor harmonic irreducibility is claimed as a new identity.

These comparisons separate three levels of assertion. The module occurrence, harmonic branching, and the ambient skew-flattening bound are classical. The all-rank restriction $\iota_q j_n$ is calculated explicitly here, using those tools; the cited branching statements alone do not specify its singular-radical kernel or its nondegenerate parity alternative. The geometric assertion is that intrinsic nonreduced normal cones recover the required coefficient data through one common projective transformation. Theorem 3.1 separates that assertion from recombination, and Theorem S7.2 verifies it for another class of Fitting schemes. This is a theorem-by-theorem boundary relative to the inspected statements, not certification of exhaustive priority.

S11.3. The restricted contraction versus classical operators. The relation with classical invariant theory is more precise than the shared occurrence of a representation. Howe’s [42, Theorem 9 and §4(a), pp. 555–557] gives harmonic irreducibility and the orthogonal decomposition through the commuting quadratic multiplication and Laplacian operators. Thus the passage to harmonic summands in the proof of Theorem S2.1 is classical, including its full-orthogonal-group interpretation. The additional step is the restriction of the particular exterior contraction to each summand, not a new harmonic decomposition.

The different operators can be compared without terminology. With a volume fixed, the Jacobian map in two variables is

$$C_2(a \wedge b) = a_{x_1} b_{x_2} - a_{x_2} b_{x_1}.$$

It is the first binary transvectant, with the normalization for quadratic inputs in [43, §2.2, Definition 2.6 and Remark 2.7]. Thus neither that covariant nor its polarization is new. Moreover, when $n = 2$, j_2 is an isomorphism onto $\wedge^2 \text{Sym}^2 V$, so for every nonzero q

$$\ker \kappa_{2,q} = j_2^{-1} \left(\bigwedge^2 \ker[q : \text{Sym}^2 V \rightarrow \mathbb{C}] \right). \quad (\text{S11.4})$$

This one-dimensional kernel is elementary exterior algebra.

For general n , a quadratic apolar differential operator has the signature $q(\partial) : \text{Sym}^n V \rightarrow \text{Sym}^{n-2} V$. In orthonormal coordinates its kernel is $\mathcal{H}_n$, of dimension $\binom{2n-1}{n} - \binom{2n-3}{n-2}$. In contrast the operator studied here has signature

$$\kappa_{n,q} : \det V \otimes \text{Sym}^n V \longrightarrow \bigwedge^{n-1} \text{Sym}^2 V, \quad \kappa_{n,q} = \iota_q j_n.$$

For nondegenerate q its kernel is zero in odd dimension and the quadratic-power line in even dimension. These are not the apolar harmonic kernels. For singular q , the proof uses its unipotent radical shears as well as its nondegenerate orthogonal factor; a full-rank harmonic branching statement alone cannot supply that argument. The normalization, nonzero harmonic coefficients, and singular-radical calculation are supplied in Section S2.

For comparison with unrestricted exterior contraction, take $0 \neq q \in W^*$, where $W = \text{Sym}^2 V$. Choose $w_0 \in W$ with $q(w_0) = 1$. The decomposition

$$\bigwedge^n W = \bigwedge^n \ker q \oplus w_0 \wedge \bigwedge^{n-1} \ker q$$

shows that $\ker \iota_q = \bigwedge^n \ker q$. Accordingly, the precise restricted problem is

$$\ker \kappa_{n,q} = j_n^{-1} \left(\text{im } j_n \cap \bigwedge^n \ker q \right). \quad (\text{S11.5})$$

Knowing the ambient kernel does not compute this intersection. In the singular case the answer is $\det V \otimes \text{Sym}^n(\text{rad } q)$; in the nondegenerate case it is zero for odd n and the quadratic-power line for even n . The calculation in Section S2 keeps the normalization $C_n j_n = 2 \text{id}$, the full orthogonal stabilizer, and its singular radical shears. These specify an equivalent-map comparison that is sharper than comparing representation names.

Finally, the universal coefficient map (4.6) is classical matrix-coefficient representation theory. What is proved geometrically is that its specified left subspaces can be recovered from an abstract failure scheme, with one common projective transformation. In the natural pencil family there is only one such subspace, and exterior duality recovers the full pencil. The classical Weierstrass–Segre classification and the pencil–hyperelliptic correspondence classify objects after a pencil has been supplied; they do not enter the proof that the abstract scheme supplies that pencil. These comparisons identify the exact additional arguments relative to the inspected statements, without claiming exhaustive historical nonanticipation.

APPENDIX S12. AN EQUALITY-LEVEL DICTIONARY FOR THE EXTERIOR OPERATOR

The exterior contraction theorem is a supporting theorem for webs; it is not an input to the principal pencil inverse theorem. Its exact relation to classical differential operators can nevertheless be stated without a novelty slogan. Fix the normalization

$$j_n(\epsilon \otimes \ell^n) = \bigwedge_{i=1}^n (\ell x_i), \quad C_n j_n = 2 \text{id}, \quad \kappa_{n,q} = \iota_q j_n.$$

For $h = x^\dagger H x$ and q represented by Q , Theorem S3.1 identifies the actual composite, on every homogeneous f of degree n , as

$$C_n \varepsilon_h \kappa_{n,q}(\epsilon \otimes f) = \epsilon \otimes \left(\frac{4}{n} \sum_{ij} (HQ)_{ij} x_i \partial_j f - \frac{2h}{n(n-1)} \sum_{ij} Q_{ij} \partial_i \partial_j f \right). \quad (\text{S12.1})$$

This formula compares operators with their coefficients and domains, not merely isomorphic source representations. It is valid for singular q and arbitrary h ; it is not inferred from a nondegenerate orthogonal change of variables.

For nondegenerate q , choose $h = \rho = q^{-1}$. On degree n , the right hand side is exactly

$$\frac{2}{n(n-1)} \epsilon \otimes (E(E+n-2) - \rho \Delta_q) f.$$

The operator in parentheses is the classical angular Casimir. The harmonic decomposition and its eigenvalues are classical, not additional representation-theoretic discoveries. In the orthonormal

normalization $q = I$, the induced symmetric-tensor norm is $\|x^\alpha\|^2 = \alpha!/k!$ in degree k . The identity $j_n^* = 2^{-n}C_n$, together with exterior adjunction, gives

$$\kappa_{n,q}^* \kappa_{n,q} = \frac{2^{1-n}}{n(n-1)} (E(E+n-2) - \rho\Delta_q).$$

Thus the singular-value statement has a specified metric and a specified scalar. Changing from the unnormalized Fischer product to symmetric-tensor norms without the factorial factors would not be the same assertion.

For rank $q = r < n$, choosing a radical square $h = b^2$ in (S12.1) first forces $\Delta_q f = 0$ for a vector in the kernel. Varying h then forces dependence only on radical variables. This is the singular-rank argument of Corollary S3.2; no inverse quadratic form or harmonic decomposition for a singular metric is used.

The closest inspected harmonic reference, [38, §1, equations (1)–(3); Definition 6.1 and Lemma 6.2], provides the Fischer decomposition and the multiplication–derivative adjunction framework. Its stated two-vector-variable operators act on polynomial spaces; the exterior maps j_n, C_n, ι_q and the normalization in (S12.1) must still be identified. The classical exterior-symmetric-square summands in [37, §2, Theorem 3.5] identify the source module but do not fix that scalar normalization. These are positive, statement-specific comparisons with the cited sources. They do not establish that no other historical source contains the mixed identity. All-rank kernel and support conclusions remain proved in full; their geometric role is confined to the web recombination branch.

APPENDIX S13. ORDINARY POWERS AND RELATIVE FLAT FLAGS

Corollary S13.1 (Ordinary powers). *For $n \geq 1$, write $D^{[n]} = V(\mathcal{I}_D^n)$ and $E^{[n]} = V(\mathcal{I}_E^n)$. Then*

$$\mathcal{I}_D^n = \mathcal{I}_\Delta^n \mathcal{I}_E^n = \mathcal{I}_\Delta^n \cap \mathcal{I}_E^{2n} \quad (\text{S13.1})$$

is an irredundant primary decomposition on U . With $\mathcal{T}_n = \text{tor}_{\eta_\Delta}(\mathcal{O}_{D^{[n]}})$,

$$0 \longrightarrow \mathcal{T}_n = \mathcal{O}_{E^{[n]}}(-n\Delta) \longrightarrow \mathcal{O}_{D^{[n]}} \longrightarrow \mathcal{O}_{n\Delta} \longrightarrow 0. \quad (\text{S13.2})$$

Its annihilator defines $E^{[n]}$. The generic multiplicity on Δ is n , the nilradical index along E is $2n$, and the length of $\mathcal{T}_n$ at η_E is $n(n+1)/2$.

Proof. In the local ring let $P = (t, f)$. The associated graded ring of P is a polynomial ring in two variables over $\mathcal{O}_E$, and the initial form of t is one of those variables. Hence $(P^{2n} : t^n) = P^n$, proving $(t^n) \cap P^{2n} = t^n P^n$. The same graded argument proves that these powers are primary. The witnesses t^n and f^{2n} at η_E prove irredundancy. Localizing at (t) contracts the ideal to (t^n) , so the generic torsion is $(t^n)/t^n P^n$, and cancellation gives its annihilator P^n . The class of t has index $2n$ at E , since t^{2n} belongs to the ideal but t^{2n-1} does not. At the generic point of Δ , f is a unit. Finally $\mathcal{O}_{U, \eta_E}/(t, f)^n$ has the $n(n+1)/2$ monomials of total degree less than n as successive graded basis elements. $\square$

Lemma S13.2 (Flat flags and arbitrary base change). *Let $U \rightarrow T$ be smooth, and let $E \subset \Delta \subset U$ be relative smooth closed subschemes, each inclusion a Cartier divisor. Put $L = \mathcal{I}_\Delta$ and $P = \mathcal{I}_E$. Then $\mathcal{O}_U/P^n$ and $\mathcal{O}_U/L^n$ are flat over T . For $D_n = V(L^n P^n)$ the sequence*

$$0 \rightarrow L^n \otimes \mathcal{O}_{E^{[n]}} \rightarrow \mathcal{O}_{D_n} \rightarrow \mathcal{O}_U/L^n \rightarrow 0 \quad (\text{S13.3})$$

is exact and remains exact after any base change. Its base-changed terms are the indicated ordinary powers and products. In particular D_n is flat over T .

Proof. The immersion $E \subset U$ is regular of codimension two and has locally free conormal module. The conormal filtration of $\mathcal{O}_U/P^n$ has quotients $\mathrm{Sym}^j(P/P^2)$ on E , for $0 \leq j < n$; these are base-flat. Induction proves flatness. The analogous principal filtration proves flatness of $\mathcal{O}_U/L^n$. Multiplication by a local equation t^n of L^n identifies the kernel in (S13.3) with $L^n \otimes \mathcal{O}_{E^{[n]}}$, since t is a non-zero-divisor. Both outer terms are base-flat, so the middle term is base-flat. For a base algebra $A \rightarrow A'$, the only possible left obstruction in tensoring this short exact sequence is $\mathrm{Tor}_1^A(\mathcal{O}_U/L^n, A')$, which is zero. The sequences defining L , P , and their powers stay exact for the same reason. Thus their images after base change are the actual ideals generated by the base-changed equations, and multiplication gives their stated products. No assertion about specialization of selected primary components is used. $\square$

Corollary S13.3. *Over G_e° , the schemes E , Δ , and U form the smooth relative flag of Lemma S13.2. All $D^{[n]}$ and $E^{[n]}$ are flat, and (S13.2) commutes with arbitrary base change. Each complex fibre has (S13.1).*

Proof. Use the universal quotient $\mathcal{S}$ and the graph bundles (S5.2)–(S5.3) over the smooth family of Proposition S4.3. The Fitting elimination works over the parameter ring, so it gives $t(t, f)$ relatively. Apply the lemma, and apply Corollary S13.1 separately on each fibre for the primary assertion. $\square$

APPENDIX S14. THE INTRINSIC NILPOTENT-DEPTH FILTRATION

We isolate the functorial mechanism that unifies the two local models. Let Z be any noetherian scheme and let $\mathcal{N}_Z$ denote the nilradical of $\mathcal{O}_Z$. Since Z is noetherian, $\mathcal{N}_Z$ is a coherent nilpotent ideal. For $j \geq 1$ define

$$\mathcal{Z}_j(Z) = V\left(\mathrm{Ann}_{\mathcal{O}_Z}(\mathcal{N}_Z^j)\right). \quad (\text{S14.1})$$

Lemma S14.1 (Functoriality). *Every isomorphism $f : Z \rightarrow Z'$ carries $\mathcal{Z}_j(Z)$ isomorphically to $\mathcal{Z}_j(Z')$ for every j . The construction commutes with restriction to open subsets.*

Proof. An isomorphism carries the nilradical to the nilradical, hence carries its j -th power and its annihilator ideal to the corresponding objects. Both formation of powers and annihilators commute with localization. $\square$

Apply this to $Z = \widehat{D}_R$. On the intrinsic open obtained by restricting to the smooth locus of Z_{red} , Proposition S5.1 identifies Z with D_R . Theorem S5.2 gives the local ring

$$A = \mathcal{O}/(t(t, f))$$

with nilradical generated by the class of t . Therefore

$$\mathrm{Ann}_A((t)) = (t, f)/(t(t, f)),$$

so (S14.1) recovers E_R scheme-theoretically.

Proposition S14.2 (Depth one is the polarized ramification support). *On the smooth-reduction locus of $\widehat{D}_R$,*

$$\mathcal{Z}_1(\widehat{D}_R) = E_R, \quad \widehat{\mathcal{N}}_R|_{E_R} = \mathcal{O}_{E_R}(-\widehat{\Delta}_R).$$

Consequently the pair

$$\left(\mathcal{Z}_1(\widehat{D}_R), \omega_{\mathcal{Z}_1(\widehat{D}_R)} \otimes (\widehat{\mathcal{N}}_R|_{\mathcal{Z}_1(\widehat{D}_R)})^{-9}\right)$$

is intrinsic when $e = 4$, and its graded section ring is the fourth Veronese of the polarized quartic section ring.

Proof. Only the last assertion is new notation. The annihilator calculation above is Theorem S5.2; the line-bundle identity and section-ring reconstruction are Proposition S8.2 and Theorem S1.2. $\square$

The point of (S14.1) is that it makes no reference to a projection chart. The closed schemes are automatically nested:

$$\widehat{\mathcal{N}}_R^{j+1} \subset \widehat{\mathcal{N}}_R^j \implies \text{Ann}(\widehat{\mathcal{N}}_R^j) \subset \text{Ann}(\widehat{\mathcal{N}}_R^{j+1}), \quad \mathcal{Z}_{j+1} \subset \mathcal{Z}_j.$$

The new issue is therefore not formal functoriality but the scheme-theoretic computation of the layers as one crosses projection rank. Theorem S21.1 solves this globally: on the whole Grassmannian the residual colon

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{D}_R} : \mathcal{I}_{\widehat{\Delta}_R})$$

determines every annihilator and every associated-graded piece, including all projection coranks and rank-drop loci. The remaining task is geometric identification of those canonical colon schemes.

On the first higher-corank chart the next power is nonzero. The calculation in §S19 gives, on the simple-contact open,

$$\widehat{\mathcal{N}}_R^3 = 0, \quad \text{Ann}(\widehat{\mathcal{N}}_R^2) = \mathcal{I}_{\Sigma_R} / \mathcal{I}_{\widehat{D}_R}.$$

Section S20 removes the simple-root assumption. Section S21 computes the two layers simultaneously across the corank-one/corank-two boundary. On the mixed-regular neighbourhood,

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{\mathcal{C}_R}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}),$$

where $\mathcal{C}_R$ is the unsplit contact cone and $\mathcal{L}$ is the Schubert conormal line. In particular $\mathcal{Z}_1$ specializes from the ramification support to its canonical contact closure with the vertex thickening forced by the Fitting algebra, while $\mathcal{Z}_2$ is the rank-two vertex. On the first tensor-rank boundary of the mixed map, $\mathcal{Z}_2$ acquires a canonical length-three nonreduced structure (Theorem S21.5).

Thus

$$\mathcal{Z}_1 \rightsquigarrow \mathcal{Z}_2$$

is not merely a comparison of two calculations on disjoint strata: it is the associated-graded specialization of one intrinsic nilpotent algebra.

Remark S14.3 (Relation with canonical filtrations on multiple structures). Nilpotent and multiple structures have long been studied through canonical filtrations and associated graded modules; ribbons are the multiplicity-two prototype, and primitive multiple schemes carry a canonical chain of lower multiplicities. See [4, 15, 11]. The construction (S14.1) is deliberately more elementary: it is available for the present Fitting schemes without assuming that they are primitive, locally complete-intersection multiple structures, or Cohen–Macaulay along every higher-corank stratum. The additional content here is that consecutive annihilator layers and their associated graded modules have a specific geometric meaning forced by multiplication: the first carries the polarized ramification K3, the second detects the rank-two Schubert support, and their specialization through contact collisions is controlled by the unsplit contact Cartier ideal rather than by a choice of primary roots.

APPENDIX S15. REYE GEOMETRY, INFINITESIMAL MODULI, AND THE NOVELTY BOUNDARY

We separate the classical moduli of webs and Reye congruences, the exact differential calculation for the polarized-K3 map, and the two nonreduced reconstruction theorems. Generic web reconstruction uses the deepest normal cone; it is not a consequence of the classical nine-dimensional count.

S15.1. The classical nine-dimensional quotient. For $e = 4$, a relation space $R \subset \text{Sym}^2 V$ is a web of quadrics. The bilinear incidence surface Z_R of §S10 is a K3 surface with its fixed-point-free factor-exchange involution; its quotient is the classical Reye Enriques surface associated with the web. This construction and the resulting congruence of lines are classical; see [7, 1, 10, 13, 9, 16].

Arrondo–Sols compute that the Hilbert scheme of Reye congruences is smooth of dimension 24 and rational, and describe it by an open Grassmannian parameter of the corresponding data. After quotienting by the 15-dimensional projective coordinate group, the expected moduli dimension is therefore nine. In characteristic zero the nodal Enriques locus is the classical nine-dimensional divisor in the ten-dimensional Enriques moduli space; the modern theorem [16] identifies every classical nodal Enriques surface as a Reye congruence in their Fano-polarized sense. The distinction from a regular classical web is recalled below.

Thus the equality

$$24 - 15 = 9$$

is not a novelty claim of this paper.

Proposition S15.1 (The classical tangent quotient). *At a general regular web with finite projective stabilizer, the tangent space to the Reye moduli locus is the quotient of the 24-dimensional web tangent space by the 15-dimensional infinitesimal $\text{PGL}(V)$ -orbit:*

$$T_{[R]}\mathcal{M}_{\text{Reye}} \simeq T_R \text{Gr}(4, \text{Sym}^2 V) / \mathfrak{pgl}(V), \quad \dim T_{[R]}\mathcal{M}_{\text{Reye}} = 9.$$

The differential of the polarized-K3 map of Theorem S17.3 factors through this quotient.

Proof. The first statement is the tangent-space form of the smooth 24-dimensional Hilbert calculation of Arrondo–Sols on the regular web locus, followed by the infinitesimal quotient by projective coordinates. The stabilizer is finite at a general point, so the orbit tangent has dimension 15.

The construction $R \mapsto Z_R \simeq Y_R$ is equivariant for $\text{PGL}(V)$. Passing from an embedded quartic to its polarized isomorphism class kills the infinitesimal projective orbit; hence the differential factors through the displayed quotient. $\square$

Proposition S15.2 (Web parameter and Reye Hilbert point). *Let*

$$\mathcal{G} = \text{Gr}(4, \text{Sym}^2 V)$$

and let $\mathcal{G}^{\text{reg}}$ be the open on which the classical Reye construction has the expected Hilbert polynomial. The assignment of a web to its Reye congruence is a $\text{PGL}(V)$ -equivariant morphism

$$\Phi : \mathcal{G}^{\text{reg}} \longrightarrow \text{Hilb}_{\text{Reye}}.$$

Under the Grassmannian parameterization of the regular Reye locus in Arrondo–Sols, Φ identifies a dense open of $\mathcal{G}^{\text{reg}}$ with the corresponding 24-dimensional Hilbert component. In particular, at a general regular web,

$$d\Phi_R : \text{Hom}(R, \text{Sym}^2 V/R) \xrightarrow{\sim} T_{[\text{Reye}(R)]} \text{Hilb}_{\text{Reye}}.$$

After quotienting by the infinitesimal projective orbit this gives the tangent quotient in Proposition S15.1.

Proof. Over $\mathcal{G}$ the universal rank-four subbundle of $\text{Sym}^2 V$ gives the universal web. The incidence equations defining the Reye congruence are polynomial in the universal Pluecker coordinates. Restrict to the open on which their Hilbert polynomial is the regular Reye polynomial. Flatness there and the Hilbert functor give the morphism Φ ; equivariance is immediate from the construction.

Arrondo–Sols describe the regular Reye Hilbert component by the open Grassmannian parameter of the defining four-dimensional system of quadrics. With the present convention this parameter is

precisely a dense open of $\mathrm{Gr}(4, \mathrm{Sym}^2 V)$ (duality only changes whether one writes the system or its annihilator). Their smoothness and dimension-24 statement therefore identifies the tangent space of the Hilbert point with the tangent space of this Grassmannian. The latter is $\mathrm{Hom}(R, \mathrm{Sym}^2 V/R)$, proving the displayed isomorphism. Since the construction is $\mathrm{PGL}(V)$ -equivariant, the orbit tangent maps to the orbit tangent and the quotient gives Proposition S15.1. $\square$

The role of Lemma S9.1 is now precise. It is not used to discover the number nine. Its exact 25×25 integral minor proves that the particular polarized K3 attached to our multiplication problem does not collapse any of the 24 embedded web directions at the explicit rational point. After projective quotient, it proves rank nine for the map to polarized K3 moduli. Since maximal rank is open, the induced map from the classical nine-dimensional Reye deformation space to our polarized K3 image is generically unramified. This is the conceptual tangent statement behind Theorem S17.3; the determinant is an exact arithmetic witness for it.

S15.2. What the finite ambiguity means. The preceding paragraph locates the Torelli ambiguity of the polarized-K3 invariant alone. The failure scheme first reconstructs the polarized K3 $(Y_R, \mathcal{O}_{Y_R}(1))$. A relation system additionally supplies the free involution coming from factor exchange on the bilinear incidence model and hence an Enriques quotient with its Reye data. Classical Reye geometry then has substantial rigidity. In particular, Dolgachev–Reider prove that for a Reye polarization on an Enriques surface the stable rank-two bundle with c_1 equal to that polarization and $c_2 = 3$ is the Reye bundle and is unique up to isomorphism; it is the restriction of the universal quotient bundle from $G(2, 4)$.

Consequently the generic finite fibre in Theorem S17.3 should not be read as an uncontrolled family of quadratic tensors. Once a compatible Enriques quotient and Reye polarization are fixed, the Grassmannian Reye bundle is rigid in the preceding sense. The remaining finite packet is concentrated in the finite choices of compatible Reye/Enriques data over the recovered polarized K3, together with the passage back to the web. For the map defined by the polarized K3 alone, we prove finiteness but do not assert a numerical degree or degree one; that enumeration is distinct from generic reconstruction using the full failure scheme. The additional normal-cone invariant of Theorem S6.1 separates this packet without identifying its K3-only degree.

There is a second useful localization. On the mixed-regular projection-corank-two locus, the unsplit contact quartic of §S20 is the restriction of the already recovered quartic Y_R to the pencil of hyperplanes containing H . Thus the generic corank-two contact cover itself is determined by the polarized K3. The following theorem turns the finite Torelli packet into a precise finite-cover statement and identifies its monodromy.

Theorem S15.3 (Generic Torelli packet as a finite étale monodromy cover). *Let $\mathcal{W}$ be the irreducible nine-dimensional coarse quotient of the regular web locus by projective equivalence on the finite-stabilizer open, and let $\mathcal{K}$ be the nine-dimensional polarized-K3 image of Theorem S17.3. There are nonempty opens*

$$\mathcal{W}^\circ \subset \mathcal{W}, \quad \mathcal{K}^\circ \subset \mathcal{K}$$

such that

$$\tau : \mathcal{W}^\circ \longrightarrow \mathcal{K}^\circ$$

is finite étale of a constant degree

$$d_{\mathrm{Tor}} = [\mathbb{C}(\mathcal{W}) : \mathbb{C}(\mathcal{K})].$$

Its geometric monodromy is transitive on the d_{Tor} points of a fibre. Equivalently, if L is the normal closure of $\mathbb{C}(\mathcal{W})/\mathbb{C}(\mathcal{K})$, the monodromy group is the permutation image of

$$\mathrm{Gal}(L/\mathbb{C}(\mathcal{K}))$$

on the embeddings of $\mathbb{C}(\mathcal{W})$ in L .

The degree-four contact incidence and its discriminant are pulled back from the polarized quartic on $\mathcal{K}^\circ$. Hence they are constant on a Torelli fibre and do not account for d_{Tor} ; any further separation of the finite packet must use web-dependent structure beyond the polarized quartic. Theorem S6.1 supplies such structure in the deepest nonreduced normal cone, and Corollary S6.15 proves separation on a nonempty open of this finite cover.

Proof. The source $\mathcal{W}$ is irreducible because it is obtained from a nonempty open of the irreducible Grassmannian $\text{Gr}(4, \text{Sym}^2 V)$ by the projective-coordinate quotient on the finite-stabilizer locus. Theorem S17.3 says that the dominant map $\tau : \mathcal{W} \dashrightarrow \mathcal{K}$ is generically finite. Thus the induced extension of function fields is finite. In characteristic zero it is separable.

Normalize a nonempty affine open of $\mathcal{K}$ in $\mathbb{C}(\mathcal{W})$. After deleting the complement of the locus on which the original rational map is defined, and then deleting the branch and nonflat loci, one obtains a finite étale morphism from a nonempty open of $\mathcal{W}$. Its degree is the function-field degree displayed above. Connectedness of the generic source, or equivalently irreducibility of $\mathcal{W}$, gives transitivity of geometric monodromy. The description by the normal closure is the standard field-theoretic realization of that action.

The contact incidence is defined solely from the recovered embedded quartic $Y \subset \mathbb{P}^3$ by

$$(H, [\alpha]) \quad \text{with} \quad H \subset \ker \alpha,$$

and the discriminant is the discriminant of its restricted binary quartic. Both constructions commute with isomorphism of the polarized quartic. They therefore descend to $\mathcal{K}^\circ$ and pull back along τ , proving the last assertion. $\square$

S15.3. Comparison with nonreduced reconstruction theory. The use of nilpotent layers has classical antecedents. A ribbon is a square-zero double structure with an invertible conormal module, and Bayer–Eisenbud developed its intrinsic and canonical geometry [4]. Manolache surveys Cohen–Macaulay nilpotent structures and their filtrations [15]. Primitive multiple schemes carry a canonical filtration by lower multiplicities in the sense of Drézet [11].

Our local corank-one model

$$t(t, f) = (t) \cap (t, f)^2$$

indeed has a square-zero nilradical line on the embedded support, and the corank-two model has one further nonzero nilpotent power. We do not claim that the existence of a nilpotent filtration is new. The specific contribution is the following chain of implications, which does not occur in the cited multiple-structure theory:

$$\text{multiplication Fitting scheme} \implies (\mathcal{Z}_1, \widehat{\mathcal{N}}|_{\mathcal{Z}_1}) \implies \omega_{\mathcal{Z}_1} \otimes \widehat{\mathcal{N}}^{-9} \implies (Y_R, \mathcal{O}_{Y_R}(1)),$$

together with

$$\mathcal{Z}_2 = \text{rank-two Schubert support}$$

on the mixed-regular higher-corank locus. The ambient failure scheme is not assumed to be a primitive multiple scheme, and the filtration is defined by annihilators of powers of its intrinsic nilradical rather than by choosing an ambient thickening.

S15.4. Ballico’s failure-locus framework. The complete theorem text of Ballico’s 1993 article [2] has not been obtained for theorem-level comparison. We therefore do not infer either anticipation or non-anticipation from its title or metadata. A theorem-level comparison is possible with Ballico’s accessible 1996 continuation [3]. There the basic data are an integral projective variety X , a very ample line bundle L , the complete embedding by $H^0(X, L)$, and multiplication maps such as

$$H^0(L) \otimes H^0(L^k) \longrightarrow H^0(L^{k+1}).$$

The principal results propagate failure of quadratic or higher normality to suitable linear sections and finite subschemes; the parameter Grassmannian is a Grassmannian of projective linear sections.

The present construction has a different input and a different scheme-theoretic output. The fixed object is the finite local cube-zero algebra B_R ; a point of X_R is a unital subspace $\mathbb{C}1 \oplus K \subset B_R$; and the failure scheme is the zeroth Fitting scheme of

$$\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}.$$

Its embedded associated supports, nilradical powers, and the polarized K3 reconstructed from their line bundles are part of the statement. Ballico's 1996 theorems are therefore an important precedent for organizing multiplication failure over a Grassmannian, but they do not by themselves supply the Fitting normal forms or the nilpotent-depth reconstruction proved here. This comparison is deliberately limited to the theorem text actually inspected; the separate 1993 comparison remains documentary work rather than a premise in a novelty claim.

S15.5. Residue of the main theorem. After these classical comparisons, the claims that remain specific to this paper are concise.

First, an abstract multiplication-failure scheme reconstructs the quartic polarization by the intrinsic line $\omega_{E_R} \otimes \mathcal{N}_R^{-9}$ and its section ring. Second, the same nonreduced scheme has a functorial annihilator filtration whose next nonzero stage recovers the first higher-corank Schubert support. Third, on the mixed-regular rank-two locus the unsplit contact ideal has a complete collision specialization law, the fifth-power vertex component remains genuinely primary, and the associated graded nilpotent algebra identifies both the contact cone and the rank-two Schubert support. The first decomposable mixed-kernel wall is also computed scheme-theoretically. Finally, Theorem S15.3 identifies the inverse problem for the K3 alone with a finite étale monodromy cover, while Theorem S6.1 proves that the actual nonreduced failure cone generically recovers the web and separates that packet. These statements are independent of whether the nine-dimensional Reye family itself is classical.

S15.6. Comparison of the inverse data. Classical inverse constructions already recover a web from suitable Reye data. In [16, Proposition 3.6], a Fano polarization H determines its Reye bundle, when that bundle exists. Theorem 3.7 of that work realizes the polarized Enriques surface in a Grassmannian. In characteristic different from two, Remark 3.8 uses the canonical K3 cover, its involution, and two exchanged line bundles L^+, L^- . A four-dimensional multiplication kernel defines symmetric $(1, 1)$ -equations and hence a web. Thus reconstruction from an Enriques polarization with its Reye structure is not asserted here as new. Nor is every nodal Enriques surface a regular classical Reye web: the distinction is explicit in [16, Remark 1.2 and Section 6].

These inputs differ from a single quartic polarization. The first-layer invariant in Theorem S1.2 recovers $(Y_R, \mathcal{O}_{Y_R}(1))$, without assuming the Reye involution or the second exchanged polarization as marked data. Theorem S15.3 therefore retains the function-field degree of this particular unmarked map; the classical reconstruction with additional data does not compute that degree. Theorem S6.1 starts from neither a marked Reye surface nor a specified web. It starts from the abstract nonreduced Fitting scheme, recovers an oriented deepest normal cone, and extracts two specified coefficient lines through Lemma S6.5. Lemma S6.8 proves that an abstract isomorphism acts on both lines through the same projective transformation. The exterior-support theorem then makes recombination unique on the entire smooth Jacobian locus, without an additional genericity restriction. This input and this extraction mechanism are the inverse assertions proved here; classical web-to-Reye and web-to-Jacobian constructions are used as comparisons, not rebranded as new results.

S15.7. Role of the relative primary theorem. The relative algebra supplies a simultaneous model for a bounded principal-colon tower. Generic freeness, flattening, associated-point technology, and effective descent are standard ingredients [27, 20, 21, 29]. Our use of them is a finite-diagram construction, not a new principle of generic freeness or primary existence. The outputs needed here are simultaneous formation of the colons and their quotients after base change, exhaustion of the geometric associated points, closed support packets on the original strata, and relative torsion modules realizing the packet lengths. The diagonal injections, their cokernels, and the regular-element witnesses are kept in the same diagram so that these outputs hold together. Individual flattening of prime supports would not imply this combination, and no theorem about canonical labelled primary decompositions is asserted. The bounded-principal-colon theorem is therefore a relative-algebra tool for the geometric results, rather than the basis for an independent foundational-priority claim. The determinant identities, calculated boundary schemes, and intrinsic inverse theorem carry the geometric content.

S15.8. Extent of the primary tables. The primary decompositions computed in full are the stated generic corank-two coefficient strata, their specified collisions, and the quotient-induced benchmark families. The first two minimal-support laws and the sharp index-five nilpotence theorem apply globally on G_4° . All their proofs and coefficient tables are retained. The seven-component two-parameter incidence calculation describes the specified pullback of an incidence resolution; it does not identify all embedded primary components of W_3 and W_4 at coranks three and four. The inverse theorem does not require such an atlas: it reads the degree-twelve maximal-minor space itself from the degree-sixteen ideal, before choosing a primary decomposition. Accordingly its hypothesis and conclusion are unchanged by this separation of the two questions.

APPENDIX S16. THE AMBIENT COMPONENT PENCIL

S16.1. Secants, tangent fibres, and flag lines. We describe the scheme-theoretic fibres of the two-component invariant, including those not covered by the rank-two certificate. This is a geometric description of the residual ambiguity of that invariant, not an assertion that every pair in such a fibre has isomorphic failure schemes. Put $X = \text{Gr}(4, W)$, and let $X^\times$ be the open where both projections are nonzero. Consider the morphism

$$\pi : X^\times \longrightarrow \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35}), \quad [w] \longmapsto ([Pw], [Qw]).$$

For $[w] \in X^\times$, set $A = Pw$, $B = Qw$, and $H_w = \langle \ell_{F,w} : F \in \mathcal{I}_{\text{Pl},2} \rangle$. The rank $r_w = \dim H_w$ is independent of a representative.

Theorem S16.1 (Exceptional fibres of the component invariant). *The fibre of π through $[w]$, as a scheme, is the intersection of X with $\mathbb{P}(\mathbb{C}A \oplus \mathbb{C}B)$, with its two endpoints removed. In coordinates $[a : b]$ on this line its closed intersection with X has the following exhaustive forms.*

- (1) *If $r_w = 2$, its saturated ideal is $(b - a)$; it is the single reduced point $[w]$.*
- (2) *If $r_w = 1$, choose a generator $\ell = ca + db$ of H_w . Its saturated ideal is $((b - a)(ca + db))$. It is the length-two divisor $[1 : 1] + [d : -c]$, with multiplicity. The residual point equals $[w]$ exactly when $c + d = 0$; it is an excluded endpoint exactly when $cd = 0$. Otherwise it gives a second, distinct reduced point of the fibre of π .*
- (3) *If $r_w = 0$, the whole line lies in X , and the fibre is $\mathbf{G}_m$. Such a line consists precisely of the four-planes R in a fixed flag*

$$A_3 \subset R \subset B_5 \subset W, \quad \dim A_3 = 3, \quad \dim B_5 = 5.$$

Equivalently, the pure endpoints are decomposable and the two four-planes they represent intersect in dimension three.

On the locally closed reduced locus of case (2) with $cd(c+d) \neq 0$, passage to the residual point is a regular involution without fixed points. The differential of π has a nonzero kernel at $[w_R]$ precisely when

$$Q_{w_R} \in \bigwedge^3 R \wedge W. \quad (\text{S16.1})$$

This is case (3), or the double-point subcase of case (2). Its kernel is then one-dimensional. In particular, a distinct secant pair does not force infinitesimal failure of reconstruction.

Proof. The ambient projection from $\mathbb{P}(E_{175} \oplus E_{35})$ to the product of projective spaces has fibre $\mathbb{P}(\mathbb{C}A \oplus \mathbb{C}B) \setminus \{[A], [B]\}$. Pullback along the closed immersion of X proves the first assertion scheme-theoretically. The Grassmannian ideal is generated by its quadratic Pluecker relations. Their restrictions generate $(b-a)H_w$. If H_w is the full two-dimensional space of linear forms, these forms have no common projective zero and the ideal sheaf is $(b-a)$. If H_w is one-dimensional, the nonzero quadratic $(b-a)\ell$ generates an already saturated principal ideal in $\mathbb{C}[a, b]$, giving its two zeros with their multiplicities. If $H_w = 0$, every Pluecker equation vanishes identically on the line. This proves the three scheme structures, including the endpoint and double-point distinctions.

We recall the flag description with its proof. Choose a point R_0 on a line in the Grassmannian and a complement to R_0 in W . On the Pluecker affine chart of R_0 , the line is parametrized by graphs of $tM : R_0 \rightarrow W/R_0$. Indeed the one-replacement Pluecker coordinates are its matrix entries and are linear in t . The two-replacement coordinates are the two-by-two minors of tM . They are also linear functions on the line, so their quadratic coefficients vanish. Thus $\text{rank } M = 1$, since the line is nonconstant. Its members share $\ker M$ of dimension three and lie in $R_0 + \text{im } M$ of dimension five; closing the affine line gives the asserted flag line. Conversely the four-planes between such a flag form a Pluecker line by wedging a fixed basis of A_3 with a varying vector of B_5/A_3 . Applied to the endpoints, this also proves the equivalent formulation.

On a chart where one coefficient of a nonzero $\ell_{F,w}$ is invertible, its coefficient pair gives the regular map

$$[w] \mapsto [dA - cB].$$

Rank one makes different choices proportional, so these maps glue. The hypotheses $cd(c+d) \neq 0$ exclude endpoints and the double point. The residual point has the same two projective components; its line of recombination is the same line, whose intersection with X has exactly the two distinct reduced points. Repeating the construction returns $[w]$. This proves regularity and the involution assertion on the stated reduced stratum.

Finally the ambient projection has a one-dimensional vertical tangent at $[w]$, represented by B modulo $\mathbb{C}w$. The affine tangent space to the Pluecker cone at w_R is $\bigwedge^3 R \wedge W$, as follows by differentiating a wedge of four basis vectors of R . This proves (S16.1). Equivalently set $a = 1$, $b = 1 + t$; each restricted equation has derivative $\ell_{F,w}(1, 1)$. All these derivatives vanish exactly when $H_w = 0$, or when its generator vanishes at $[1 : 1]$. The rank-two case cannot have this property. The vertical tangent is nonzero because A, B are independent. This proves the last claims. $\square$

Corollary S16.2 (Candidate bounds without a genericity certificate). *On $X^\times$, away from the flag-line locus, the two-component invariant has fibres of length at most two. Failure of the rank-two certificate can represent either a double point, a genuine second candidate, or a discarded pure endpoint. Only the flag-line case allows a positive-dimensional fibre. For an abstract failure-scheme isomorphism these are necessary candidate alternatives after one common projective change of coordinates.*

Proof. Remove the two endpoints from the three closed intersections in the theorem, and apply Lemma S6.8 for the last assertion. No converse from equality of component pairs to isomorphism of failure schemes is used. $\square$

The theorem describes every fibre configuration on $X^\times$. Theorem S6.13 below determines their incidence with the smooth Jacobian open: only the rank-two case occurs. No ambient irreducible-component or codimension classification is implicit in the fibre classification itself. It separates positive-dimensional flag ambiguity from tangent degeneracy and from two distinct candidates; these distinctions are invisible in the single rank-failure condition.

Example S16.3 (A nonempty flag-line stratum). In the monomial notation of Appendix S26 put

$$A = b_1 \wedge b_2 \wedge b_3 \wedge b_5, \quad B = b_1 \wedge b_2 \wedge b_3 \wedge b_4.$$

These are highest-weight vectors of weights $(4, 3, 1, 0)$ and $(5, 1, 1, 1)$; hence $A \in E_{175}$, $B \in E_{35}$. Every $aA + bB$ is the decomposable vector $b_1 \wedge b_2 \wedge b_3 \wedge (ab_5 + bb_4)$. Thus the flag-line stratum is nonempty in $X^\times$, and this example supplies an entire $\mathbf{G}_m$ -fibre. It is not an example on G_4° : the first three quadrics have the common factor x_1 , and the fourth quadric has a zero on $x_1 = 0$.

S16.2. Algebraic rank strata and Fitting ramification. The pencil classification admits a global scheme-theoretic formulation. On $X^\times$, let $\mathcal{L} = \mathcal{O}_X(-1)$, and let $\mathcal{A} = P\mathcal{L}$, $\mathcal{B} = Q\mathcal{L}$ be its two image line bundles in $E \otimes \mathcal{O}_X$. Set $\mathcal{M} = \mathcal{A} \oplus \mathcal{B}$. The tautological line $\mathcal{L} \subset \mathcal{M}$ is the diagonal under the isomorphisms induced by P and Q . Restricting the Pluecker quadrics gives a morphism

$$\mathcal{I}_{\mathbb{P}^1, 2} \otimes \mathcal{O}_{X^\times} \xrightarrow{\rho} \mathcal{K} := \ker(\mathrm{Sym}^2 \mathcal{M}^* \longrightarrow (\mathcal{L}^*)^{\otimes 2}). \quad (\text{S16.2})$$

The target has rank two. Multiplication identifies it with $(\mathcal{M}/\mathcal{L})^* \otimes \mathcal{M}^*$. The map lands in this kernel because each quadric vanishes on the original Pluecker point. A local generator of $\mathcal{L}$ identifies ρ with the family $(b - a)\ell_{F,w}(a, b)$.

Lemma S16.4 (Residual Cartier sections over the rank-one scheme). *Endow $T = D_1 \setminus D_0$ with the determinantal locally closed scheme structure, not its reduction. On a bundle chart the restricted Pluecker ideal on $\mathbb{P}(\mathcal{M}_T)$ is generated by*

$$(b - a)(ca + db), \quad (c, d) = \mathcal{O}_T.$$

Let $T_{\mathrm{sep}} \subset T$ be the open where the residual section avoids both endpoints and the diagonal. In these coordinates it is $T \setminus V(cd(c + d))$. The closed pencil intersection over this open is the disjoint union of the effective Cartier sections

$$s = [1 : 1], \quad r = [d : -c].$$

All these statements commute with arbitrary base change, including to a nonreduced base. The residual map $\tau : T_{\mathrm{sep}} \rightarrow T_{\mathrm{sep}}$ is a regular involution and its fixed-point scheme is empty.

Proof. On $D_1 \setminus D_0$ some matrix entry of ρ is invertible. Its two-by-two minors vanish in the structure sheaf; row and column operations over that sheaf therefore put the matrix in rank-one normal form. The coefficient rows of the residual linear forms generate a line direct summand, represented by a unimodular pair (c, d) . Thus the displayed generator is an ideal-sheaf equality over T , not just a fibrewise equation.

A unimodular linear form on a projective-line bundle cuts out a Cartier section over any base. For example on an open where its second coefficient is a unit, its equation on the relevant affine chart is a unit multiple of a monic degree-one polynomial. Such a polynomial is a non-zero-divisor over an arbitrary coefficient ring. This proves the Cartier assertions for $b - a$ and $ca + db$. The two forms have determinant $-(c + d)$. Where $c + d$ is a unit their zero sections are disjoint, so their ideal sheaves are comaximal. Their product equals their intersection, and the Chinese remainder isomorphism identifies the divisor with $s(T_{\mathrm{sep}}) \amalg r(T_{\mathrm{sep}})$. The additional units c, d exclude the two pure endpoints. This construction and these unit conditions are unchanged by any base change.

The section r lies in the Grassmannian as a scheme: substitution in every restricted Pluecker equation gives zero. It has both Schur components nonzero. At this point the component frames

are changed from (Pw, Qw) to $(dPw, -cQw)$, an invertible diagonal change on T_{sep} . Writing $a = da', b = -cb'$, its equation becomes

$$cd(b' - a')(da' + cb').$$

Thus the new residual coefficient pair is (d, c) , up to a unit; its section $[c : -d]$ in the new frames represents $cd(Pw + Qw)$, which is exactly s . Consequently its residual coefficient matrix again has rank one as a matrix over the base: the remaining linear factor is unimodular, and at least one generating coefficient is a unit. Thus r factors through $D_1 \setminus D_0$, including its determinantal scheme structure. Its residual section is s and is disjoint from it and from the endpoints, so the map factors through T_{sep} . Changing trivializations multiplies these equations by units and does not change their sections; the local morphisms therefore glue. Division by the other Cartier factor a second time gives $\tau^2 = \text{id}$ as morphisms. The disjointness of the sections gives an empty equalizer of τ and the identity, proving the fixed-point-scheme assertion. $\square$

Proposition S16.5 (The rank stratification and its finite part). *Let D_1 be the closed subscheme defined by the two-by-two minors of ρ , and D_0 the one defined by its entries. The rank strata*

$$X^\times \setminus D_1, \quad D_1 \setminus D_0, \quad D_0$$

have ranks two, one and zero, respectively. On the rank-one stratum the residual point is a regular section of $\mathbb{P}(\mathcal{M})$. Deleting its endpoint and diagonal loci gives a regular fixed-point-free involution. The restriction of π to $X^\times \setminus D_0$ is quasi-finite.

Put $Y = \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35})$ and $\mathcal{R} = V(\text{Fitt}_0 \Omega_{X^\times/Y})$. We call $\mathcal{R}|_{X^\times \setminus D_0}$ the Fitting ramification locus of the quasi-finite part. This convention records the failure to be unramified, equivalently the nonvanishing of the relative differential module [33, Lemma 29.36.14]; it does not assert a globally finite or flat double cover. The same Fitting ideal is defined on all of $X^\times$. Its local formula is

$$(\ell_{F,w}(1, 1) : F \in \mathcal{I}_{\text{Pl},2}). \quad (\text{S16.3})$$

On the rank-one finite stratum it is exactly the double-point subscheme, defined locally by $c + d = 0$ when $\ell = ca + db$. It is empty on the rank-two stratum. In particular π is unramified on G_4° .

Proof. The determinantal definitions are invariant under changes of bundle frames. Where a rank-one matrix entry is invertible, its two-column coefficient vectors are all multiples of one unimodular vector (c, d) ; its image is a line subbundle of $\mathcal{K}$. Dividing by the equation of the diagonal gives the line of residual linear forms, whose zero is a regular section of $\mathbb{P}(\mathcal{M})$. This also works on the indicated scheme structures, not merely on their closed points. The residual section lies in X , because every restricted Pluecker quadric vanishes on it. The Cartier-section and involution assertions, with their nonreduced scheme structures, follow from Lemma S16.4. On the complement of D_0 , all geometric fibres have length at most two. The morphism is of finite type, hence quasi-finite there.

Let $U \subset \mathbb{P}(E)$ be the open where both components are nonzero. The projection $U \rightarrow Y$ is a $\mathbf{G}_m$ -bundle. In a local trivialization its relative differential module is generated by the torus differential. The conormal exact sequence for $X^\times \subset U$ presents $\Omega_{X^\times/Y}$ as the quotient of this line bundle by the relative derivatives of the Pluecker quadrics. At a point represented by w , put $a = 1, b = 1 + t$. The derivative of $(b - a)\ell_{F,w}(a, b)$ in the vertical direction is $\ell_{F,w}(1, 1)$. This proves (S16.3), including its scheme structure. At rank one one coefficient multiplier is a unit, so the ideal restricts to $(c + d)$. At rank two two independent linear forms cannot both vanish at the diagonal point. The last assertion now follows from Theorem S6.13. $\square$

For clarity, the lengths in the ambient closed pencil and in a fibre of π are different in the endpoint case:

Rank	Type	Closed pencil intersection	Fibre after endpoint deletion
2	reduced point	length 1	length 1
1	distinct secant	two reduced points	length 2
1	double point	length 2, nonreduced	length 2, nonreduced
1	pure endpoint	two reduced points	length 1
0	flag line	$\mathbb{P}^1$	$\mathbf{G}_m$

The first row is the whole of G_4° , a nonempty smooth open of $\mathrm{Gr}(4, 10)$ of dimension 24 and codimension zero in that Grassmannian. Every other row has empty intersection with G_4° . Thus there is no positive-codimension exceptional stratum in this geometric family whose dimension or generic singularity remains to be computed. The determinantal schemes may have further geometry on the binary-Jacobian boundary; the preceding assertions do not assign an expected codimension to those ambient schemes or make π into a global double cover.

S16.3. An alternative coordinate character calculation. The following direct calculation is independent of the character decomposition in Lemma S6.9; it is not needed for the main proof. Use an orthonormal basis and write $\rho = \sum_i x_i^2$.

Any possible image of ρ^2 , with its determinant twist, must transform by the determinant character of O_4 in $\wedge^3 \mathrm{Sym}^2 V$. There is no such vector. To see this directly, invariance with this character under the four coordinate reflections requires each variable to have odd total degree in any nonzero exterior monomial. Its total degree is six, so the multidegree must be a permutation of $(3, 1, 1, 1)$. For a fixed variable x_i of degree three, the possible monomials are

$$\omega_i = \bigwedge_{j \neq i} x_i x_j, \quad x_i^2 \wedge (x_i x_j) \wedge (x_k x_l) \quad (\{i, j, k, l\} = \{1, 2, 3, 4\}).$$

The coefficient of the second kind is zero: the transposition of x_k, x_l fixes that monomial but has determinant minus one. Only a combination of the ω_i remains. Applying the infinitesimal rotation $x_i \partial / \partial x_j - x_j \partial / \partial x_i$, the coefficient of $x_i^2 \wedge (x_i x_k) \wedge (x_i x_l)$ comes, up to sign, only from ω_i . Invariance forces its coefficient to vanish, for every i . This proves the assertion.

APPENDIX S17. BOUNDARY STRUCTURE AND FAMILY THEOREMS

The following statements organize the independent boundary and family results proved in the subsequent appendices. They retain the full primary and specialization theory while separating it from the logical spine of the inverse theorem.

S17.1. Sharp global laws. The general exponent has a sharper geometric form on the open G_4° . Theorem S24.2 proves

$$\widehat{\mathcal{N}}_R^5 = 0, \quad \widehat{\mathcal{N}}_R^4 \neq 0 \text{ at every point of projection corank three or four.}$$

The reason is global rather than semicontinuous: the determinant-four identity in the full matrix coordinate ring restricts to every Schur chart. The needed coordinate-ring Pieri projection is calculated with the Fischer form; its coefficient is $1/35$, not an unspecified straightening coefficient. The lower bound at corank three is the Schur-symbol exclusion proved below. The coefficient-space tables are retained in full, but Proposition S24.3 determines necessary rank conditions for their occurrence on the smooth, basepoint-free geometric open.

There is also a global geometric description beyond determinant depth. Let Z_R be the image of the incidence vector bundle $\mathrm{Hom}(V, \mathcal{H})$ over the Jacobian K3 surface, and let D_2 be the rank-at-most-two matrix variety. Theorem S24.4 identifies these as the unique minimal supports of the first and second layers. The incidence resolution has fibres given by linear sections of the same K3: four contact points, a plane quartic of genus three, and the entire surface. Theorem S24.6

realizes this sequence in one two-parameter family and computes its seven reduced components. Proposition S24.7 computes the intrinsic graded residue-field algebras across the same corank transitions. These statements do not assert a complete list of all embedded transverse primes of the third and fourth layers.

The relative theorem is proved by primary modules, rather than by an inference from open-containment constructibility. A diagonal injection into filtered primary quotients excludes every untracked associated point; associated-element injections give the opposite inclusion. A regular element on the quotient by a coherent torsion module prevents new packet torsion after specialization. Both kinds of injections survive by flattening their actual cokernels. In the proper quotient-induced family, the classical Bruns–Vasconcelos identity [25] further gives $dJ = (d) \cap P^{a+1}$, an exact exponent and explicit embedded lengths (Theorem S24.9). The classical minor identity itself is not a novelty claim of this paper.

The argument has two scales. The first is universal: determinant completion bounds the principal nilpotence exponent in every symmetric power. The second is relative: once such an exponent is bounded, the entire colon tower is finite and can be stratified simultaneously. Combining the two produces a uniform theorem on the parameter space of surjections $\gamma : \text{Sym}^r V \rightarrow S$: after finite stratification, all residual colons, geometric associated-point packets, embedded multiplicities, and multiplication ranks are compatible with arbitrary base change. The bounded principal-colon theorem proved below makes kernel formation commute with arbitrary base change before any associated-point argument is invoked. Actual primary quotients then exhaust the geometric associated set. Orbit-ideal descent and a power-certified construction on the original base produce the closed packets; a relative torsion module controls their embedded multiplicities. For the $(1, 4, 6)$ problem this mechanism is supplemented by exact rational-function-field calculations on transverse stabilizer slices and by explicit boundary degenerations.

S17.2. The intrinsic graded boundary packet. Write $\widehat{\mathcal{N}}_R$ for the nilradical of $\mathcal{O}_{\widehat{D}_R}$. The schemes

$$\mathcal{Z}_j(\widehat{D}_R) = V\left(\text{Ann}(\widehat{\mathcal{N}}_R^j)\right)$$

and the products among $\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1}$ are invariant under every abstract isomorphism of $\widehat{D}_R$. On a Schur chart the reduced equation is $d = \det T$ and

$$\mathcal{I}_{\widehat{D}_R} = dJ.$$

The residual-colon theorem of Section S21 identifies

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad W_j = V((d) + (J : d^{j-1})). \quad (\text{S17.1})$$

The coordinate expression changes from chart to chart; the modules and schemes in (S17.1) do not.

Theorem S17.1 (Boundary structure). *For $R \in G_4^\circ$ the following hold.*

- (1) *On every projection-rank chart,*

$$d^5 \in J, \quad \widehat{\mathcal{N}}_R^6 = 0.$$

This is the $(e, r) = (4, 2)$ case of the universal determinant completion theorem, whose exponent $\binom{e+r-1}{r-1}$ is sharp in the universal class.

- (2) *The multiplication of the positive graded pieces is the canonical quotient multiplication*

$$\mathcal{O}_{W_i} \otimes \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}$$

with the expected conormal-line twist. No additional scalar multiplication constants occur.

- (3) *If the pure block A_H is injective at projection corank two, the mixed block has nine orbit types. On the generic coefficient stratum of every orbit, the complete nonzero graded supports through W_3 , their Hilbert functions, radicals, associated supports, and vertex-primary corrections are given in Theorems S18.4 and S23.2. Genericity is proved over rational function fields on transverse stabilizer slices.*
- (4) *If A_H drops rank at projection corank two, every generic rank type (a, b) has an explicit primary signature. Besides the Segre quadric, the only positive-dimensional generic supports that occur are reduced divisors of two or four lines in one ruling. The remaining layers are vertex-primary; in the $(a, b) = (1, 3)$ row a length-three embedded vertex accompanies the two ruling lines.*
- (5) *At projection corank three, $d^3 \notin J$ and $d^4 \in J$; the index is exactly five. The upper bound is the restriction of the full-matrix identity, as proved in Theorem S24.2. The lower bound is the functorial Schur-symbol exclusion.*
- (6) *At projection corank four,*

$$d^3 \notin J, \quad d^4 \in J.$$

The second membership is proved from the actual Cauchy–Pieri multiplication in $\mathbb{C}[\text{End}(V)]$: the product of the $(4, 4, 4, 0)$ - and (4) -components has nonzero $(4, 4, 4, 4)$ -projection. Since that line is spanned by d^4 , the local nilpotency index is exactly five.

Thus the phrase *boundary atlas* refers to intrinsic graded data, not to the chosen coefficient matrices. The Macaulay matrices form a finite presentation atlas. Any two such presentation atlases have a common refinement and compute the same intrinsic primary signature.

S17.3. Primary specialization. The preceding finite calculations are promoted to family statements in Section S23. The key homological step is explicit. If $B = P_S/J$, $C_j = B/d^{j-1}B$, and $I_j = d^{j-1}B$, we first flatten B and C_j . The exact sequence $0 \rightarrow I_j \rightarrow B \rightarrow C_j \rightarrow 0$ then makes I_j flat, and $0 \rightarrow K_j \rightarrow P_S \rightarrow I_j \rightarrow 0$ remains exact after arbitrary base change. Only after the equality $(K_j)_s = (J_s : d_s^{j-1})$ has been established do we apply relative assassins. Finite primary models and the two opposite associated-point inclusions give exact geometric support packets. A regular element on the quotient by a coherent torsion module excludes new torsion after specialization, and a finite power filtration computes the local lengths. The closed packets descend as ideals with an explicit cocycle in characteristic zero. In arbitrary characteristic their ideals are constructed downstairs first and their geometric components are certified by finite power inclusions on a splitting cover. The relative torsion module is also constructed downstairs. Noetherian induction then handles the complement.

The same section writes explicit one-parameter degenerations of the mixed block. For example an off-Segre kernel point specializes to a Segre point; a secant kernel line specializes to a tangent or either ruling line; rank-two kernels specialize to the two rank-one types; and both rank-one types specialize to the zero mixed map. The graded schemes in the two fibres are computed, so the specialization records the birth of vertex-primary and ruling components rather than merely changing orbit labels.

S17.4. The simple-contact locus. The general atlas contains the earlier primary theorem as its split form. We retain the following statement because it is the geometric entry point to the collision analysis.

Theorem S17.2 (Corank-two primary structure on the simple-contact open). *There is a nonempty invariant open $G_4^\dagger \subset G_4^\circ$ such that, for each $R \in G_4^\dagger$, a dense open set of rank-two images $H \subset V$*

has, after an étale extension and smooth auxiliary coordinates,

$$\mathcal{I}_{\widehat{D}_R} = (\delta) \cap P_1^2 \cap P_2^2 \cap P_3^2 \cap P_4^2 \cap \mathfrak{m}^5. \quad (\text{S17.2})$$

The rank-two stratum is an additional associated support and $\widehat{\mathcal{N}}_R^3 = 0 \neq \widehat{\mathcal{N}}_R^2$.

Repeated roots, the containment locus, the decomposable-kernel wall, and all remaining mixed-kernel orbits are treated without splitting the contact quartic.

S17.5. Variation and finite ambiguity.

Theorem S17.3 (Variation and finite ambiguity). *For $e = 4$, the image of*

$$R \longmapsto (Y_R, \mathcal{O}_{Y_R}(1))$$

in quartic K3 moduli has dimension nine. On a nonempty invariant open the induced map from projective equivalence classes of relation spaces to its image is generically finite.

The nine-dimensional count is classical Reye geometry, not a novelty claim. The polarized K3 alone is not identified with the original web. Theorem S6.1 instead uses the deepest nonreduced normal cone to recover every smooth Jacobian web, and Corollary S6.15 separates the finite packet. The degree of the map defined by the K3 alone is not asserted to be one. The sharp exponent, the two minimal supports, and the explicit cross-corank geometry of Section S24 remain independent global consequences of the primary-boundary analysis.

APPENDIX S18. UNIVERSAL DETERMINANT COMPLETION AND THE INTRINSIC BOUNDARY ATLAS

The residual-colon filtration of Theorem S21.1 is intrinsic at every projection rank. We first isolate the determinant-completion mechanism in arbitrary dimension and symmetric degree. We then compute the projection-corank-two boundary on genuine generic coefficient strata, including all pure-block rank drops, and finally analyze projection coranks three and four. The finite coefficient matrices used below form a presentation atlas; the schemes W_j , their associated points, generic lengths, and graded multiplication form the intrinsic boundary packet.

Keep the notation of Theorem S20.1. Thus, at a point of projection rank r , put

$$H = \text{im}(\mathcal{K} \rightarrow V), \quad W = V/H, \quad c = \dim W = 4 - r,$$

choose a Schur chart

$$M = I_H \oplus T, \quad d = \det T,$$

and write

$$\mathcal{I}_{\widehat{D}_R} = dJ, \quad J = I_6 \left[A_H, B_H(1_H \otimes T), C_H \text{Sym}^2 T \right]. \quad (\text{S18.1})$$

The local nilradical is generated by the class of d .

Theorem S18.1 (Universal symmetric-power determinant completion). *Let V be a complex vector space of dimension e , let $r \geq 1$, and let*

$$\gamma : \text{Sym}^r V \rightarrow E$$

be a surjection, with $s = \dim E$. On $\text{End}(V)$ let M be the generic endomorphism, put $d = \det M$, and set

$$J_{\gamma,r} = I_s(\gamma \text{Sym}^r M).$$

Then

$$d^{\kappa(e,r)} \in J_{\gamma,r}, \quad \kappa(e,r) = \binom{e+r-1}{r-1}. \quad (\text{S18.2})$$

The exponent is sharp in the universal class: if $E = \text{Sym}^r V$ and γ is an isomorphism, then $J_{\gamma,r} = (d^{\kappa(e,r)})$.

For $(e, r) = (4, 2)$, $\kappa(4, 2) = 5$. Thus the global depth bound below is a specialization of a dimension-free determinant-completion principle, rather than an isolated identity for the Hilbert function $(1, 4, 6)$.

Theorem S18.2 (Finite all-corank depth and multiplication). *For every $R \in G_4^o$ and every projection-rank chart,*

$$d^5 \in J. \quad (\text{S18.3})$$

Consequently

$$\widehat{\mathcal{N}}_R^6 = 0$$

on the whole Grassmannian. In particular

$$W_j = \emptyset \quad (j \geq 6),$$

and the special fibre of the intrinsic normal-cone specialization has at most five positive graded layers.

For $i, j \geq 1$, $i + j \leq 5$, the multiplication map

$$\frac{\widehat{\mathcal{N}}^i}{\widehat{\mathcal{N}}^{i+1}} \otimes \frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \longrightarrow \frac{\widehat{\mathcal{N}}^{i+j}}{\widehat{\mathcal{N}}^{i+j+1}}$$

is, under the identifications

$$\frac{\widehat{\mathcal{N}}^j}{\widehat{\mathcal{N}}^{j+1}} \simeq \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

the tensor twist of the canonical quotient multiplication

$$\mathcal{O}_{W_i} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j} \longrightarrow \mathcal{O}_{W_{i+j}}. \quad (\text{S18.4})$$

In particular it is surjective. Thus no additional multiplication constant is hidden after the schemes W_j are known.

The coefficient-space tables below concern surjective quadratic quotients on their stated generic coefficient strata. Their intersection with the smooth, basepoint-free web open is subject to Proposition S24.3. In particular the ambient index-six rows are not asserted to occur on that geometric open. Theorem S24.2 strengthens the global bound there to nilpotency index five.

We next sharpen the corank-two analysis. Assume A_H is injective. Put

$$E_H = S_R / \text{im } A_H, \quad \beta_H = \overline{B}_H : H \otimes W \longrightarrow E_H, \quad \chi_H = \overline{C}_H : \text{Sym}^2 W \longrightarrow E_H.$$

Here $\dim H = \dim W = 2$ and $\dim E_H = 3$. Let

$$Q = \mathbb{P}(H \otimes W)_{\text{rk}=1} \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

be the Segre quadric.

Definition S18.3 (Mixed-kernel orbit type). The *mixed-kernel orbit type* of (R, H) is the rank $b = \text{rank } \beta_H$, together with the incidence type of $\mathbb{P}(\ker \beta_H)$ with Q .

For $b = 3$ this is a point off Q or on Q . For $b = 2$ it is a secant line, a tangent line, an H -ruling $\mathbb{P}(h_0 \otimes W)$, or a W -ruling $\mathbb{P}(H \otimes w_0)$ contained in Q . For $b = 1$, the one-dimensional image of $\beta_H^* : E_H^* \rightarrow (H \otimes W)^*$ is generated by a tensor of rank two or rank one; equivalently the kernel plane cuts Q in a smooth conic or in a pair of ruling lines. The final type is $b = 0$.

These are exactly the nine $GL(H) \times GL(W) \times GL(E_H)$ -orbits for the mixed block. The first two are the mixed-regular and decomposable-kernel cases already encountered in Section S21; the remaining seven are higher-dimensional mixed kernels or failures of mixed surjectivity.

For a homogeneous ideal $K \subset P = \mathbb{C}[a, b, c, d]$, write

$$H_{P/K}(n) = \dim_{\mathbb{C}}(P/K)_n.$$

For each orbit, the *generic coefficient stratum* is the dense open obtained from a transverse stabilizer slice by deleting the explicit denominator and leading-coefficient divisors in Theorem S23.1. Equivalently it is the locus on which the rational-function-field Groebner bases used there retain their initial ideals. The isolated integral matrices listed in the proof are base points of those slices; they are not used as a substitute for genericity.

Theorem S18.4 (Corank-two mixed-kernel atlas). *Assume A_H is injective. Then the first positive graded support is determined without any mixed-regularity assumption:*

$$\mathcal{O}_{W_1} \simeq P/(\delta, g_0, \dots, g_4), \quad \delta = ad - bc, \quad (\text{S18.5})$$

where, on $P/(\delta)$ with $T = uv^t$, the degree-four classes span

$$h_H(u) \operatorname{Sym}^4(\mathbb{C}_v^2).$$

Thus, if $h_H \neq 0$,

$$\operatorname{Proj} W_1 = V(h_H) \times \mathbb{P}_v^1 \subset Q,$$

whereas $h_H = 0$ gives $W_1 = V(\delta)$.

On the generic coefficient stratum of each mixed-kernel orbit, the remaining graded supports and the exact local nilpotency index are as follows.

b	mixed-kernel type	index	W_2	W_3
3	point off Q	3	length 1	empty
3	point on Q	3	length 3, $H = (1, 2)$	empty
2	secant line	3	length 8, $H = (1, 4, 3)$	empty
2	tangent line	3	length 8, $H = (1, 4, 3)$	empty
2	H -ruling $\mathbb{P}(h_0 \otimes W)$	4	length 15, $H = (1, 4, 6, 4)$	length 1
2	W -ruling $\mathbb{P}(H \otimes w_0)$	3	$H(n) = n + 1$ for $n \geq 4$	empty
1	dual tensor rank 2	4	$H(n) = 2n + 2$ for $n \geq 3$	length 1
1	dual tensor rank 1	4	$H(n) = 3n + 3$ for $n \geq 3$	length 3
0	zero mixed map	4	$V(\delta)$	$V(\delta)$

For $b = 0$, surjectivity of the original quotient multiplication forces χ_H to be an isomorphism and

$$J = (\delta^3); \quad (\text{S18.6})$$

hence the last row is an exact identity, not only a generic one.

In the H -ruling and $b \leq 1$ rows one has $h_H = 0$. On the W -ruling row the contact quartic is nonzero on the transverse open and has the root forced by w_0 . For $b = 1$, the projectivization of W_2 has degree two in the dual-rank-two case and degree three in the dual-rank-one case. The length-one and length-three third layers are respectively the reduced rank-two vertex and the length-three vertex thickening already appearing on the decomposable-kernel wall.

The complement of the generic coefficient stratum is itself part of the atlas, rather than an omitted exceptional set. We make this finite refinement explicit.

Definition S18.5 (Macaulay valuation atlas). Let $P_c = \mathbb{C}[t_{ij} : 1 \leq i, j \leq c]$, $d = \det T$, and let J be the residual ideal on a fixed projection-rank chart. For $1 \leq k \leq 5$, let

$$\mathbf{M}_k(J)$$

be the coefficient matrix whose columns are all products

$$m q, \quad q \in \mathcal{G}(J), \quad \deg m + \deg q = ck,$$

written in the monomial basis of $(P_c)_{ck}$, where $\mathcal{G}(J)$ is the displayed set of maximal minors in (S18.1). Let v_k be the coefficient vector of d^k . Define

$$\varepsilon_k(J) = \operatorname{rank}[\mathbf{M}_k(J) \mid v_k] - \operatorname{rank} \mathbf{M}_k(J) \in \{0, 1\}.$$

Then

$$\varepsilon_k(J) = 0 \iff d^k \in J.$$

The locally closed loci obtained by fixing the ranks of the matrices M_k , their augmented matrices, and the analogous matrices for the kernels

$$K_j = (J : d^{j-1}), \quad 1 \leq j \leq 5,$$

are called the *Macaulay valuation atlas*. All of their equations are minors of explicit finite coefficient matrices.

Theorem S18.6 (Bounded principal-colon primary stratification). *Let S be noetherian, let A be a finitely presented S -algebra which is flat over S , let $J \subset A$ be a finitely generated ideal, and let $f \in A$. Assume that $f^N \in J$ for some $N \geq 1$. Put*

$$B = A/J, \quad K_q = (J : f^q), \quad M_q = A/(f, K_q), \quad 0 \leq q < N.$$

Then S admits a finite locally closed stratification $S = \coprod_{\alpha} S_{\alpha}$ with the following properties.

For every α , all the assertions below are preserved by an arbitrary base change $T \rightarrow S_{\alpha}$. Here J on a stratum means the image ideal, and each K_q is formed anew over that stratum. No commutation across different strata is asserted.

- (1) *Formation of J and of every K_q commutes with base change. The modules*

$$B, \quad C_q = A/(J, f^q), \quad I_q = f^q B, \quad M_q$$

are flat over S_{α} .

- (2) *For each q there are finitely many closed relative support packets $Z_{q,\nu} \subset \text{Spec } A_{S_{\alpha}}$, flat after a further refinement, such that for every geometric point $\bar{s} \rightarrow S_{\alpha}$ the associated points of $(M_q)_{\bar{s}}$ are exactly the generic points of the irreducible components of the nonempty $(Z_{q,\nu})_{\bar{s}}$. Geometric splitting of a packet is allowed. Dimensions, incidence, and minimal-versus-embedded status are constant on the stratum.*

- (3) *If p is one of these geometric associated points, the canonical embedded multiplicity*

$$\mu_p(M_q) = \text{length}_{\mathcal{O}_{\text{Spec } A_{\bar{s}},p}} H_{p, \mathcal{O}_{\text{Spec } A_{\bar{s}},p}}^0((M_q)_p)$$

is constant along the corresponding packet.

- (4) *Any prescribed finite collection of maps between the M_q may be included in the same refinement so that the relevant source, target, image and cokernel modules are flat, and their defining exact diagrams commute with arbitrary base change. When grading is present, degreewise numerical ranks are given by Lemma S23.12.*

Thus a bounded principal power converts the a priori infinite primary problem into a finite, base-change-compatible invariant.

Corollary S18.7 (Universal primary finiteness for symmetric-power quotients). *Fix $e, r \geq 1$ and $1 \leq s \leq \dim \text{Sym}^r V$, and fix a target vector space E of dimension s . Let $\mathcal{U}$ be the open parameter space of surjections $\gamma : \text{Sym}^r V \twoheadrightarrow E$. On a coefficient affine chart write*

$$A = \mathcal{O}_{\mathcal{U}}[t_{ij}], \quad \deg t_{ij} = 1, \quad \deg \mathcal{O}_{\mathcal{U}} = 0, \quad d = \det T, \quad J = I_s(\gamma \text{Sym}^r T), \quad N = \binom{e+r-1}{r-1}.$$

There is a finite locally closed stratification of $\mathcal{U}$ on which all the colons $K_q = (J : d^q)$, $0 \leq q < N$, commute with arbitrary base change within a stratum. The graded modules

$$M_q = A/(d, K_q), \quad L_j = M_{j-1}(-ej), \quad 1 \leq j \leq N,$$

are flat. Their geometric associated-support packets, containment posets, minimal and embedded status, and canonical local torsion lengths are constant. Their Hilbert functions, and therefore their Hilbert polynomials, are constant on each connected final stratum.

The finite family of natural multiplication maps meant here is

$$\mu_{i,j} : L_i \otimes_A L_j \longrightarrow L_{i+j}, \quad i, j \geq 1, \quad i + j \leq N. \quad (\text{S18.7})$$

These maps are surjective homogeneous maps; their sources, targets, and images are flat and their ranks in every degree commute with base change and are constant. A prescribed finite list of additional maps between fixed coefficient-degree pieces can also be included in the refinement. No simultaneous rank assertion for an unspecified infinite family of such additional maps is intended.

Proof. The universal determinant-completion identity gives $d^N \in J$. Apply Theorem S18.6 to the finite list of homogeneous modules M_q . The primary models and relative torsion construction in Propositions S23.5 and S23.6 give the associated-support and multiplicity assertions. Apply Lemma S23.12 to obtain the Hilbert functions and polynomials. This is an additional graded consequence, not a clause of an ungraded flattening theorem.

The colon ideals form an increasing chain. Thus, with the indicated shifts, $L_i \otimes_A L_j$ is the quotient $M_{\max(i-1, j-1)}$, and the multiplication in (S18.7) is the canonical quotient to M_{i+j-1} . Its source and target are among the already flat modules and its cokernel is zero. Lemma S23.12 applies in every degree. Finally, any additionally prescribed finite list of maps between finite coefficient-degree modules is handled by flattening its finite list of images and cokernels. Affine coefficient charts admit a common finite refinement. $\square$

Definition S18.8 (Primary signature and primary components). In the universal statements a *primary signature* is the package of residual modules, associated-support packets, their containment poset and local torsion lengths, graded Hilbert functions, and the finite family (S18.7). This package is intrinsic. A primary decomposition means an actual intersection of primary ideals or primary submodules. Such noncanonical decompositions are constructed on finite splitting covers by Proposition S23.5; they need not descend as individually labelled components and are not asserted to be canonical or explicitly classified throughout the universal family.

The theorem is independent of the special $(1, 4, 6)$ model. In the present paper the universal determinant-completion theorem supplies $N = 5$ on every projection-rank chart. Consequently the residual colon algebra, its associated-point packets, and all graded multiplication maps admit one finite primary stratification before any orbit-specific computation is made. Because of Theorem S18.2, no matrix with $k > 5$ occurs. We refine these rank strata simultaneously by the flattening strata of the finite modules

$$\mathcal{M}_j = P_c / ((d) + K_j), \quad 1 \leq j \leq 5,$$

by the flattening strata of the cokernels of (S18.4), and by the associated-point stratification in the following proposition.

Proposition S18.9 (Finite intrinsic primary-signature refinement). *On every finite-type coefficient chart for (A_H, B_H, C_H) , the presentation atlas admits a finite locally closed refinement such that the following statements hold for every geometric fibre.*

- (1) *Formation of each residual colon $K_j = (J : dj^{-1})$, $1 \leq j \leq 5$, commutes with base change, and*

$$\mathcal{M}_j = P_c / ((d) + K_j)$$

is flat over the stratum with constant Hilbert polynomial.

- (2) *There are finitely many relative closed supports $Z_{j,\nu}$ whose nonempty fibrewise generic points are exactly $\text{Ass}((\mathcal{M}_j)_s)$. Their dimensions, minimal-versus-embedded status, incidence relations, and generic lengths are constant on the stratum.*
- (3) *The cokernels of the graded multiplication maps are flat, so their ranks are constant and commute with base change.*

Consequently the radicals, associated supports, generic multiplicities, nilpotency index, Hilbert polynomials, and multiplication ranks of all positive graded layers have constant intrinsic primary signature on each final stratum.

The refinement is not asserted to be canonical as a collection of coordinate strata. The primary signature is intrinsic; different Schur charts, generating sets, or monomial orders admit a common finite refinement and compute the same signature.

This refinement is used below for rank drops. The proof in Section S23 is the application of Theorem S18.6: it flattens both P_S/J and $P_S/(J, d^{j-1})$, deduces flatness of the image of multiplication by d^{j-1} , and only then invokes relative assassins. Thus coefficient collisions are recorded by actual specialization of the intrinsic graded algebra, not by relabelling a primary decomposition.

There is also a useful degree bound that detects many strata before any Macaulay matrix is formed. Put

$$a = \text{rank } A_H, \quad q = 6 - a,$$

and, after quotienting by $\text{im } A_H$,

$$b = \text{rank } \overline{B}_H.$$

Every nonzero residual q -minor uses at most b linear mixed columns and therefore has transverse degree at least

$$\ell(a, b) = 2q - b. \tag{S18.8}$$

Proposition S18.10 (Rank-drop depth bounds at corank two). *At projection corank two,*

$$\min\{k : d^k \in J\} \geq \kappa_0(a, b) := q - \left\lfloor \frac{b}{2} \right\rfloor. \tag{S18.9}$$

Together with $d^5 \in J$, this leaves only the following finite possibilities when A_H drops rank:

a	q	possible $(b; k)$
2	4	$(4; 2, 3, 4, 5), (3; 2; 3, 4, 5), (1; 4, 5),$
1	5	$(4; 3, 4, 5), (3; 2; 4, 5),$
0	6	$(4; 4, 5), (3; 5).$

Here $k = \min\{j : d^j \in J\}$, so the nilpotency index is $k + 1$. The omitted values of b cannot occur under surjectivity of the original quadratic quotient, because the quadratic block contributes at most three independent target directions. Each comma-separated choice of k is cut out exactly by $\varepsilon_k(J) = 0$ and $\varepsilon_{k-1}(J) = 1$. In particular the extreme rank-drop type $(a, b) = (0, 3)$ has exact nilpotency index six.

At projection corank three a stronger representation-theoretic exclusion reduces the atlas to one binary choice.

Theorem S18.11 (Projection corank three). *For $R \in G_4^\circ$, at every projection-corank-three point,*

$$d^3 \notin J, \quad d^4 \in J. \tag{S18.10}$$

The local nilpotency index is exactly five, and $\varepsilon_4(J) = 0$ identically on this geometric open. For arbitrary surjective coefficient data outside this open, the universal bounds $d^3 \notin J$, $d^5 \in J$ and the membership test for d^4 remain valid; the smooth-web theorem eliminates that binary alternative on G_4° .

At the deepest Schubert stratum the remaining condition is identified geometrically.

Theorem S18.12 (Projection corank four and the Jacobian quartic). *Let $R \in G_4^\circ$. At a projection-corank-four point, $H = 0$, $T \in \text{Mat}_{4 \times 4}$, and*

$$J = I_6(\gamma_R \text{Sym}^2 T).$$

Then

$$d^3 \notin J, \quad d^4 \in J, \quad d = \det T. \quad (\text{S18.11})$$

Consequently the local nilradical has exact index five.

More precisely, as a $GL(V)$ -module,

$$\bigwedge^6 \text{Sym}^2 V \simeq \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (\text{S18.12})$$

The absence of $(\det V)^3$ in (S18.12) gives the first nonmembership in (S18.11). Under the canonical duality

$$\bigwedge^6 \text{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \text{Sym}^2 V \right)^*,$$

the second summand in (S18.12) is dual, up to determinant twist, to

$$\det V \otimes \text{Sym}^4 V,$$

and the corresponding covariant of $\bigwedge^4 R$ is the Jacobian quartic F_R . Since $F_R \neq 0$ on G_4° , the determinant line $(\det V)^4$ occurs in the degree-sixteen part of J , which is exactly the second membership in (S18.11).

Theorems S18.2, S18.4, S18.11, and S18.12, together with the generic primary signature and specialization theorems of Section S23, form the boundary classification used here. The residual-colon formulas are coordinate-free identities; the presentation matrices are finite computational devices. At corank two the generic supports and their primary corrections are geometrically identified, including pure-block rank drops. At higher corank the exact depth statements remain intrinsic and are coupled to the finite primary-signature refinement.

APPENDIX S19. FIXED TENSORS ACROSS PROJECTION CORANK TWO

We now work with $e = 4$ and $p = 6$. All calculations below concern a fixed quotient tensor γ_R , while its generating plane varies. They are identities of ideals, before taking radicals.

Definition S19.1. A pair (R, H) , where $R \in G_4^\circ$ and $H \subset V$ is a plane, is called admissible for the corank-two calculation if the following conditions hold. The map $A_H = \gamma_R|_{\text{Sym}^2 H}$ is injective. For a complement W to H , put $Q_H = S_R / \text{im } A_H$ and let

$$B_H : H \otimes W \longrightarrow Q_H, \quad C_H : \text{Sym}^2 W \longrightarrow Q_H$$

be induced by multiplication. The map B_H is surjective and its one-dimensional kernel is generated by a rank-two tensor in $H \otimes W$. Finally, the line of hyperplanes containing H meets Y_R in four distinct points.

The first-order map B_H is intrinsically a map from $H \otimes (V/H)$: changing a lift of a vector in V/H adds an element of $\text{Sym}^2 H$, which is zero in Q_H . Thus these conditions are independent of the chosen complement. The last condition is transverse intersection with a projective line, not smoothness of a parameter-selected polynomial without a geometric interpretation.

Lemma S19.2 (Nonemptiness of the corank-two open). *The admissible pairs form a nonempty open subset of $G_4^\circ \times \text{Gr}(2, V)$. Its projection is a nonempty invariant open $G_4^\dagger$. For each R in that image, the admissible H form a dense open subset of $\text{Gr}(2, V)$.*

Proof. Injectivity and surjectivity are open rank conditions. After taking the line kernel, tensor rank two is the nonvanishing of its determinant. On this locus the pencil restriction of the determinant is a binary quartic; nonzero discriminant is open. These conditions are invariant. Projection from a product over $\mathbb{C}$ is open, and $\text{Gr}(2, V)$ is irreducible, giving the density claim once nonemptiness is proved.

For C_* in (S9.2), take $H = \langle x_0, x_1 \rangle$ and $W = \langle x_2, x_3 \rangle$. In the square-first, cross-second basis, $\gamma = (-C_* \mid I_6)$. The three columns for x_0^2, x_0x_1, x_1^2 have rank three. After quotienting their image, the remaining maps are

$$B_H = \begin{pmatrix} -1/2 & -1 & 1 & 0 \\ -1/2 & -1 & 0 & 1 \\ -3/2 & -2 & 0 & 0 \end{pmatrix}, \quad C_H = \begin{pmatrix} -9/2 & 0 & -1/2 \\ -5/2 & 0 & 1/2 \\ -13/2 & 1 & 7/2 \end{pmatrix}. \quad (\text{S19.1})$$

The four columns of B_H correspond to $x_0x_2, x_0x_3, x_1x_2, x_1x_3$; the three of C_H correspond to x_2^2, x_2x_3, x_3^2 . The kernel vector is $(-4, 3, 1, 1)$, whose 2×2 determinant is -7 . For $ux_2 + vx_3$, the final square column is $C_H(u^2, 2uv, v^2)^t$. The determinant of that column and the two mixed columns is

$$h(u, v) = \frac{u^4 - 8u^3v - u^2v^2 - 4uv^3 - 9v^4}{2}, \quad \text{disc } h(u, 1) = -\frac{632301}{4} \neq 0. \quad (\text{S19.2})$$

Its leading coefficient is nonzero, so there is no root at infinity. This determinant is the restriction determinant on $\text{Sym}^2(H + \mathbb{C}(ux_2 + vx_3))$ after eliminating A_H , hence defines the pencil intersection with Y_R . Lemma S9.1 places this witness in G_4^o . $\square$

Lemma S19.3 (The residual matrix). *Near a plane K whose projection has admissible rank-two image H , there are smooth auxiliary parameters and four transverse coordinates a, b, c, d in which $\mathcal{I}_{\widehat{D}_R} = \delta J$, with*

$$\begin{aligned} T &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc, \quad \mathbf{m} = (a, b, c, d), \\ L(T) &= \begin{pmatrix} a & b & 0 & 0 \\ c & d & a & b \\ 0 & 0 & c & d \end{pmatrix}, \quad T^{(2)} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad + bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}, \\ J &= I_3[L(T), C T^{(2)}]. \end{aligned} \quad (\text{S19.3})$$

Here C is a 3×3 matrix depending only on the auxiliary parameters, not on a, b, c, d .

Proof. Use Proposition S5.1 and an invertible projection two-minor to write $M = \text{diag}(I_2, T)$. The basis of its first image plane and a complement can be chosen independently of the Schur block T : the Gaussian row operations depend on the first two columns, and the remaining column operations remove their components from the last two columns. This proves the asserted separation of parameters.

The quadratic columns divide into $\text{Sym}^2 H$, $H \otimes W$, and $\text{Sym}^2 W$. Eliminate the injective rank-three A_H block. The residual matrix is $[B_H(I_H \otimes T), C_H T^{(2)}]$, and its maximal minors generate the residual ideal: the eliminated identity block removes a free direct summand of the cokernel. A rank-two generator of $\ker B_H$ can be put in the form $h_1 \otimes w_2 - h_2 \otimes w_1$ by invertible changes of bases. The quotient is then the symmetrization map from two two-dimensional spaces, with the displayed matrix $L(T)$. These basis changes involve only the auxiliary parameters. Finally $\det M$ is a unit times δ . This proves the ideal identity, not just its leading term. $\square$

Let

$$h(u_1, u_2) = \det \left[\begin{pmatrix} u_1 \\ u_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ u_1 \\ u_2 \end{pmatrix}, C \begin{pmatrix} u_1^2 \\ 2u_1u_2 \\ u_2^2 \end{pmatrix} \right]. \quad (\text{S19.4})$$

It is the binary quartic of the pencil intersection. After an étale extension of the auxiliary coefficient ring, its four roots split as distinct line subbundles $\ell_i \subset W$. Write P_i for the ideal of $\text{im } T \subset \ell_i$; it is generated by the two entries of the composite $\mathbb{C}^2 \xrightarrow{T} W \rightarrow W/\ell_i$.

Lemma S19.4 (A rank-one-cone computation). *Under the simple-root hypothesis, in the local coefficient ring of Lemma S19.3 one has*

$$I := \bigcap_{i=1}^4 P_i = (\delta, g_0, \dots, g_4), \quad J = I \cap \mathfrak{m}^3 = \delta \mathfrak{m} + (g_0, \dots, g_4). \quad (\text{S19.5})$$

The degree-four polynomials g_j can be chosen with restrictions $h(u_1, u_2)v_1^{4-j}v_2^j$ on $T = uv^t$.

Proof. Give a, b, c, d degree one and the auxiliary coefficients degree zero. The coordinate ring of the rank-one cone is the diagonal subring

$$A_0[a, b, c, d]/(\delta) \simeq \bigoplus_{r \geq 0} \text{Sym}^r(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v, \quad T = uv^t, \quad (\text{S19.6})$$

where A_0 denotes the auxiliary ring. This follows directly from $a = u_1v_1$, $b = u_1v_2$, $c = u_2v_1$, $d = u_2v_2$ and the single relation $ad - bc$; the monomials on either side give the same basis. In degree r , vanishing on the four distinct u -lines means divisibility by h in the u variables. The resulting module is zero for $r < 4$, and for $r \geq 4$ it is

$$h \text{Sym}^{r-4}(A_0^2)_u \otimes \text{Sym}^r(A_0^2)_v.$$

The simple-root discriminant is a unit, so the elementary divisibility and interpolation assertions used here hold over A_0 as well as over its residue field. Multiplication of binary forms shows that this ideal is generated in diagonal degree four by $hv_1^{4-j}v_2^j$. Lifting those generators proves the first identity.

The four 3-minors of the linear block $L(T)$ generate $\delta \mathfrak{m}$, by direct expansion. Modulo δ , its four columns are

$$v_1B(h_1u), \quad v_2B(h_1u), \quad v_1B(h_2u), \quad v_2B(h_2u),$$

and the three quadratic columns are $v_1^2C(u^2)$, $v_1v_2C(u^2)$, and $v_2^2C(u^2)$. Here u^2 has coefficient vector $(u_1^2, 2u_1u_2, u_2^2)^t$. A minor using one column from each of the two linear groups and one quadratic column is h times a degree-four monomial in v ; these span all five such monomials. Thus they furnish the lifts g_j . A minor with at least two quadratic columns vanishes modulo δ , and its degree is at least five, so it already lies in $\delta \mathfrak{m}$. All other minors have been accounted for. This proves $J = \delta \mathfrak{m} + (g_j)$ exactly.

Finally I is generated by its degree-two element δ and the g_j of degree four. Its intersection with $\mathfrak{m}^3$ deletes only the scalar multiple of δ in degree two, proving the second equality. Localizing in the auxiliary parameters or taking the indicated étale extension preserves these explicit identities. $\square$

Theorem S19.5 (The corank-two primary structure). *For an admissible pair the ideal δJ has the decomposition (S17.2). It is irredundant. In the local failure ring its nilradical $\widehat{\mathcal{N}}$ satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathfrak{m}/(\delta J). \quad (\text{S19.7})$$

Consequently the rank-two stratum is intrinsic to this local scheme as the annihilator support of $\widehat{\mathcal{N}}^2$.

Proof. Work on a connected regular étale auxiliary chart, so that its coefficient ring is a domain. Each P_i is a prime generated by two independent linear coordinates. Its ordinary square is P_i -primary; the same argument applies to $\mathfrak{m}^5$. The ideal (δ) is prime. At the generic point of P_i , the determinant has order one, because the other row after choosing ℓ_i as a coordinate line is not zero. Therefore

$$(P_i^2 : \delta) = P_i, \quad (\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

The second equality also follows immediately from the homogeneous order two of δ and the polynomial associated graded ring of $\mathfrak{m}$. Intersecting with (δ) and cancelling δ now gives

$$(\delta) \cap \bigcap_i P_i^2 \cap \mathfrak{m}^5 = \delta \left(\bigcap_i P_i \cap \mathfrak{m}^3 \right) = \delta J$$

by Lemma S19.4.

At the generic point of the determinant divisor all the other ideals are units, so (δ) is necessary. At the generic point of P_i the other P_j and $\mathfrak{m}$ become units, and the local ideal is $\delta P_i = (\delta) \cap P_i^2$; its two components are necessary. To see that $\mathfrak{m}^5$ is necessary, note that δ^2 lies in (δ) and in every P_i^2 , but is not in δJ : $\delta \notin J$ because its degree is two. On the other hand $\mathfrak{m}\delta^2 \subset \delta J$ since $\delta\mathfrak{m} \subset J$. This also proves that the annihilator of the class of δ^2 is exactly $\mathfrak{m}$; a nonzero auxiliary coefficient cannot kill that degree-four class.

The radical of the failure ideal is (δ) , so the nilradical is generated by the class of δ . Its square is the nonzero class just considered and its cube vanishes, since $\delta^2 \in \delta\mathfrak{m} \subset J$. This proves (S19.7). Over the auxiliary ring the square is free of rank one on the rank-two stratum. Intrinsically its transition functions are those of $\mathcal{I}_{\Delta}^2$ restricted to that stratum; thus it is $\mathcal{O}_{\Sigma}(-2\hat{\Delta})$ on the admissible open Σ . The generic transverse length is one. $\square$

Proof of Theorem S17.2. Lemma S19.2 supplies the asserted nonempty open class. Lemmas S19.3 and S19.4 and Theorem S19.5 supply the fixed-tensor calculation on every indicated frame chart. The graph coordinates are smooth auxiliary variables, so no transverse component has been discarded. The individual P_i are allowed to permute under monodromy; the intersection and the intrinsic annihilator in (S19.7) descend. Four local branches need not be four distinct global irreducible components. This is why the primary decomposition was stated étale locally rather than with globally labelled roots. $\square$

APPENDIX S20. RESIDUAL PRESENTATIONS AND THE RANK-TWO CONTACT BOUNDARY

We first record the projection-rank block presentation inherited from the full Fitting formula, and then isolate the rank-two structure to which it gives rise. The distinction between a presentation theorem and a structure theorem is important: the block formula below is exact at every projection rank, whereas the associated-prime, collision, and nilpotent-filtration theorems are proved on the loci explicitly stated. Section S21 pushes the structure theorem through all mixed-regular contact collisions, through the containment locus, and across the first non-mixed-regular tensor-rank wall.

For $0 \leq r \leq 3$, let $\Sigma_R^{(r)} \subset X_R$ be the locally closed stratum on which the projection $\mathcal{K} \rightarrow V$ has rank exactly r . If $K \in \Sigma_R^{(r)}$ has image $H \subset V$, choose a complement $V = H \oplus W$, where $c = \dim W = 4 - r$. On a frame chart around K , an invertible r -minor and Schur reduction put the projection block in the form

$$M = \begin{pmatrix} I_H & 0 \\ 0 & T \end{pmatrix}, \quad T \in \text{Mat}_{c \times c}.$$

Decompose the quotient multiplication tensor according to

$$\text{Sym}^2 V = \text{Sym}^2 H \oplus (H \otimes W) \oplus \text{Sym}^2 W$$

and write the corresponding blocks as A_H, B_H, C_H .

Theorem S20.1 (Projection-rank residual block presentation). *For every $R \in G_4^\circ$, every $0 \leq r \leq 3$, and every $K \in \Sigma_R^{(r)}$, one has on the preceding frame chart, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\det T) I_6[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (\text{S20.1})$$

The formula is independent of any injectivity, mixed-rank, tensor-rank, or contact-transversality assumption. Thus every projection-rank stratum has the same exact block presentation. No claim of a primary or associated-prime classification on the non-mixed-regular corank-three or corank-four boundary is made by this formula alone; the separate structure theorems below derive additional information where the residual algebra can be determined.

Proof. Proposition S5.1 gives on every frame chart

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \operatorname{Sym}^2 M).$$

The Schur operations placing M in the displayed block form are invertible and therefore change neither the maximal-minor ideal nor the local scheme. On $\operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$ the induced map $\operatorname{Sym}^2 M$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Since $\det M$ is a unit times $\det T$, substitution proves (S20.1). $\square$

We now specialize this exact all-corank formula to $r = 2$. Write $\Sigma_R = \Sigma_R^{(2)}$. Then $c = 2$, and after choosing bases

$$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = \det T = ad - bc.$$

Decompose the quotient multiplication tensor according to

$$\operatorname{Sym}^2 V = \operatorname{Sym}^2 H \oplus (H \otimes W) \oplus \operatorname{Sym}^2 W$$

and write

$$A_H : \operatorname{Sym}^2 H \rightarrow S_R, \quad B_H : H \otimes W \rightarrow S_R, \quad C_H : \operatorname{Sym}^2 W \rightarrow S_R.$$

Theorem S20.2 (Exact presentation on every rank-two chart). *In the preceding coordinates one has, before taking radicals,*

$$\mathcal{I}_{\widehat{D}_R} = (\delta) I_6[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]. \quad (\text{S20.2})$$

The formula is valid at every projection-rank-two point; no injectivity, mixed-rank, tensor-rank, or simple-contact hypothesis is required.

Proof. Proposition S5.1 gives

$$\mathcal{I}_{\widehat{D}_R} = (\det M) I_6(\gamma_R \operatorname{Sym}^2 M).$$

For $M = I_H \oplus T$, the induced map on the three summands of $\operatorname{Sym}^2(H \oplus W)$ is respectively

$$I_{\operatorname{Sym}^2 H}, \quad 1_H \otimes T, \quad \operatorname{Sym}^2 T.$$

Substitution gives (S20.2). All row and column operations used to reach the block form are invertible, so the maximal-minor ideal is unchanged. $\square$

Thus the exceptional rank-two locus is included in the exact presentation. Determining radicals, associated primes, and nilpotent layers of its residual factor is a separate problem. The mixed-regular case and its first tensor-rank boundary are solved below and in Section S21.

Definition S20.3. A pair (R, H) is *mixed-regular* if A_H is injective and, after quotienting S_R by its image, the induced map

$$\overline{B}_H : H \otimes (V/H) \longrightarrow S_R / \text{im } A_H$$

is surjective with one-dimensional kernel generated by a rank-two tensor. No hypothesis is imposed on the multiplicities of the four ramification contacts.

These are precisely the first three conditions in Definition S19.1; the discriminant condition has been removed. On this locus the same Gaussian elimination as in Lemma S19.3 is valid and yields

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = I_3[L(T), CT^{(2)}],$$

with $L(T)$ and $T^{(2)}$ as in (S19.3).

Lemma S20.4 (Root-free residual normal form). *Over the local auxiliary coefficient ring of a mixed-regular pair,*

$$J = \delta \mathbf{m} + (g_0, \dots, g_4), \quad \mathbf{m} = (a, b, c, d), \quad (\text{S20.3})$$

where each g_i is homogeneous of transverse degree four. Modulo (δ) , under the Segre parametrization $T = uv^t$, their classes span

$$h_H(u) \cdot \text{Sym}^4(\mathbb{C}_v^2).$$

Here h_H is the binary quartic obtained by restricting the ramification equation to the pencil of hyperplanes containing H . The identity is valid with repeated roots, and also when the discriminant of h_H vanishes.

Proof. The four 3-minors of the linear block $L(T)$ generate $\delta \mathbf{m}$, exactly as in Lemma S19.4. Modulo δ , write $T = uv^t$. The two pairs of linear columns are scalar multiples, by v_1, v_2 , of the two mixed columns determined by u , while the three quadratic columns are scalar multiples, by $v_1^2, v_1 v_2, v_2^2$, of $C(u^2)$. A minor containing one column from each linear pair and one quadratic column is therefore $h_H(u)$ times a degree-four monomial in v ; the five choices span $\text{Sym}^4(\mathbb{C}_v^2)$.

Every minor containing at least two quadratic columns vanishes on the rank-one cone, hence is divisible by the prime equation δ . Its transverse degree is at least five, so the quotient by δ lies in $\mathbf{m}$. The remaining minors are the already listed linear and mixed ones. This proves (S20.3). No factorization of h_H was used. When its roots are distinct, factorizing h_H recovers Lemma S19.4. $\square$

Theorem S20.5 (Depth two across contact collisions). *At every mixed-regular projection-rank-two point, the radical of $\mathcal{I}_{\widehat{D}_R}$ is (δ) and the nilradical $\widehat{\mathcal{N}}$ of the local failure ring satisfies*

$$\widehat{\mathcal{N}}^3 = 0, \quad \widehat{\mathcal{N}}^2 \neq 0, \quad \text{Ann}(\widehat{\mathcal{N}}^2) = \mathbf{m}/(\delta J).$$

Consequently $\mathcal{Z}_2(\widehat{D}_R)$ is the rank-two stratum on the mixed-regular locus, even when the contact quartic has multiple roots. Moreover

$$\widehat{\mathcal{N}}^2|_{\Sigma_R} \simeq \mathcal{I}_{\Delta_R}^2|_{\Sigma_R}.$$

Proof. By Lemma S20.4, $J \subset \mathbf{m}^3$ and $\delta \mathbf{m} \subset J$. Since $\delta \in \mathbf{m}^2$, we have $\delta^2 \in J$, hence $\delta^3 \in \delta J$. On the other hand $\delta \notin J$ by transverse degree, so $\delta^2 \notin \delta J$. Since $\delta J \subset (\delta)$ and a power of δ belongs to δJ , the radical is exactly (δ) . Thus the nilradical is generated by the class of δ and has exact index three.

For the annihilator, cancellation in the regular local coefficient ring gives

$$(\delta J : \delta^2) = (J : \delta).$$

The inclusion $\mathfrak{m} \subset (J : \delta)$ follows from $\delta\mathfrak{m} \subset J$. Conversely, use the transverse grading: J has no component of transverse degree two, whereas δ has degree two. If the transverse degree-zero part of r were nonzero, the degree-two part of $r\delta$ could not lie in J . Hence $r \in \mathfrak{m}$, proving $(J : \delta) = \mathfrak{m}$.

Finally δ is a local equation of the Schubert divisor $\hat{\Delta}_R$. Therefore the generator δ^2 of the square of the nilradical transforms as the square of the conormal line. This gives the displayed line bundle after restriction to the rank-two stratum. $\square$

We next globalize the four contacts without labelling them. Put

$$G = \text{Gr}(2, V)$$

and let $\mathcal{Q}$ be its universal rank-two quotient bundle. The quartic equation F_R of $Y_R \subset \mathbb{P}(V^*)$ restricts to every line of hyperplanes containing H . Equivalently it defines a global binary quartic

$$h_R \in H^0(G, \text{Sym}^4 \mathcal{Q}). \quad (\text{S20.4})$$

Define the contact incidence

$$\mathcal{R}_R = \{(H, [\alpha]) \in G \times Y_R : H \subset \ker \alpha\}. \quad (\text{S20.5})$$

Proposition S20.6 (The global contact cover). *The projection $\mathcal{R}_R \rightarrow Y_R$ is a $\mathbb{P}^2$ -bundle; in particular $\mathcal{R}_R$ is smooth and irreducible. Over the open set $G_R^{\text{fin}} \subset G$ on which the binary quartic (S20.4) is not identically zero, the map*

$$\mathcal{R}_R \longrightarrow G$$

is finite flat of degree four. There is a globally defined discriminant divisor

$$\mathfrak{D}_R = V(\text{Disc}(h_R)), \quad \text{Disc}(h_R) \in H^0(G, (\det \mathcal{Q})^{12}).$$

Whenever this section is not identically zero, $[\mathfrak{D}_R] = 12H$ in the Pluecker class group. On the finite-flat locus G_R^{fin} , the support of the branch divisor is the restriction of $\mathfrak{D}_R$; away from $\mathfrak{D}_R$ the degree-four cover is finite etale. The monodromy action on the four sheets over that etale locus is transitive.

Proof. For $[\alpha] \in Y_R$, the fibre of $\mathcal{R}_R \rightarrow Y_R$ consists of the two-planes in the three-dimensional hyperplane $\ker \alpha$, hence is $\text{Gr}(2, \ker \alpha) \simeq \mathbb{P}^2$. These fibres form the relative Grassmannian of the rank-three hyperplane bundle on Y_R , proving smoothness and irreducibility.

For fixed H , the other projection has fibre $Y_R \cap \mathbb{P}(H^\perp)$, the zero scheme of the restricted quartic. On the locus where the restriction is nonzero this is a relative effective Cartier divisor of degree four in the $\mathbb{P}^1$ -bundle $\mathbb{P}(\mathcal{Q}^*)$, hence finite flat of degree four.

The discriminant of a binary quartic is homogeneous of degree six in its coefficients and is a relative invariant of determinant weight twelve. Applied to the section of $\text{Sym}^4 \mathcal{Q}$ in (S20.4), it is therefore a section of $(\det \mathcal{Q})^{12}$. Since $\det \mathcal{Q}$ is the Pluecker line bundle up to the constant factor $\det V$, its zero divisor has class $12H$ whenever it is nonzero. On G_R^{fin} the fibre is a degree-four Cartier divisor on a projective line, so its non-etale locus is exactly the multiple-root locus, i.e. the restriction of the discriminant locus. The total incidence is irreducible; therefore the monodromy action of the etale restriction on a geometric fibre is transitive. $\square$

For R in the nonempty open $G_4^\dagger$ of Lemma S19.2, the discriminant in Proposition S20.6 is not identically zero. Let $G_R^{\text{mr}} \subset G$ denote the mixed-regular locus of Definition S20.3. The contact sheets themselves split etale locally over $G_R^{\text{fin}} \setminus \mathfrak{D}_R$. Their interpretation as the four primary ideals P_i of Theorem S19.5, however, also requires the independent mixed-regular rank conditions. Thus the exact locus for that primary decomposition is

$$G_R^{\text{mr}} \cap (G_R^{\text{fin}} \setminus \mathfrak{D}_R),$$

not the discriminant complement by itself. On this intersection the P_i are the four etale contact sheets and their permutation is the monodromy of the canonical cover (S20.5).

The collision divisor has the five nonzero fibre types corresponding to the partitions

$$1 + 1 + 1 + 1, \quad 2 + 1 + 1, \quad 2 + 2, \quad 3 + 1, \quad 4$$

of the degree of a binary quartic, together with the separate containment locus $h_H \equiv 0$. Theorem S20.5 does not change across these collision types. What changes is the primary splitting of the contact part, so we do not force the simple-root intersection formula onto the boundary.

The ordinary fifth power of the rank-two Schubert ideal is of course globally defined. The stronger assertion that it remains the actual $\mathfrak{m}$ -primary component of the failure ideal through every mixed-regular collision, including the zero-quartic containment locus, is proved in Theorem S21.2; it is not inferred merely from the existence of the global ideal. The same theorem gives the exact collided primary types and separates the finite contact geometry from the containment geometry.

Corollary S20.7 (Rank-two part of Theorem S17.1). *The contact-incidence and collision assertions of Theorem S17.1 hold on the stated mixed-regular locus.*

Proof. Theorem S20.5 identifies depth two and its line bundle, while Proposition S20.6 globalizes the contact incidence and discriminant. The global residual-colon filtration, the simultaneous identification of the first two graded pieces, the collided primary components, the containment locus, and the decomposable-kernel wall are proved in Section S21. $\square$

APPENDIX S21. THE STRATIFIED NILPOTENT ALGEBRA AT PROJECTION CORANK TWO

The root-free formula of Lemma S20.4 contains more scheme structure than the nilpotency-index statement extracted from it in the preceding local calculations. We record that structure before passing to the exceptional mixed-kernel boundary.

Theorem S21.1 (Global residual-colon filtration). *Let $X = X_R$, let $\widehat{\Delta}_R \subset X$ be the reduced Schubert divisor, and put*

$$\mathcal{J}_R^{\text{res}} = (\mathcal{I}_{\widehat{\Delta}_R} : \mathcal{I}_{\widehat{\Delta}_R}) \subset \mathcal{O}_X.$$

Then

$$\mathcal{I}_{\widehat{\Delta}_R} = \mathcal{I}_{\widehat{\Delta}_R} \mathcal{J}_R^{\text{res}}. \quad (\text{S21.1})$$

For every $j \geq 1$,

$$\text{Ann}_{\mathcal{O}_{\widehat{\Delta}_R}}(\widehat{\mathcal{N}}_R^j) = \frac{(\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1})}{\mathcal{I}_{\widehat{\Delta}_R}}. \quad (\text{S21.2})$$

Define the closed subscheme of the reduced divisor

$$W_j = V_{\widehat{\Delta}_R} \left((\mathcal{J}_R^{\text{res}} : \mathcal{I}_{\widehat{\Delta}_R}^{j-1}) + \mathcal{I}_{\widehat{\Delta}_R} \right). \quad (\text{S21.3})$$

If

$$\mathcal{L} = \mathcal{I}_{\widehat{\Delta}_R} / \mathcal{I}_{\widehat{\Delta}_R}^2$$

is the conormal line of the Schubert divisor, then there are canonical isomorphisms

$$\widehat{\mathcal{N}}_R^j / \widehat{\mathcal{N}}_R^{j+1} \simeq \mathcal{L}^{\otimes j} \otimes_{\mathcal{O}_{\widehat{\Delta}_R}} \mathcal{O}_{W_j}. \quad (\text{S21.4})$$

Consequently

$$\text{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{\Delta}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j \geq 1} \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j}, \quad (\text{S21.5})$$

with the multiplication induced by multiplication of powers of $\mathcal{I}_{\widehat{\Delta}_R}$. The sum is finite.

These formulas hold on the whole Grassmannian, including every projection corank and every rank-drop locus. On a Schur chart with local equation $d = \det T$ and residual ideal J , they become

$$\text{Ann}(N^j) = (J : d^{j-1})/(dJ), \quad N^j/N^{j+1} \simeq A/((d) + (J : d^{j-1})). \quad (\text{S21.6})$$

In particular the nilpotency index at the point is

$$1 + \min\{k \geq 0 : d^k \in J\}. \quad (\text{S21.7})$$

Proof. Proposition S5.1 shows locally that $\mathcal{I}_{\widehat{D}_R} = dJ$, where d is a local equation of the Cartier divisor $\widehat{\Delta}_R$. Since d is a non-zero-divisor in the regular local ring of X ,

$$(dJ : d) = J.$$

Thus J is the local expression of the intrinsic colon $\mathcal{J}_R^{\text{res}}$, and the local identities glue to (S21.1).

In $A/(dJ)$ the nilradical is the image of (d) , because the reduction is $A/(d)$. Therefore

$$\text{Ann}(N^j) = (dJ : d^j)/(dJ) = (J : d^{j-1})/(dJ),$$

which proves (S21.2). For the graded piece, the map

$$A \longrightarrow N^j/N^{j+1}, \quad r \longmapsto rd^j$$

has kernel

$$(d) + (J : d^{j-1}).$$

Indeed

$$rd^j \in d^{j+1} + dJ \iff rd^{j-1} \in d^j + J,$$

and the latter is equivalent to writing $r = ds + t$ with $td^{j-1} \in J$. This proves the local form of (S21.4). On overlaps local equations of the Cartier divisor differ by units, and d^j transforms by the j -th tensor power of the conormal transition function. Hence the local identifications glue to (S21.4) and (S21.5). Nilpotence of the nilradical makes the sum finite, and the same cancellation gives (S21.7). $\square$

The theorem converts the globally defined annihilator filtration into an exact residual-colon filtration. What remains is to identify the colon schemes W_j geometrically on the successive rank strata. For projection corank one, $W_1 = E_R$ and all higher W_j are empty. The next two results compute W_1, W_2 through the entire mixed-regular rank-two boundary and then across its first tensor-rank wall.

Work on a mixed-regular rank-two chart. Let A_0 be the regular auxiliary coefficient ring, put

$$P = A_0[a, b, c, d], \quad \mathfrak{m} = (a, b, c, d), \quad \delta = ad - bc,$$

and localize, or complete, at the point under consideration. Thus

$$\mathcal{I}_{\widehat{D}_R} = \delta J, \quad J = \delta \mathfrak{m} + (g_0, \dots, g_4),$$

where the g_i have transverse degree four. Define the *unsplit contact ideal*

$$I_H = (\delta, g_0, \dots, g_4). \quad (\text{S21.8})$$

Modulo δ , the Segre parametrization $T = uv^t$ identifies the homogeneous coordinate ring with

$$P/(\delta) \simeq \bigoplus_{n \geq 0} \text{Sym}^n(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2),$$

and the image of I_H in degree n is zero for $n < 4$ and

$$h_H \text{Sym}^{n-4}(\mathbb{C}_u^2) \otimes \text{Sym}^n(\mathbb{C}_v^2) \quad (n \geq 4). \quad (\text{S21.9})$$

Theorem S21.2 (Saturated contact ideal and the fixed vertex component). *At every mixed-regular rank-two point one has the exact identities*

$$J = I_H \cap \mathfrak{m}^3 \quad (\text{S21.10})$$

and

$$\mathcal{I}_{\widehat{D}_R} = \delta I_H \cap \mathfrak{m}^5. \quad (\text{S21.11})$$

If $h_H \not\equiv 0$, the ideal I_H is $\mathfrak{m}$ -saturated. Put $\mathcal{C}_H = V(I_H)$ for the affine contact cone. Its projectivization on the Segre quadric

$$Q = \mathbb{P}_u^1 \times \mathbb{P}_v^1 \subset \mathbb{P}^3$$

is the effective Cartier divisor

$$C_H^{\text{proj}} = V(h_H) \times \mathbb{P}_v^1 \quad (\text{S21.12})$$

of class $(4, 0)$. Consequently δI_H is also $\mathfrak{m}$ -saturated, and $\mathfrak{m}^5$ in (S21.11) is an actual irredundant $\mathfrak{m}$ -primary component, not merely a globally defined ideal.

If

$$h_H = \prod_{\nu=1}^s \ell_\nu^{e_\nu}, \quad \sum_\nu e_\nu = 4,$$

over an etale coefficient extension, and P_ν denotes the height-two contact prime corresponding to ℓ_ν , then

$$\text{Ass}(P/\mathcal{I}_{\widehat{D}_R}) = \{(\delta), P_1, \dots, P_s, \mathfrak{m}\}. \quad (\text{S21.13})$$

At the generic point of P_ν , choose a second regular parameter q_ν so that $P_\nu = (\delta, q_\nu)$. The P_ν -primary part of δI_H has the exact two-parameter normal form

$$Q_{\nu, e_\nu} = (\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1}). \quad (\text{S21.14})$$

For $e_\nu = 1$ this is P_ν^2 ; for a collided contact it is the corresponding non-ordinary primary thickening.

If $h_H \equiv 0$, then

$$I_H = (\delta), \quad J = \delta \mathfrak{m}, \quad \mathcal{I}_{\widehat{D}_R} = \delta^2 \mathfrak{m} = (\delta^2) \cap \mathfrak{m}^5. \quad (\text{S21.15})$$

In particular the rank-two vertex component remains $\mathfrak{m}^5$ on the containment locus.

Proof. The equality (S21.10) is a graded calculation. The ideal I_H is generated in transverse degrees two and four, its only degree-two generator being δ . Intersecting with $\mathfrak{m}^3$ therefore removes exactly the scalar multiple of δ and leaves $\delta \mathfrak{m} + (g_0, \dots, g_4) = J$.

Since the initial form of δ has degree two and is nonzero in the polynomial associated graded ring,

$$(\mathfrak{m}^5 : \delta) = \mathfrak{m}^3.$$

Cancellation in the regular ring gives

$$\delta I_H \cap \mathfrak{m}^5 = \delta(I_H \cap (\mathfrak{m}^5 : \delta)) = \delta(I_H \cap \mathfrak{m}^3) = \delta J,$$

which proves (S21.11).

Assume first $h_H \neq 0$. Formula (S21.9) is precisely the section ideal of $\mathcal{O}_Q(-4, 0) \subset \mathcal{O}_Q$, cut out by the nonzero section h_H of $\mathcal{O}_Q(4, 0)$. Hence it is saturated and defines the Cartier divisor (S21.12). The inverse image $I_H \subset P$ is therefore $\mathfrak{m}$ -saturated. The product δI_H is saturated as well: if $\mathfrak{m}^N f \subset \delta I_H$, choose a coordinate not divisible by the prime element δ to see that $\delta \mid f$; after cancelling δ , saturation of I_H applies. Thus $\mathfrak{m} \notin \text{Ass}(P/\delta I_H)$. On the other hand $\delta^2 \in \delta I_H$ has transverse degree four and does not belong to $\mathfrak{m}^5$, so the $\mathfrak{m}^5$ factor in (S21.11) is irredundant.

The divisor C_H^{proj} has components $e_\nu(\{\ell_\nu = 0\} \times \mathbb{P}^1)$. Hence P/I_H is Cohen–Macaulay of pure codimension two and its associated primes are exactly the P_ν . Multiplication by δ gives an exact sequence

$$0 \longrightarrow P/I_H \xrightarrow{\delta} P/\delta I_H \longrightarrow P/(\delta) \longrightarrow 0.$$

It follows, and localization shows equality, that the associated primes of $P/\delta I_H$ are $(\delta), P_1, \dots, P_s$. Intersecting with the irredundant $\mathfrak{m}$ -primary ideal $\mathfrak{m}^5$ gives (S21.13).

At the generic point of P_ν , the Cartier divisor has equation $q_\nu^{e_\nu}$, so

$$(I_H)_{P_\nu} = (\delta, q_\nu^{e_\nu}), \quad (\delta I_H)_{P_\nu} = (\delta^2, \delta q_\nu^{e_\nu}).$$

The ideal

$$(\delta^2, \delta q_\nu^{e_\nu}, q_\nu^{e_\nu+1})$$

is P_ν -primary, and its colon by δ is $(\delta, q_\nu^{e_\nu})$. Therefore

$$(\delta) \cap Q_{\nu, e_\nu} = \delta(\delta, q_\nu^{e_\nu}),$$

which proves the asserted generic primary normal form.

If $h_H = 0$, every g_i vanishes modulo δ . Since it has transverse degree four, $g_i = \delta q_i$ with $q_i \in \mathfrak{m}^2$. Thus $I_H = (\delta)$ and $J = \delta \mathfrak{m}$. Finally

$$(\delta^2) \cap \mathfrak{m}^5 = \delta^2(\mathfrak{m}^5 : \delta^2) = \delta^2 \mathfrak{m},$$

again by the polynomial associated graded ring. This is (S21.15). $\square$

The preceding theorem gives the requested specialization law without labelling roots. A collision changes the Cartier multiplicities in C_H and hence the primary type (S21.14); it does not change the vertex component. The zero quartic is the unique point at which the Cartier contact divisor ceases to be a divisor and fills the whole Segre quadric.

Theorem S21.3 (Associated graded nilpotent algebra across coranks one and two). *On a mixed-regular rank-two chart let*

$$A = P/(\delta J), \quad N = \sqrt{0_A}.$$

Then $N = (\delta)/(\delta J)$, $N^3 = 0 \neq N^2$, and

$$N/N^2 \simeq P/I_H, \tag{S21.16}$$

$$N^2 \simeq P/\mathfrak{m}. \tag{S21.17}$$

Equivalently, with $\mathcal{L}$ the conormal line of the Schubert divisor, the associated graded algebra has the intrinsic form

$$\text{gr}_N \mathcal{O}_{\widehat{D}_R} \simeq \mathcal{O}_{\widehat{\Delta}_R} \oplus (\mathcal{L} \otimes \mathcal{O}_{C_H}) \oplus (\mathcal{L}^{\otimes 2} \otimes \mathcal{O}_{\Sigma_R}) \tag{S21.18}$$

on the mixed-regular neighbourhood, with multiplication induced by the incidence of the affine contact cone C_H with the rank-two vertex Σ_R . Moreover

$$\text{Ann}_A(N) = J/(\delta J), \quad \text{Ann}_A(N^2) = \mathfrak{m}/(\delta J), \tag{S21.19}$$

so

$$\mathcal{Z}_1 = V(I_H \cap \mathfrak{m}^3), \quad \mathcal{Z}_2 = V(\mathfrak{m}).$$

On the punctured determinant cone, i.e. on the adjacent projection-corank-one locus, $\mathcal{Z}_1$ is exactly the ramification support E_R . Thus C_H is its scheme-theoretic closure away from the vertex, while $\mathcal{Z}_1$ records in addition the canonical third-order vertex thickening forced by the Fitting ideal.

Proof. The radical of δJ is (δ) , so the nilradical is generated by the class of δ . The isomorphism (S21.16) follows by cancelling one factor:

$$N/N^2 = \frac{(\delta)}{(\delta^2) + (\delta J)} \simeq \frac{P}{(\delta) + J} = \frac{P}{I_H}.$$

For the square, multiplication by δ^2 identifies its kernel with

$$(\delta J : \delta^2) = (J : \delta) = \mathfrak{m}$$

by Theorem S20.5; hence (S21.17). The transition functions of the local generator δ are those of the conormal line, giving (S21.18).

Cancellation also gives

$$(\delta J : \delta) = J,$$

while the preceding colon calculation gives the second formula in (S21.19). Away from the vertex $\mathfrak{m}$ becomes the unit ideal, so $J = I_H$. On the smooth rank-one part of the determinant divisor, the corank-one normal form $t(t, f)$ identifies this contact equation with the ramification equation. Hence the restriction is precisely E_R . $\square$

We finally cross the first non-mixed-regular wall. This is the rank-one tensor wall in the one-dimensional mixed kernel, explicitly requested by the referee.

Definition S21.4. Assume A_H is injective and $\overline{B}_H$ is surjective with one-dimensional kernel generated by a decomposable tensor $h_1 \otimes w_1$. Choose h_2, w_2 completing bases and put

$$q_1 = \overline{B}_H(h_1 \otimes w_2), \quad q_2 = \overline{B}_H(h_2 \otimes w_1).$$

We call the point *transverse decomposable-kernel regular* if

$$\det[q_1, q_2, \overline{C}_H(w_1^2)] \neq 0.$$

This condition is independent of the chosen completing vectors. It says that the contact root forced by the decomposable kernel is simple in the direction transverse to the kernel wall.

Theorem S21.5 (The decomposable mixed-kernel wall). *At a transverse decomposable-kernel regular rank-two point there are bases for which*

$$L_{\text{dec}}(T) = \begin{pmatrix} c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}. \quad (\text{S21.20})$$

Let

$$J_{\text{dec}} = I_3[L_{\text{dec}}(T), C T^{(2)}], \quad K_{\text{dec}} = (c, d, a^2, ab, b^2).$$

Then

$$(J_{\text{dec}} : \delta) = K_{\text{dec}}. \quad (\text{S21.21})$$

Hence the local failure ring still has nilradical of exact index three, but its second intrinsic depth support is no longer reduced:

$$\mathcal{Z}_2 = V(K_{\text{dec}}). \quad (\text{S21.22})$$

Its support is the rank-two vertex, and its transverse local length is three. Thus crossing the tensor-rank-two wall changes the scheme structure of $\mathcal{Z}_2$ before it changes its set-theoretic support.

Proof. The stated normal form is obtained by taking $\ker \overline{B}_H = \mathbb{C}(h_1 \otimes w_1)$ and using

$$q_1, q_2, q_3 = \overline{B}_H(h_2 \otimes w_2)$$

as target basis. Applying $1_H \otimes T$ gives (S21.20). Its maximal minors generate

$$I_3(L_{\text{dec}}) = \delta(c, d).$$

After replacing the first quadratic target column by its component modulo $\langle q_1, q_2 \rangle$, the transversality assumption allows us to normalize $\overline{C}_H(w_1^2) \equiv q_3$. The minors containing one column from each mixed group and this quadratic column give

$$\delta a^2, \quad \delta ab, \quad \delta b^2 \in J_{\text{dec}}.$$

Together with $I_3(L_{\text{dec}})$ this proves $\delta K_{\text{dec}} \subset J_{\text{dec}}$, hence $K_{\text{dec}} \subset (J_{\text{dec}} : \delta)$.

For the reverse inclusion use the transverse grading. The degree-three part of J_{dec} is exactly $\delta(c, d)$; every minor using a quadratic column has degree at least four. Modulo K_{dec} , every class is represented by

$$\lambda + \mu a + \nu b.$$

If $r\delta \in J_{\text{dec}}$, the degree-two term rules out $\lambda \neq 0$, while the degree-three identity

$$(\mu a + \nu b)\delta \in \delta(c, d)$$

forces $\mu = \nu = 0$. Thus $r \in K_{\text{dec}}$, proving (S21.21).

Since $\delta \in (c, d)$, one has $\delta^2 \in \delta(c, d) \subset J_{\text{dec}}$, whereas $\delta \notin J_{\text{dec}}$ by degree. Therefore the nilradical of $P/(\delta J_{\text{dec}})$ has exact index three. Finally

$$\text{Ann}(N^2) = (\delta J_{\text{dec}} : \delta^2) = (J_{\text{dec}} : \delta) = K_{\text{dec}}.$$

The quotient by K_{dec} has basis $1, a, b$, proving the length assertion. $\square$

Proposition S21.6 (The containment locus is the finite line scheme). *For fixed $R \in G_4^2$, the locus*

$$\mathcal{C}_R = \{H \in \text{Gr}(2, V) : h_H \equiv 0\}$$

is canonically the Fano scheme $F_1(Y_R)$ of lines on the smooth quartic Y_R . It is finite and reduced. At its mixed-regular points the local failure ideal is (S21.15), and the contact morphism $\mathcal{R}_R \rightarrow \text{Gr}(2, V)$ has fibre $\mathbb{P}^1$, rather than a finite length-four fibre.

Proof. The line of hyperplanes containing H is $\mathbb{P}(H^\perp) \subset \mathbb{P}(V^*)$. Its quartic restriction is h_H , so $h_H = 0$ identically exactly when $\mathbb{P}(H^\perp) \subset Y_R$. This identifies $\mathcal{C}_R$ with $F_1(Y_R)$.

If $L \subset Y_R$ is such a line, the normal sequence in $\mathbb{P}^3$ gives

$$\deg N_{L/Y_R} = \deg N_{L/\mathbb{P}^3} - \deg N_{Y_R/\mathbb{P}^3}|_L = 2 - 4 = -2.$$

Since N_{L/Y_R} has rank one, $N_{L/Y_R} \simeq \mathcal{O}_{\mathbb{P}^1}(-2)$, and therefore $H^0(N_{L/Y_R}) = 0$. The Zariski tangent space of the Fano scheme vanishes at every point. The Fano scheme is projective and zero-dimensional; a zero-dimensional local algebra with zero tangent space is a field by Nakayama. Hence it is finite reduced. The final two assertions follow from Theorem S21.2 and the definition of the incidence fibre. $\square$

APPENDIX S22. PROOFS FOR THE INTRINSIC BOUNDARY ATLAS

We prove the assertions of Section S18. All calculations are made in the regular coefficient ring of a Schur chart. Since the identities are invariant under changes of frames, they descend by the same argument as the Fitting identities in Proposition S5.1.

S22.1. Universal determinant completion.

Proof of Theorem S18.1. Put $N = \dim \text{Sym}^r V$ and $s = \dim S$. Choose a vector space K of dimension $N - s$ and a linear map $\rho : \text{Sym}^r V \rightarrow K$ such that

$$G = (\gamma, \rho) : \text{Sym}^r V \xrightarrow{\sim} S \oplus K$$

is an isomorphism. For the generic endomorphism M ,

$$\det(\mathrm{Sym}^r M) = (\det M)^{\kappa(e,r)}, \quad \kappa(e,r) = \frac{r}{e} \binom{e+r-1}{r} = \binom{e+r-1}{r-1}.$$

The eigenvalue identity proves the formula on diagonal matrices and hence as a polynomial identity on $\mathrm{End}(V)$.

Expand

$$\det(G \mathrm{Sym}^r M) = \det(G) d^{\kappa(e,r)}$$

by Laplace expansion along the first s rows. Every summand is an s -minor of $\gamma \mathrm{Sym}^r M$ times a complementary minor of $\rho \mathrm{Sym}^r M$. Therefore the determinant belongs to $J_{\gamma,r}$. Since $\det G \neq 0$, $d^{\kappa(e,r)} \in J_{\gamma,r}$.

If γ is an isomorphism, the ideal of maximal minors is principal and generated by $\det(\gamma) \det(\mathrm{Sym}^r M)$, so the exponent is sharp. $\square$

S22.2. The depth bound and graded multiplication.

Proof of Theorem S18.2. Apply Theorem S18.1 with $e = 4$, $r = 2$, $\gamma = \gamma_R$, and $M = I_H \oplus T$. Since $\kappa(4, 2) = 5$, $d^5 \in I_6(\gamma_R \mathrm{Sym}^2 M) = J$. Now $\mathcal{I}_{\widehat{D}_R} = dJ$. Therefore $d^6 \in dJ$, so the nilradical generated by the class of d satisfies $N^6 = 0$. Formula (S21.4) then gives $W_j = \emptyset$ for $j \geq 6$.

For the multiplication statement work locally in $A = P/(dJ)$. The class of rd^i in N^i/N^{i+1} depends on r modulo

$$(d) + (J : d^{i-1}),$$

and multiplication sends

$$(rd^i)(sd^j) = rs d^{i+j}.$$

Because

$$(J : d^{i-1}) + (J : d^{j-1}) \subset (J : d^{i+j-1}),$$

this factors through the canonical restriction to W_{i+j} . The image of $1 \otimes 1$ is the generator d^{i+j} , hence the map is surjective. The transition functions of d supply the factor $\mathcal{L}^{\otimes(i+j)}$, proving (S18.4). $\square$

S22.3. The first layer at projection corank two. Assume throughout this subsection that A_H is injective. After eliminating its image, the residual ideal is the ideal of 3-minors of

$$\Psi_H(T) = \left[\beta_H(1_H \otimes T), \chi_H \mathrm{Sym}^2 T \right], \quad \dim E_H = 3. \quad (\text{S22.1})$$

Lemma S22.1 (The contact symbol without mixed regularity). *Modulo (δ) , the residual ideal of (S22.1) is generated in degree four and equals*

$$h_H(u) \mathrm{Sym}^4(\mathbb{C}_v^2)$$

under $T = uv^t$. In particular

$$J + (\delta) = (\delta, g_0, \dots, g_4)$$

with the notation of Theorem S18.4.

Proof. On the rank-one cone write $T = uv^t$. The four mixed columns are scalar multiples, with coefficients v_1, v_2 , of the two vectors

$$\beta_H(h_1 \otimes u), \quad \beta_H(h_2 \otimes u),$$

and the three quadratic columns are scalar multiples, with coefficients $v_1^2, v_1 v_2, v_2^2$, of the single vector $\chi_H(u^2)$. A 3-minor containing three mixed columns vanishes because the mixed span has dimension at most two. A minor containing at least two quadratic columns vanishes because

those columns are proportional. Thus only minors with two mixed columns and one quadratic column survive. Their common u -factor is

$$\det[\beta_H(h_1 \otimes u), \beta_H(h_2 \otimes u), \chi_H(u^2)],$$

a binary quartic. The five possible v -monomials span $\text{Sym}^4(\mathbb{C}_v^2)$. By the same determinant comparison used in Lemma S20.4, this binary quartic is the restriction h_H of the Jacobian quartic to the pencil of hyperplanes containing H . This proves the assertion without a rank assumption on β_H . $\square$

The orbit classification in Definition S18.3 is elementary projective geometry. We record normal forms because they are also used in the colon calculation. Order the basis of $H \otimes W$ as

$$h_1 w_1, h_1 w_2, h_2 w_1, h_2 w_2.$$

Up to $GL(H) \times GL(W) \times GL(E_H)$, the mixed block can be taken to be one of the following matrices:

type	β	
3, off Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}$	
3, on Q	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	
2, secant	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, tangent	$\begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, H -ruling	$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
2, W -ruling	$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
1, dual rank 2	$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
1, dual rank 1	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$	
0	0.	(S22.2)

For rank two, a projective kernel line meets the smooth quadric in two reduced points, one double point, or is contained in one of the two ruling families. For rank one, the dual generator has tensor rank two or one. These alternatives are invariant and exhaustive, proving the orbit statement.

Lemma S22.2 (Exact transverse models for the nine mixed-kernel orbits). *Let*

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For each normal form in (S22.2) there is a nonempty Zariski-open set of quadratic blocks χ on which the Hilbert functions of

$$P/((\delta) + (J : \delta^{j-1}))$$

are those displayed in Theorem S18.4.

Proof. The two rank-three rows are Theorem S21.3 and Theorem S21.5. For the remaining rows, the assertion is a homogeneous determinant calculation. We give integer witnesses and the resulting colons; this simultaneously proves nonemptiness of the relevant rank opens.

For the secant normal form take

$$\chi = \begin{pmatrix} 1 & 2 & -2 \\ -2 & -1 & 1 \\ -2 & 1 & -2 \end{pmatrix}.$$

A direct expansion of the 3-minors followed by cancellation by δ gives

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - c^2, ab - cd, b^2 - d^2, \\ ac - 2c^2, bc - 2cd, ad - 2cd, bd - 2d^2),$$

and $(J : \delta^2) = P$. The first ideal contains δ , and the quotient has Hilbert function $(1, 4, 3)$, hence length eight.

For the tangent normal form take

$$\chi = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ -1 & -2 & 2 \end{pmatrix}.$$

Then

$$(J : \delta) = (c^3, c^2d, cd^2, d^3, a^2 - 10c^2, ab - 10cd, b^2 - 10d^2, \\ ac + 2c^2, bc + 2cd, ad + 2cd, bd + 2d^2),$$

again with $(J : \delta^2) = P$. The same Hilbert function $(1, 4, 3)$ follows.

For the H -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 0 \end{pmatrix}.$$

One obtains

$$(J : \delta) = (c^4, c^3d, c^2d^2, cd^3, d^4, 5ac^2 - c^3, 5bc^2 - c^2d, \\ 5acd - c^2d, 5bcd - cd^2, 5ad^2 - cd^2, 5bd^2 - d^3, \\ a^2 - ac, 2ab - ad - bc, b^2 - bd),$$

while

$$(J : \delta^2) = (a, b, c, d).$$

Here $\delta \notin (J : \delta)$, so $\delta^2 \notin J$; on the other hand $\delta \in (J : \delta^2)$, so $\delta^3 \in J$. After adjoining δ , the second-layer quotient has Hilbert function

$$(1, 4, 6, 4)$$

and length fifteen, while the third layer has length one.

For the W -ruling normal form take

$$\chi = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -2 \end{pmatrix}.$$

Then

$$(J : \delta) = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3), \quad (J : \delta^2) = P.$$

This ideal already contains δ . The Hilbert function of the second layer is

$$1, 4, 6, 4, 5, 6, 7, 8, \dots,$$

hence its Hilbert polynomial is $n + 1$ from degree four onward.

For the dual-rank-two rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 1 & -1 \\ 0 & 0 & -2 \\ -2 & 0 & -1 \end{pmatrix}.$$

After adjoining δ , a homogeneous basis for the second colon may be chosen as

$$(\delta, a^2c, ac^2, a^2d, abd, b^2d, acd, ad^2, bd^2).$$

Its Hilbert function is

$$1, 4, 9, 8, 10, 12, 14, 16, \dots,$$

so the Hilbert polynomial is $2n + 2$ from degree three onward. Moreover

$$(J : \delta^2) = (a, b, c, d),$$

and the third layer is the reduced vertex.

For the dual-rank-one normal form take

$$\chi = \begin{pmatrix} -2 & 2 & -1 \\ 1 & 1 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

A homogeneous basis after adjoining δ is

$$(\delta, a^3 + 4a^2c + 3ac^2, a^2b + 4a^2d + 3acd, \\ ab^2 + 4abd + 3ad^2, b^3 + 4b^2d + 3bd^2),$$

with Hilbert function

$$1, 4, 9, 12, 15, 18, 21, \dots$$

The next colon is, after an invertible linear change,

$$(c^2, cd, d^2, a, b),$$

so the third layer has Hilbert function $(1, 2)$ and length three.

Finally, when $\beta = 0$, surjectivity of $[\beta, \chi]$ forces χ to be invertible. Hence

$$J = I_3(\chi \operatorname{Sym}^2 T) = (\det \chi) \det(\operatorname{Sym}^2 T) = (\delta^3),$$

because $\det(\operatorname{Sym}^2 T) = \delta^3$ in dimension two. This proves the last row exactly.

All displayed Hilbert functions are computed from the displayed homogeneous bases by counting standard monomials. The role of these integral points is only to provide convenient base points. Genericity is proved separately in Theorem S23.1: the stabilizer orbit of each normal form is supplemented by an explicit transverse affine slice, and the residual colons are computed over the rational function field of that slice. The resulting initial ideals give exactly the Hilbert functions displayed above. Hence the open conditions in the statement are genuine dense-open conditions on the whole orbit family, not constant-rank loci inferred from one fibre. $\square$

Proof of Theorem S18.4. Lemma S22.1 gives the first layer and its contact interpretation. The orbit list follows from (S22.2). Lemma S22.2 gives the second and third layers on each coefficient-transverse open. The nilpotency index is

$$1 + \min\{k : \delta^k \in J\}$$

by Theorem S21.1; the displayed colons give the values in the table.

If $b \leq 1$, two mixed columns never span two independent target directions, so the determinant in the proof of Lemma S22.1 vanishes identically and $h_H = 0$. If the kernel is the H -ruling $h_0 \otimes W$, one of the two mixed vectors vanishes for every u , so again $h_H = 0$. For the W -ruling, the mixed span drops only at $u = w_0$, giving the forced contact root but not identical vanishing on the transverse open. This proves the remaining assertions. $\square$

S22.4. Rank-drop depth and the finite coefficient atlas.

Proof of Proposition S18.10. Eliminate the rank- a block A_H and put $q = 6 - a$. The remaining target has dimension q . A maximal residual minor uses at most b independent mixed columns, each of transverse degree one. Every remaining column must come from the quadratic block and has transverse degree two. Therefore every nonzero generator of J has degree at least

$$b + 2(q - b) = 2q - b.$$

If $2k < 2q - b$, the homogeneous polynomial δ^k cannot lie in J , proving

$$k \geq q - \lfloor b/2 \rfloor.$$

Theorem S18.2 gives $k \leq 5$.

The quadratic block has only three source dimensions when $c = 2$. Surjectivity of the original quotient after eliminating A_H therefore implies $b + 3 \geq q$. Combining this with $b \leq 4$ gives exactly the table in the statement. The equivalence between $d^k \in J$ and the augmented-rank condition $\varepsilon_k(J) = 0$ is ordinary linear algebra in the degree $2k$ homogeneous component. In the type $(a, b) = (0, 3)$ the lower bound already gives $k \geq 5$, while $d^5 \in J$, so $k = 5$ exactly. $\square$

The same coefficient construction works for $c = 3, 4$. Because $d^5 \in J$, it terminates at $k = 5$. Colon ideals are kernels of the maps

$$(P_c)_m \xrightarrow{\cdot d^{j-1}} (P_c)_{m+c(j-1)} / J_{m+c(j-1)},$$

hence are obtained by finite syzygy elimination. Noetherianity gives finite homogeneous generating sets. Rank stratification of the resulting coefficient matrices, followed by the usual flattening stratification of the finitely presented graded quotients, gives the finite atlas in Definition S18.5. This is the finite algebraic replacement for an unspecified infinite colon tower.

S22.5. Projection corank three.

Lemma S22.3 (Functorial symbol containment under mixed-rank drop). *Fix the rank of A_H at projection corank three. In every homogeneous degree, the right $GL(W)$ -module generated by the residual maximal minors for a specialization of the mixed block is a quotient of the corresponding universal maximal-mixed-rank symbol module. After multiplication by coordinate functions the same is true degree by degree.*

Proof. After eliminating $\text{im } A_H$, a residual minor with u mixed columns and $q - u$ quadratic columns is the image of the natural alternating map

$$\bigwedge^u (H \otimes W) \otimes \bigwedge^{q-u} \text{Sym}^2 W \longrightarrow \bigwedge^q E_H.$$

The construction is polynomial and functorial in the mixed map $H \otimes W \rightarrow E_H$. Replacing that map by one of smaller rank factors the displayed map through the corresponding quotient of $H \otimes W$; it cannot create a new right $GL(W)$ -constituent. Tensoring with homogeneous coordinate functions and taking images preserves this quotient relation. $\square$

Lemma S22.4 (No cubic determinant power at corank three). *At projection corank three, $d^3 \notin J$.*

Proof. Here $\dim W = 3$ and $\deg d = 3$. First suppose A_H has rank one. After eliminating its image the residual target has dimension five. The maximal-rank mixed case has rank $\overline{B}_H = 3$. By Lemma S22.3, every mixed-rank drop is a quotient of the symbol module treated below, degree by degree. Thus a determinant representation absent from the universal symbol cannot be created by specialization.

The degree-seven, degree-eight and degree-nine symbols of the five-minors use respectively three, two and one mixed columns. As right $GL(W)$ -representations they are contained in

$$\bigwedge^3 W \otimes \bigwedge^2 \text{Sym}^2 W, \quad \bigwedge^2 W \otimes \bigwedge^3 \text{Sym}^2 W, \quad W \otimes \bigwedge^4 \text{Sym}^2 W.$$

The elementary Schur decompositions are

$$\begin{aligned} \bigwedge^2 \text{Sym}^2 W &= \mathbb{S}_{(3,1,0)} W, \\ \bigwedge^3 \text{Sym}^2 W &= \mathbb{S}_{(4,1,1)} W \oplus \mathbb{S}_{(3,3,0)} W, \end{aligned}$$

and

$$\bigwedge^4 \text{Sym}^2 W = \mathbb{S}_{(4,3,1)} W.$$

Consequently the degree-eight symbol is a sum of

$$\mathbb{S}_{(5,2,1)}, \mathbb{S}_{(4,4,0)}, \mathbb{S}_{(4,3,1)}, \mathbb{S}_{(4,2,2)},$$

and the degree-nine symbol is a sum of

$$\mathbb{S}_{(5,3,1)}, \mathbb{S}_{(4,4,1)}, \mathbb{S}_{(4,3,2)}.$$

The degree-seven symbol is $\mathbb{S}_{(4,2,1)}$. To obtain a degree-nine element of the ideal one may multiply the degree-seven symbol by degree-two coordinate functions, the degree-eight symbol by degree-one coordinate functions, or take the degree-nine symbol itself. None can contain

$$(\det W)^3 = \mathbb{S}_{(3,3,3)} W :$$

every displayed highest weight has first part at least four, and the Littlewood–Richardson rule only adds boxes. Since d^3 spans this determinant representation, it is not in J .

If $A_H = 0$, the residual target has dimension six. A degree-nine minor can occur only when the mixed block has rank three and uses all three mixed columns and three quadratic columns. Its right type is contained in

$$\det W \otimes \bigwedge^3 \text{Sym}^2 W = \mathbb{S}_{(5,2,2)} W \oplus \mathbb{S}_{(4,4,1)} W,$$

again with no $(3,3,3)$ summand. If the mixed rank drops, the minimal degree is larger than nine. This proves the assertion in all cases. $\square$

Proof of Theorem S18.11. Lemma S22.4 gives $d^3 \notin J$ for all the stated coefficient data. The polynomial restriction argument in Theorem S24.2, using only the corank-four theorem and this already proved exclusion, gives $d^4 \in J$ on G_4° . The nilpotency index is one plus the first determinant exponent, hence exactly five. Outside this open the universal exponent-five bound and the definition of ε_4 still give the stated membership test. $\square$

S22.6. Projection corank four. We use one standard plethysm calculation (see, for example, [18] for Schur functors and character methods). The identity may also be checked directly by the Weyl character formula:

$$\bigwedge^6 \text{Sym}^2 V = \mathbb{S}_{(5,4,2,1)} V \oplus \mathbb{S}_{(4,4,4,0)} V. \quad (\text{S22.3})$$

Both summands occur with multiplicity one.

Lemma S22.5 (Equivariance of the corank-four residual ideal). *Let $A = \text{Sym}(V^* \otimes V) = \mathbb{C}[\text{End}(V)]$ and $J = I_6(\gamma_R \text{Sym}^2 T)$ on the projection-corank-four chart. For the right action $T \mapsto Tg$ of $GL(V)$, the ideal J is stable. Hence every homogeneous piece $J_m \subset A_m$ is a $GL(V)$ -subrepresentation.*

Proof. For $g \in GL(V)$,

$$\gamma_R \operatorname{Sym}^2(Tg) = \gamma_R(\operatorname{Sym}^2 T)(\operatorname{Sym}^2 g).$$

The last factor is invertible. Maximal minors are therefore changed only by an invertible linear combination of the columns, so their ideal is unchanged. Homogeneity gives the assertion degree by degree. $\square$

Lemma S22.6 (Effective Cauchy–Pieri multiplication in the matrix ring). *Let $A = \operatorname{Sym}(V^* \otimes V) = \mathbb{C}[\operatorname{End}(V)]$. In the Cauchy decomposition of A , put*

$$\lambda = (4, 4, 4, 0), \quad \mu = (4, 0, 0, 0), \quad \nu = (4, 4, 4, 4).$$

The actual polynomial multiplication

$$A_{12}[\lambda] \otimes A_4[\mu] \longrightarrow A_{16}$$

has nonzero projection to $A_{16}[\nu]$. This projection has multiplicity one on both the left and right Schur factors and, after the identifications

$$\mathbb{S}_\lambda V \simeq (\det V)^4 \otimes (\operatorname{Sym}^4 V)^*, \quad \mathbb{S}_\mu V = \operatorname{Sym}^4 V,$$

is a nonzero scalar multiple of the evaluation pairing.

Proof. The skew diagram ν/λ is a horizontal four-strip, so $c_{\lambda, \mu}'' = 1$. This alone does not prove nonvanishing of polynomial multiplication. Lemma S24.1 calculates that the product $[123 \mid 123]^4 x_{44}^4$ has isotypic projection $(\det X)^4/35$. Its factors belong to the indicated Cauchy summands, so actual multiplication has nonzero projection. Multiplicity one on the left and right factors and Schur’s lemma identify the resulting maps with nonzero multiples of the evaluation pairings under $\mathbb{S}_{(4,4,4,0)} V = (\det V)^4 \otimes (\operatorname{Sym}^4 V)^*$. They are nondegenerate. In particular an arbitrary nonzero Jacobian vector, not only the extremal vector used in the calculation, pairs nontrivially with a suitable degree-four coefficient. The coefficient $1/35$ is the Fischer isotypic normalization; no coefficient-one straightening identity is needed. $\square$

Lemma S22.7 (The Jacobian projection). *Under*

$$\bigwedge^6 \operatorname{Sym}^2 V \simeq (\det V)^5 \otimes \left(\bigwedge^4 \operatorname{Sym}^2 V \right)^*,$$

the summand $\mathbb{S}_{(4,4,4,0)} V$ is dual, up to the displayed determinant twist, to

$$\det V \otimes \operatorname{Sym}^4 V \subset \bigwedge^4 \operatorname{Sym}^2 V.$$

The projection

$$\bigwedge^4 \operatorname{Sym}^2 V \longrightarrow \det V \otimes \operatorname{Sym}^4 V$$

sends $q_1 \wedge \cdots \wedge q_4$, up to a nonzero scalar, to

$$\det \left(\frac{\partial q_i}{\partial x_j} \right).$$

Proof. Duality of Schur functors gives

$$(\det V)^5 \otimes \mathbb{S}_{(4,4,4,0)} V^* \simeq \det V \otimes \operatorname{Sym}^4 V.$$

The Jacobian determinant is alternating in the four quadrics and is $GL(V)$ -equivariant with exactly this target weight. It is nonzero, for instance on $x_1^2, x_2^2, x_3^2, x_4^2$. Multiplicity one in (S22.3) therefore identifies it with the stated projection. $\square$

Proof of Theorem S18.12. At $H = 0$ the ideal J is generated in degree twelve by the 6-minors of $\gamma_R \text{Sym}^2 T$. Under right multiplication of T , their span is a quotient of the representation $\bigwedge^6 \text{Sym}^2 V$. The polynomial d^3 spans $(\det V)^3 = \mathbb{S}_{(3,3,3,3)} V$, which does not occur in (S22.3). Hence $d^3 \notin J$.

Let

$$\lambda_R = \bigwedge^6 \gamma_R.$$

By Lemma S22.7, the component of λ_R on the $\mathbb{S}_{(4,4,4,0)}$ -summand is nonzero exactly when the Jacobian quartic F_R is nonzero. For $R \in G_4^\circ$, the quartic $Y_R = V(F_R)$ is smooth, so this component is nonzero.

Let $A = \mathbb{C}[\text{End}(V)]$. The nonzero Jacobian component just identified gives a nonzero vector in the $\lambda = (4, 4, 4, 0)$ Cauchy component of J_{12} . By Lemma S22.6, multiplication by the $\mu = (4)$ component of A_4 has a nonzero $\nu = (4, 4, 4, 4)$ -projection. Because the induced pairing

$$\mathbb{S}_{(4,4,4,0)} V \otimes \text{Sym}^4 V \longrightarrow (\det V)^4$$

is nondegenerate, one may choose the degree-four matrix coefficient to pair nontrivially with the specific nonzero Jacobian vector.

By Lemma S22.5, J_{16} is a right $GL(V)$ -subrepresentation of A_{16} . Over $\mathbb{C}$ it is completely reducible, so the equivariant projection onto any right isotypic summand preserves J_{16} . Apply the projection onto right type $\nu = (4, 4, 4, 4)$ to the product just constructed. Its image is nonzero by Lemma S22.6 and still belongs to J_{16} .

In the Cauchy decomposition of the full matrix coordinate ring, right type ν is paired with the same left type. Both $\mathbb{S}_{(4,4,4,4)} V$ and $\mathbb{S}_{(4,4,4,4)} V^*$ are one-dimensional determinant characters. Hence this entire Cauchy summand is the one-dimensional line

$$\mathbb{C}(\det T)^4 = \mathbb{C}d^4.$$

A nonzero element of this line lying in J_{16} therefore implies

$$d^4 \in J.$$

Thus ideal membership follows from three explicit ingredients: equivariance of the maximal-minor ideal, semisimple projection in characteristic zero, and the nonzero coordinate-ring Pieri coefficient. No projection is taken outside the ideal. Together with $d^3 \notin J$, this gives nilpotency index five. $\square$

S22.7. Specialization and associated supports. The coefficient matrices above have polynomial entries, but this fact alone does not justify fibrewise statements about embedded associated points or base change of colons. The required order of argument is carried out in Section S23. There we first flatten the finite presentations controlling each kernel ($J : d^{j-1}$) and its cokernel, then apply relative assassins, generic freeness at the associated generic points, and Noetherian induction. Proposition S18.9 follows from that argument.

The same section computes the generic corank-two primary signatures over rational function fields and gives explicit one-parameter orbit degenerations. Thus the specialization statements used by the paper are consequences of exact family ideals and relative associated-point theory, not of generic flatness alone.

APPENDIX S23. RELATIVE PRIMARY SIGNATURES AND SPECIALIZATION LAWS

This section supplies the two family arguments that are not consequences of a single exact witness: genericity on coefficient orbits and fibrewise control of associated points. It also extracts the geometric primary signatures of the corank-two layers and writes explicit orbit degenerations.

S23.1. Stabilizer slices and generic Groebner bases. Keep

$$P = \mathbb{C}[a, b, c, d], \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \delta = ad - bc.$$

For a mixed normal form β , let $G_\beta \subset GL(H) \times GL(W) \times GL(E_H)$ be its stabilizer. If (X, Y, Z) lies in its Lie algebra, then

$$Z\beta - \beta(X \otimes 1 + 1 \otimes Y) = 0, \quad (\text{S23.1})$$

and its infinitesimal action on the quadratic block is

$$\dot{\chi} = Z\chi - \chi \operatorname{Sym}^2 Y. \quad (\text{S23.2})$$

These are linear equations over $\mathbb{Q}$ at the integral base points of Lemma S22.2.

Theorem S23.1 (Generic slice certification for the mixed-kernel atlas). *For each of the seven mixed-kernel rows not already covered by the mixed-regular and decomposable-kernel theorems, the Hilbert functions in Theorem S18.4 are the generic Hilbert functions on the entire orbit family.*

More precisely, at the integral base points used in Lemma S22.2, the ranks of the infinitesimal G_β -orbits in the nine-dimensional χ -space and one choice of transverse affine slice are

type	$\operatorname{rank} T_\chi(G_\beta \chi)$	slice entries
secant	6	$\chi_{11}, \chi_{21}, \chi_{31}$
tangent	7	χ_{11}, χ_{21}
H -ruling	7	χ_{11}, χ_{13}
W -ruling	9	$\emptyset$
dual rank 2	9	$\emptyset$
dual rank 1	9	$\emptyset$
zero mixed map	9	$\emptyset$.

On the first three slices, with coordinates s_i , Groebner bases of all required colons over $\mathbb{Q}(s_i)$ give respectively

type	$\min\{k : \delta^k \in J\}$	H_{W_2}	H_{W_3}
secant	2	$(1, 4, 3)$	0
tangent	2	$(1, 4, 3)$	0
H -ruling	3	$(1, 4, 6, 4)$	(1) .

The only nonconstant denominators in reduced bases may be chosen to divide

$$s_2 - 2 \quad (\text{secant}), \quad s_0 + 2s_1 - 5 \quad (H\text{-ruling}),$$

and neither vanishes at the base point $s = 0$; the tangent slice requires no nonconstant denominator. In the remaining four rows the stabilizer orbit is already open in χ -space, so the exact base-point calculation is the generic calculation.

Proof. Solving (S23.1) and applying (S23.2) gives the ranks in the table. Adjoining the displayed coordinate directions raises the tangent rank to nine. Thus the morphism from the stabilizer times the affine slice to χ -space has surjective differential at the base point. Over characteristic zero it is therefore dominant and smooth on a neighbourhood of that point.

Compute the residual minors and successive colons in $\mathbb{Q}(s_i)[a, b, c, d]$. For the secant and tangent slices the second-layer generic initial ideal is

$$(c, d)^3 + (a, b)^2 + (a, b)(c, d),$$

which has Hilbert function $(1, 4, 3)$, and the next colon is the unit ideal. For the H -ruling slice a generic initial ideal of the second layer is generated by

$$c^4, c^3d, c^2d^2, cd^3, d^4, ac^2, acd, ad^2, bd^2, a^2, ab, b^2, bc,$$

with Hilbert function $(1, 4, 6, 4)$; the third-layer initial ideal is (a, b, c, d) . These initial ideals are unchanged after specialization away from the displayed denominator divisors.

For the four open-orbit rows the exact Groebner bases give the functions printed in Theorem S18.4. The calculation is reproduced, over exact rational function fields rather than floating-point samples, by `checks/generic_boundary_atlas.py`. Since the slice map is dominant, these are the generic initial ideals on the orbit families. $\square$

S23.2. Geometric primary signatures for the nine mixed-kernel types. Let $Q = V(\delta)$ be the affine Segre cone and let $\mathfrak{m} = (a, b, c, d)$ be its vertex. Write R_r for a reduced divisor consisting of r distinct lines in one ruling of Q . Write V_ℓ for a homogeneous $\mathfrak{m}$ -primary scheme of length ℓ . Finally, $R_r + E_\ell$ means a scheme whose saturation is R_r and whose saturation quotient is an embedded vertex module of length ℓ . This notation records associated supports and generic lengths; it does not choose a noncanonical embedded primary component.

Theorem S23.2 (Generic primary signatures for the mixed-kernel orbits). *On the generic coefficient stratum of each A_H -injective mixed-kernel orbit, the positive layers have the following primary signatures:*

<i>mixed-kernel type</i>	W_2	W_3
3, point off Q	V_1	$\emptyset$
3, point on Q	V_3	$\emptyset$
2, secant	V_8	$\emptyset$
2, tangent	V_8	$\emptyset$
2, H -ruling	V_{15}	V_1
2, W -ruling	$R_1 + E_5$	$\emptyset$
1, dual rank 2	$R_2 + E_3$	V_1
1, dual rank 1	R_3	V_3
0	Q	Q .

In particular, every associated support and every generic length in these layers is determined. The symbols R_r have r minimal ruling-line primes, each of generic length one. The only additional associated prime in $R_1 + E_5$ and $R_2 + E_3$ is the embedded vertex $\mathfrak{m}$.

Proof. The finite rows are homogeneous zero-dimensional schemes supported at the vertex, hence are $\mathfrak{m}$ -primary; their lengths are the Hilbert-function sums. For the W -ruling witness the second layer is

$$I = (ac, bc, ad, bd, c^3, c^2d, cd^2, d^3).$$

Its saturation is (c, d) , and $(c, d)/I$ has length five. Thus its associated primes are (c, d) and $\mathfrak{m}$, with an embedded vertex contribution of length five.

For the dual-rank-two witness, saturation gives

$$(a, b) \cap (c, d),$$

the union of two ruling lines, and the saturation quotient has length three. For the dual-rank-one witness, modulo δ , the second-layer ideal is exactly

$$\bigcap_{\lambda \in \{0,1,3\}} (a + \lambda c, b + \lambda d),$$

so it is a reduced union of three ruling lines and has no embedded point. The third layer is the length-three vertex thickening already computed in Lemma S22.2. The zero mixed-map identity $J = (\delta^3)$ gives $W_2 = W_3 = Q$.

Theorem S23.1 transports these exact signatures to the generic coefficient strata. Associated supports are preserved by the stabilizer action, and the relative-primary argument below prevents an isolated base-point calculation from being used beyond its certified open stratum. $\square$

S23.3. Pure-block rank drop at projection corank two. When A_H has rank $a < 3$, put $q = 6 - a$ and $b = \text{rank } \overline{B}_H$. For $b = 3, 2, 1$ the phrase *generic mixed kernel* means respectively an off-Segre kernel point, a secant kernel line, and a kernel plane whose dual tensor has rank two.

Theorem S23.3 (Generic primary signatures for every A_H -rank-drop type). *At projection corank two, every generic rank type allowed by Proposition S18.10 has the following complete list of nonzero positive layers:*

a	b	nilpotency index	$(W_1, W_2, W_3, W_4, W_5)$
2	4	3	(Q, V_{22})
2	3	4	(Q, V_{27}, V_1)
2	2	4	(Q, R_4, V_8)
2	1	5	(Q, Q, Q, V_1)
1	4	5	(Q, Q, V_{11}, V_1)
1	3	5	$(Q, Q, R_2 + E_3, V_1)$
1	2	5	(Q, Q, Q, R_2)
0	4	6	(Q, Q, Q, R_2, V_1)
0	3	6	(Q, Q, Q, Q, V_1) .

Here every Q is the reduced Segre quadric with generic length one; every R_r is a reduced union of r ruling lines with generic length one on each component; and every V_ℓ is $\mathfrak{m}$ -primary of the indicated length. In the unique row $(a, b) = (1, 3)$ carrying E_3 , the associated primes of that layer are the two ruling-line primes together with the embedded vertex.

Proof. Choose the generic normal forms of the mixed block described above and let χ vary. The stabilizer-slice codimensions in χ -space for

$$(a, b) = (2, 4), (2, 3), (2, 2), (2, 1), (1, 4), (1, 3), (1, 2), (0, 4), (0, 3)$$

are respectively

$$5, 4, 1, 0, 3, 1, 0, 0, 0.$$

On explicit complementary coordinate slices the successive colons are computed over the rational function fields of those slices. Their Hilbert functions are

a	b	non-quadric Hilbert functions
2	4	$(1, 4, 9, 8)$
2	3	$(1, 4, 9, 8, 5); (1)$
2	2	$4n + 4 \ (n \geq 4); (1, 4, 3)$
2	1	(1)
1	4	$(1, 4, 6); (1)$
1	3	$2n + 2 \ (n \geq 3); (1)$
1	2	$2n + 2 \ (n \geq 1)$
0	4	$2n + 2 \ (n \geq 1); (1)$
0	3	(1) .

Every omitted earlier layer has Hilbert function $(n + 1)^2$, hence equals Q .

To identify the positive-dimensional supports, pull the layer ideals back to the rank-one parametrization $T = uv^t$, but now perform the factorization over the same rational-function fields as the Groebner calculation.

For $(a, b) = (2, 2)$, with slice parameter s , the second layer has common rank-one factor, up to a unit of $\mathbb{Q}(s)$,

$$u_0 u_1 \left((s + 9)u_0^2 + (3s + 6)u_0 u_1 - 7u_1^2 \right).$$

The quadratic discriminant is

$$9s^2 + 64s + 288.$$

On the open set $(s + 9)(9s^2 + 64s + 288) \neq 0$, its two roots are distinct and different from the two roots $u_0u_1 = 0$. Thus the generic support is a reduced divisor of four distinct ruling lines. Its Hilbert polynomial agrees with the computed layer, so there is no generic embedded correction.

For $(a, b) = (1, 3)$, with slice parameter t , the third layer has common factor, again up to a unit of $\mathbb{Q}(t)$,

$$q(u) = 10u_0^2 + u_0u_1 - 4u_1^2.$$

Its discriminant is 161, so it gives two distinct ruling lines over the algebraic closure of $\mathbb{Q}(t)$. The exact generic-field calculation also computes the saturation. Writing $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the saturated ruling ideal is

$$I_{\text{sat}} = (\delta, 10a^2 + ac - 4c^2, 10ab + ad - 4cd, \\ 10b^2 + bd - 4d^2).$$

The Groebner bases over $\mathbb{Q}(t)$ give

$$H_{M_3} = (1, 4, 9, 8, 10, 12, 14, \dots), \quad H_{P/I_{\text{sat}}} = (1, 4, 6, 8, 10, 12, 14, \dots).$$

The quotient I_{sat}/I_{M_3} is therefore concentrated in degree two and has length three. This is the claimed embedded vertex module, so the generic signature is $R_2 + E_3$.

For the two open-orbit rows $(1, 2)$ and $(0, 4)$, the previously displayed base-point factors remain generic by orbit transport:

$$u_0u_1, \quad 6u_0^2 + u_0u_1 + 4u_1^2,$$

and the second quadratic has nonzero discriminant. Thus every positive-dimensional rank-drop support in the table is certified on its generic coefficient field, rather than inferred from an isolated integral point. The exact stabilizer ranks, slice coordinates, Groebner leading monomials, denominator checks, and Hilbert counts are reproduced by `checks/generic_boundary_atlas.py`. Because each slice map is dominant at its base point, the calculation is at the generic point of the rank type rather than at a special witness. $\square$

S23.4. Relative assassins and base change for the colon tower. We prove the general bounded principal-colon theorem before applying it to the determinant d . The order of the argument is essential.

S23.4.1. Finite diagrams and primary models. We give an algebraic construction which distinguishes persistence of an associated point from exhaustion of the associated set. Every restriction to a base stratum in this construction is followed by formation of the indicated kernels on that stratum. We do not identify these kernels with the pullback of a kernel from an earlier, unflattened stratum.

Lemma S23.4 (Spreading a finite exact diagram). *Let D be a noetherian domain with fraction field K , let A be a finite type D -algebra, and fix a finite diagram of finite A_K -modules and maps, including finitely many short exact sequences. After replacing D by D_g , the diagram has coherent models for which all the specified short exact sequences remain exact after every base change over D_g . The same assertion holds for a diagram over a finite extension of K , after a finite faithfully flat cover of a nonempty open of $\text{Spec } D$.*

Proof. Choose presentations, maps, and the finitely many identities expressing commutativity. Clear their denominators. Kernels and homology modules are finite over A . A finite module which vanishes after tensoring with K is killed by a nonzero element of D , so a further localization makes the specified sequences exact. Apply generic freeness to their finitely many terms and cokernels [27, Tag 051S]. In a short exact sequence with flat cokernel, tensoring with any base

algebra preserves the injection; right exactness preserves the remaining terms. This proves the claim without any four-term kernel inference.

For a finite field extension L/K , choose finitely many generators of L , clear denominators in their monic equations, and let D' be the finite D_g -algebra which they generate. After another localization, D' is finite free of positive rank over D_g . It is therefore faithfully flat. Apply the preceding construction over D' . A further shrinking upstairs descends to a shrinking downstairs: the image of its closed complement is closed under the finite morphism and misses the generic point. No separability is needed for this step. $\square$

Proposition S23.5 (Primary quotients on a splitting cover). *Let A be a finite type algebra over a noetherian domain D , and let M be a finite A -module. There is a nonempty open of $\text{Spec } D$ and a finite faithfully flat cover of that open on which there exist ideals $\mathfrak{p}_i$, modules Q_i , and maps*

$$M \longrightarrow Q_i, \quad M \longrightarrow \bigoplus_i Q_i, \quad A/\mathfrak{p}_i \longrightarrow M$$

with the following properties. On every geometric fibre the first maps are surjective, the last two displayed maps are injective, the rings $A/\mathfrak{p}_i$ are nonzero integral domains, and each Q_i is a nonzero $\mathfrak{p}_i$ -primary module. The primes may be taken distinct on every geometric fibre. Consequently

$$\text{Ass}(M_{\bar{s}}) = \{\mathfrak{p}_{i,\bar{s}}\}_i, \quad 0 = \bigcap_i \ker(M_{\bar{s}} \longrightarrow Q_{i,\bar{s}})$$

is an irredundant primary decomposition. All exact diagrams used to construct these maps are preserved by arbitrary base change on the chosen cover.

Proof. On the geometric generic fibre choose an irredundant primary decomposition $0 = \bigcap_i N_i$, and put $Q_i = M/N_i$, with $\text{Ass } Q_i = \{\mathfrak{p}_i\}$. Choose also an associated element for each $\mathfrak{p}_i$; it gives an injection $A/\mathfrak{p}_i \hookrightarrow M$. The diagonal map $M \hookrightarrow \bigoplus_i Q_i$ is injective.

We must spread the assertion that a quotient is primary, not merely its support. Choose n_i with $\mathfrak{p}_i^{n_i} Q_i = 0$, and set

$$F_{i,j} = (0 :_{Q_i} \mathfrak{p}_i^j), \quad 0 \leq j \leq n_i, \quad E_{i,j} = F_{i,j}/F_{i,j-1}.$$

Each $E_{i,j}$ is killed by $\mathfrak{p}_i$ and is torsion-free over the domain $A/\mathfrak{p}_i$. Indeed, if $h \notin \mathfrak{p}_i$ and $hv \in F_{i,j-1}$, then $h\mathfrak{p}_i^{j-1}v = 0$; multiplication by h is injective on the primary module Q_i , so $v \in F_{i,j-1}$. Clearing fraction-field denominators now gives injections

$$E_{i,j} \hookrightarrow (A/\mathfrak{p}_i)^{r_{i,j}}.$$

These injections, the filtration sequences, the diagonal map, the quotient maps, and the associated-element injections form a finite diagram. All its coefficients descend to a finite extension of the generic field. The prime quotients can be descended at the same time with geometrically integral generic fibres. After shrinking, their geometric fibres are integral and nonempty: apply generic flatness, spreading of geometric reducedness and irreducibility, and the fact that the image of a finite-presentation morphism containing the generic point contains a nonempty open [28, Sections 37.26–37.27]. Apply Lemma S23.4 to the whole finite diagram.

In a geometric fibre, every $E_{i,j}$ embeds in a free module over an integral prime quotient. It therefore has no associated prime other than $\mathfrak{p}_i$. The filtration gives $\text{Ass } Q_i \subset \{\mathfrak{p}_i\}$; the surviving quotient map and associated-element data imply nonvanishing whenever that index occurs. More explicitly, the diagonal injection gives the inclusion

$$\text{Ass } M_{\bar{s}} \subset \bigcup_i \text{Ass } Q_{i,\bar{s}} \subset \{\mathfrak{p}_{i,\bar{s}}\}_i.$$

The associated-element injections give the reverse inclusion. This is the exhaustion argument: an additional embedded prime anywhere in the fibre would contradict the diagonal injection into these filtered primary modules. It is not an open-neighbourhood containment argument.

For every generic noncontainment $\mathfrak{p}_i \not\subset \mathfrak{p}_j$, choose $h_{ij} \in \mathfrak{p}_i \setminus \mathfrak{p}_j$. Include multiplication by h_{ij} on $A/\mathfrak{p}_j$ and its cokernel in the finite diagram. Fibrewise injectivity preserves that noncontainment. Generic containments spread by finite generation. Thus distinctness and the containment poset persist. The kernels of the quotient maps have zero intersection by the diagonal injection. No component can be omitted: otherwise the associated-prime inclusion for the remaining diagonal injection would exclude its distinct associated prime. This proves irredundance as well as primaryness. $\square$

S23.4.2. *A relative module for the local torsion length.*

Proposition S23.6 (No new packet torsion after refinement). *In the situation of Proposition S23.5, fix a tracked prime $\mathfrak{p}$. After shrinking the base there are an integer n , a coherent submodule $T \subset M$, and an element $h \in \mathfrak{p}$ such that, after every base change on this open,*

$$T = (0 :_M \mathfrak{p}^n) = H_{\mathfrak{p}}^0(M), \quad h : M/T \longrightarrow M/T \text{ is injective.}$$

At each generic point p of a geometric packet fibre,

$$T_p = H_{\mathcal{O}_p}^0(M_p).$$

There are nonnegative integers $r_0, \dots, r_{n-1}$, independent of that geometric fibre, for which

$$\text{length}_{\mathcal{O}_p} H_{\mathcal{O}_p}^0(M_p) = \sum_{j=0}^{n-1} r_j.$$

Here H_I^0 denotes elements killed by some power of the ideal I ; the notation does not denote a sheaf cohomology group.

Proof. On the split generic fibre the annihilator chain stabilizes. Choose n with $T = (0 :_M \mathfrak{p}^n) = H_{\mathfrak{p}}^0(M)$, and put $Q = M/T$. This quotient has no $\mathfrak{p}$ -power torsion. Equivalently, no associated prime of Q contains $\mathfrak{p}$. Prime avoidance therefore supplies $h \in \mathfrak{p}$ outside all the associated primes of Q . Multiplication by h is injective on Q . When $Q = 0$ this assertion is vacuous and no choice is needed.

Spread the sequence $0 \rightarrow T \rightarrow M \rightarrow Q \rightarrow 0$, the relation $\mathfrak{p}^n T = 0$, and the injection $h : Q \rightarrow Q$. Flatten Q and the cokernel of this injection, together with the other finite diagram terms. They remain exact after arbitrary base change. An element of the base-changed M killed by $\mathfrak{p}^a$ has image in Q killed by h^a . The image is zero, so the element belongs to T . The converse follows from $\mathfrak{p}^n T = 0$. This proves the asserted equality for the entire relative torsion module, not only for a submodule chosen at the generic point. Localization gives the displayed local equality.

Consider the finite filtration

$$T \supset \mathfrak{p}T \supset \dots \supset \mathfrak{p}^n T = 0, \quad G_j = \mathfrak{p}^j T / \mathfrak{p}^{j+1} T.$$

Include its exact sequences and the quotients $T/\mathfrak{p}^j T$ in the finite diagram. Their flatness makes this the same power filtration on every fibre. On the generic integral packet each G_j becomes free of rank r_j after inverting an element $g_j \notin \mathfrak{p}$. Spread such a free presentation and its inverse. Include multiplication by g_j on $A/\mathfrak{p}$ and its cokernel in the diagram. It remains injective, so g_j is nonzero at the generic point of every geometric packet fibre. Thus $(G_j)_p$ is an r_j -dimensional vector space over the residue field at p . Additivity of length gives the sum. All these operations were performed over the splitting cover, hence also after geometric field extension. In particular, inseparable residue field extension is not silently treated as an ordinary rational point. $\square$

S23.4.3. *Descent of the closed packets.* A packet is a closed subscheme on the original base whose geometric irreducible components are a full conjugacy orbit of tracked associated points. Its ideal, rather than its finite image as a set, is the object that must be constructed. We give an effective descent construction in characteristic zero and a power-certified construction in arbitrary characteristic. The latter retains the noetherian-base generality of Theorem S18.6; it does not require a packet to be geometrically reduced.

Lemma S23.7 (Effective Galois descent of an orbit ideal). *Let D be a noetherian domain with fraction field K of characteristic zero. Let A be a finite type D -algebra. After shrinking $\text{Spec } D$, a finite Galois extension L/K with group G can be modelled by a finite étale G -torsor D'/D . Put $B = A \otimes_D D'$. Suppose that a finite G -stable set of generic primes has closures $P_i \subset B$. For a full orbit O , set $I'_O = \bigcap_{i \in O} P_i$. There is a unique ideal $I_O \subset A$ with its natural inclusion in A such that*

$$I_O B = I'_O, \quad I_O = \{a \in A : a \otimes 1 \in I'_O\}. \quad (\text{S23.3})$$

If the B/P_i have nonempty geometrically integral fibres, a further shrinking makes every geometric fibre of $V(I_O)$ exactly the reduced union of the intended components. The orbit-ideal identity and the fibrewise intersection identity persist after every base change on this open.

Proof. Choose a finite D -subalgebra of L , include the conjugates of its generators, and invert one nonzero element of D . The resulting algebra D' is finite free of positive rank. The generic Galois isomorphism

$$D' \otimes_D D' \longrightarrow \prod_{g \in G} D', \quad b \otimes c \longmapsto (bg(c))_g$$

is an isomorphism after another shrinking: its kernel and cokernel are finite D -modules with zero generic fibre. The torsor identities, or equivalently this isomorphism together with faithful flatness, give a finite étale G -torsor. No normalization or excellence hypothesis on D is used.

Schematic closure commutes with each automorphism g , so $g(P_i) = P_{gi}$ and $g(I'_O) = I'_O$. On the double overlap $\text{Spec } B \times_{\text{Spec } A} \text{Spec } B$, identify the two pullbacks of I'_O on the component indexed by g by the restriction of this semilinear action. On a triple overlap the two successive identifications are multiplication of the group elements. They therefore equal the direct identification. This is the cocycle condition, as an identity of maps of ideal sheaves, not just of their supports. The inclusion $I'_O \hookrightarrow B$ is equivariant. Effective fpqc descent for quasi-coherent modules, together with full faithfulness, descends this inclusion to a submodule of A [29, Tag 023T and Section 35.6]. Multiplication preserves the submodule after the faithfully flat extension, hence before it; the descended submodule is an ideal. Faithful flatness also proves both identities in (S23.3).

To identify the fibres, include the actual exact sequence

$$0 \longrightarrow B/I'_O \longrightarrow \bigoplus_{i \in O} B/P_i \longrightarrow C_O \longrightarrow 0 \quad (\text{S23.4})$$

in the finite diagram and flatten C_O and its terms over D' . After arbitrary base change the diagonal map stays injective. Consequently the extended ideal I'_O is the intersection of the extended P_i ; this conclusion would not follow from flatness of the individual prime quotients alone. On a geometric fibre each P_i is prime, so this intersection is radical. Distinctness is protected by the noncontainment witnesses of Proposition S23.5. The torsor over an algebraically closed field is a disjoint union of copies of that field. Pulling the descended packet to any one of these copies identifies its fibre with precisely this intersection. Thus no additional component is introduced downstairs. All subsequent pullbacks of (S23.3) remain equalities; the flattened sequence gives the asserted intersection compatibility. $\square$

Lemma S23.8 (Packets without a separability hypothesis). *Let D , K , and A be as above, without a restriction on the characteristic. A full orbit of geometric generic associated points has a closed*

packet on $\text{Spec } A$, after shrinking the base, with exactly those geometric fibre components supplied by the split primary models. These packets and their intersections can be refined and flattened on the original base. Their component identification remains valid after every base change in the final stratum.

Proof. Let $\mathfrak{q} \subset A_K$ be the contraction of a member of the orbit. Geometric components over $\mathfrak{q}$ form one conjugacy orbit: over the maximal separable subextension this is the usual transitivity of primes in a Galois extension, and purely inseparable extension is a universal homeomorphism. The associated point formula for extension of the ground field identifies these components with the tracked geometric associated points [21, Section 31.7].

Construct the ideal on the original ring first:

$$Q = \ker(A \longrightarrow A_K/\mathfrak{q}).$$

This is the schematic-closure ideal and is finite because A is noetherian. Take a common finite splitting extension L/K and a finite faithfully flat model D'/D as in Lemma S23.4. On $B = A \otimes_D D'$, let P_i be the closures of the generic geometric primes in this orbit and put $I' = \bigcap_i P_i$. On the generic fibre

$$\sqrt{\mathfrak{q}A_L} = \bigcap_i (P_i)_L.$$

Finite generation therefore gives an integer $n \geq 1$ with $(I'_L)^n \subset \mathfrak{q}A_L \subset I'_L$. Clear the finitely many denominators in these two inclusions. After shrinking on the original base, they become actual ideal inclusions

$$(I')^n \subset QB \subset I'. \quad (\text{S23.5})$$

As before, a shrinking needed upstairs is made downstairs by deleting the finite image of its bad closed complement. Finite images are used only to shrink a base on which an ideal has already been constructed; they are not used as a substitute for descent.

Flatten the diagonal sequence $0 \rightarrow B/I' \rightarrow \bigoplus_i B/P_i \rightarrow C \rightarrow 0$, and arrange the nonempty geometrically integral fibres and distinctness of the P_i by the split primary models. For every geometric fibre Ω , the intersection identity and (S23.5) now give

$$\sqrt{(QB)_\Omega} = \bigcap_i (P_i)_\Omega. \quad (\text{S23.6})$$

Indeed, products of finitely generated ideals commute with extension of scalars, while the flattened diagonal sequence supplies exactly the intersection identity that is needed here. The right side is an intersection of distinct primes. A geometric point of $\text{Spec } D$ has a lift to $\text{Spec } D'$ over the same algebraically closed field: the nonzero finite fibre algebra has a residue field equal to that field. At this lift, the fibre of A and that of B are the same algebra. Hence (S23.6) identifies the components of the packet $V(Q)$ on the original base itself. This argument does not assert that a nonreduced inseparable fibre is reduced or that a rational local length is unchanged by inseparable extension.

Schematic closures glue on affine overlaps. In a finite affine cover, use also a finite affine refinement of the overlaps and one common shrinking. A closure whose generic restriction to a chart is empty has empty restriction after shrinking, by constructibility of the image of that finite-presentation restriction. Thus the chartwise component statements glue without an untracked boundary component. For any finite collection of packets, their scheme-theoretic intersections are defined downstairs by sums of their ideals. Flatten these finitely many quotient algebras and the packet algebras on the original base. Sums commute with base change, and all previous exact sequences and inclusions survive this further restriction. Stratifying their images and fibre dimensions fixes nonemptiness and dimensions of their incidences. This proves the assertion. $\square$

Proposition S23.9 (The torsion module on the original base). *Let $I \subset A$ be any one of the packet ideals just constructed and let M be a finite A -module. On a nonempty open of the integral base there are a finite submodule $T \subset M$ and an integer n such that, after every base change on this open,*

$$T = (0 :_M I^n) = H_I^0(M). \quad (\text{S23.7})$$

At the generic point p of each geometric component of the packet, $T_p = H_{p\mathcal{O}_p}^0(M_p)$. After common refinement with the split primary models, these geometric local lengths are constant and equal on the components of each packet.

Proof. On the generic fibre choose n for which $T_K = (0 :_{M_K} I_K^n) = H_{I_K}^0(M_K)$ and set $Q_K = M_K/T_K$. There is no associated prime of Q_K containing I_K ; otherwise an associated element would give nonzero I_K -torsion. Prime avoidance gives an element $h_K \in I_K$ regular on Q_K . If $Q_K = 0$, omit this map. Spread the submodule, the relation $I^n T = 0$, and

$$0 \rightarrow T \rightarrow M \rightarrow Q \rightarrow 0, \quad h : Q \rightarrow Q.$$

Flatten the quotient Q and the cokernel of the second map. After every base change the first injection and regularity of h remain true. An element killed by a power of I has image in Q killed by a power of h , hence has zero image. This proves (S23.7) downstairs, with no descent assertion about a chosen primary decomposition.

At a minimal prime p of a geometric packet, the radical of $I\mathcal{O}_p$ is the maximal ideal $p\mathcal{O}_p$. Zeroth local torsion depends only on this radical, which proves the local equality. On the splitting cover apply the finite power filtrations and free generic presentations of Proposition S23.6 to every component. They give a constant integer for each component. Conjugate generic points have equal lengths, so the integers agree within an orbit. At a geometric lift over the same algebraically closed field, the local ring and module are literally the same as downstairs. Consequently their lengths agree. This supplies both the relative module and the length statement on the original base. $\square$

Lemma S23.10 (Finite geometric-assassin stratification). *Let $X \rightarrow S$ be of finite presentation with S noetherian, and let $\mathcal{F}$ be a finitely presented coherent module flat over S . There is a finite locally closed stratification with finitely many closed packets Z_ν on each stratum such that*

$$\text{Ass}(\mathcal{F}_{\bar{s}}) = \bigcup_{\nu} \{\text{generic points of the irreducible components of } (Z_\nu)_{\bar{s}}\}$$

for every geometric fibre. The packets may split geometrically. After further finite refinement they are flat, and their dimensions, incidences, minimal-versus-embedded status, and local torsion lengths at associated generic points are constant. On finite splitting covers of affine charts the packets are realized by the actual primary quotients of Proposition S23.5.

Proof. We first construct the required data on a dense open of an integral base. Choose a finite affine cover of X . Apply Propositions S23.5 and S23.6 on these charts, using one common finite extension of the generic field and one common shrinking. Associated points are local for an affine open cover. Consequently every fibrewise associated point is one of the primes in those finite primary models. This proves exhaustion on the whole fibre.

Take the schematic closures on the original base of the contractions of the tracked geometric generic points. Apply Lemma S23.8 to each conjugacy orbit, with one common splitting cover and refinement. It constructs the ideal of each packet downstairs and proves exact equality of its geometric component set using the power sandwich and a base-change-exact diagonal sequence. In characteristic zero, the stronger reduced orbit-ideal construction of Lemma S23.7 gives the explicit fpqc datum and its cocycle. On a geometric lift over the same algebraically closed field, the already proved associated-point equality upstairs is an equality for the original fibre. Thus

the generic points of these descended packets are exactly the associated points, in both directions. The overlap construction in Lemma S23.8 proves the same statement across the affine charts.

The noncontainment witnesses in Proposition S23.5 fix the containment poset, hence minimal and embedded status, on the splitting cover. Conjugate generic points have equal dimensions and torsion lengths; these equalities persist by Proposition S23.6 and give the downstairs lengths by Proposition S23.9. That proposition also constructs the coherent relative torsion module on the original base, not just on its cover. Generic flatness for the finitely many packet closures fixes their fibre dimensions. For intersections, apply the same finite-presentation dimension and image stratifications to the finitely many intersections of packet closures. This fixes nonemptiness and dimensions of their incidences as well. Finally flatten their structure sheaves.

There are finitely many irreducible components of the reduced base. Carry out the preceding construction on dense opens of those components, remove these opens, and repeat on the proper closed complement, with the modules and kernels re-formed there. Noetherian induction gives a finite stratification. Nilpotents in the base do not alter its geometric fibres; reduced locally closed strata suffice. The exact-diagram constructions hold after every base change within a final stratum. They make no assertion that a kernel or a torsion module commutes with a specialization crossing two different strata. $\square$

Remark S23.11 (Why a torsion construction is necessary). Over $D = \mathbb{C}[t]$ the module

$$M = D[x, y]/(x^2, x(y^2 - t)) = D[y] \oplus xD[y]/(y^2 - t)$$

is flat. The second summand is its finite torsion contribution along $(x, y^2 - t)$. For $t \neq 0$ it splits into two geometric embedded points of length one; at $t = 0$ they form one embedded point of length two. The quotient $D[y]$ is protected from new torsion by multiplication by $y^2 - t$. Flatness of the total module alone would neither identify this torsion summand nor keep the length of an individual geometric component constant across the collision.

S23.4.4. Grading and the finite family of multiplication maps.

Lemma S23.12 (Graded consequences of a finite flat diagram). *Let $A = D[x_1, \dots, x_m]$ have its standard grading, and let L be a finite graded A -module flat over a noetherian ring D . Every L_n is a finite locally free D -module. On a connected base its Hilbert function, and therefore its Hilbert polynomial, is constant. A prescribed finite family of homogeneous A -linear maps whose sources, targets, images, and cokernels are flat has base-change-compatible, constant rank in every degree on such a base.*

Proof. Every graded component is a direct summand of the underlying flat D -module and is finite over D . It is thus finite flat, hence locally free. Its rank is locally constant and is constant on a connected base. This applies simultaneously to all degrees because flatness of the whole graded module has already been obtained; it does not require an infinite sequence of independent refinements. Apply the same observation to the image and cokernel of each map, and base-change the two short exact sequences defining its image. $\square$

Proof of Theorem S18.6. Fix $0 \leq q < N$ and put

$$B = A/J, \quad C_q = B/f^q B = A/(J, f^q), \quad I_q = f^q B.$$

There are two short exact sequences

$$0 \longrightarrow I_q \longrightarrow B \longrightarrow C_q \longrightarrow 0, \quad 0 \longrightarrow K_q \longrightarrow A \xrightarrow{a \mapsto af^q} I_q \longrightarrow 0. \quad (\text{S23.8})$$

There is also

$$0 \longrightarrow J \longrightarrow A \longrightarrow B \longrightarrow 0.$$

By generic flatness, simultaneously for the finite list $q = 0, \dots, N - 1$, followed by Noetherian induction on the complement, refine S so that B and every C_q are flat. Then the first sequence in (S23.8) shows that every I_q is flat. Since A is flat, the second sequence remains exact after *arbitrary* base change. Since B is flat, the sequence defining J does as well.

Let $T \rightarrow S_\alpha$ be any base change. Exactness identifies

$$B_T = A_T/J_T, \quad (I_q)_T = f_T^q B_T,$$

and the kernel of $A_T \rightarrow (I_q)_T$ is consequently

$$(K_q)_T = (J_T : f_T^q).$$

This is the desired colon base-change identity. Notice that flatness of the final cokernel C_q alone is not being used to preserve a kernel; the image module I_q has first been proved flat.

Now refine once more so that every $M_q = A/(f, K_q)$, and the sources, targets and cokernels of the finite list of multiplication maps under consideration, are flat. The specified exact diagrams then commute with arbitrary base change. For graded coefficient algebras, Lemma S23.12 gives constant Hilbert functions and degreewise ranks on connected strata. Apply Lemma S23.10 to each M_q , and take a common finite refinement. This produces the support packets, incidences, minimal/embedded status and embedded multiplicities asserted in the theorem. $\square$

Proof of Proposition S18.9. Apply Theorem S18.6 to the finite-type coefficient chart with

$$A = P_S, \quad f = d, \quad N = 5.$$

The hypothesis $d^5 \in J$ is Theorem S18.2, itself a specialization of the universal symmetric-power determinant-completion theorem. Include the finitely many graded multiplication maps in the same refinement. The theorem gives base change for every residual colon, flatness and constant Hilbert polynomial for every $\mathcal{M}_j$, the exact geometric fibrewise associated-point packets with constant embedded multiplicities, and constant multiplication ranks. These are precisely the three clauses of the proposition.

The construction of the coefficient strata is noncanonical, but its fibrewise output is the intrinsic graded algebra, its geometric assassin, and the canonical torsion lengths at associated generic points. Two presentation atlases therefore admit a common refinement carrying the same primary signature. $\square$

S23.5. Explicit orbit closures and graded specialization. Let

$$e_1 = h_1 w_1, \quad e_2 = h_1 w_2, \quad e_3 = h_2 w_1, \quad e_4 = h_2 w_2$$

be the standard basis of $H \otimes W$. The following families are written before applying a general quadratic block χ_t .

Theorem S23.13 (Corank-two orbit closure and specialization laws). *The closure relations needed to pass between all nine mixed-kernel types are realized by the following one-parameter families:*

(1) *rank three, off Q to on Q :*

$$\beta_t = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -t & 0 & 0 & 1 \end{pmatrix}, \quad \ker \beta_t = \langle e_1 + t e_4 \rangle;$$

(2) *secant to tangent:*

$$\ker \beta_t = \langle e_1 + e_4, e_2 - t e_3 \rangle,$$

for which $\delta|_{\mathbb{P}(\ker \beta_t)} = x^2 + t y^2$;

(3) *secant to the two ruling families:*

$$\langle e_1, e_2 + te_4 \rangle \rightsquigarrow \langle e_1, e_2 \rangle, \quad \langle e_1, e_3 + te_4 \rangle \rightsquigarrow \langle e_1, e_3 \rangle;$$

(4) *tangent to dual-rank-two:*

$$\beta_t = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & t \\ 0 & 0 & 0 & 0 \end{pmatrix};$$

(5) *either ruling to dual-rank-one, by multiplying the second nonzero row in the corresponding rank-two normal form by t ;*

(6) *either rank-one type to the zero map, by replacing β with $t\beta$.*

For a general polynomial choice of χ_t , the generic fibre and special fibre lie in the generic coefficient strata of the indicated orbit types. The corresponding primary signatures specialize as follows:

<i>degeneration</i>	<i>change of positive layers</i>
<i>off $Q \rightarrow$ on Q</i>	$V_1 \rightarrow V_3$
<i>secant $\rightarrow$ tangent</i>	$V_8 \rightarrow V_8$
<i>secant $\rightarrow$ H-ruling</i>	$V_8 \rightarrow (V_{15}, V_1)$
<i>secant $\rightarrow$ W-ruling</i>	$V_8 \rightarrow (R_1 + E_5)$
<i>tangent $\rightarrow$ dual rank 2</i>	$V_8 \rightarrow (R_2 + E_3, V_1)$
<i>H-ruling $\rightarrow$ dual rank 1</i>	$(V_{15}, V_1) \rightarrow (R_3, V_3)$
<i>W-ruling $\rightarrow$ dual rank 1</i>	$(R_1 + E_5) \rightarrow (R_3, V_3)$
<i>dual rank 2 $\rightarrow$ 0</i>	$(R_2 + E_3, V_1) \rightarrow (Q, Q)$
<i>dual rank 1 $\rightarrow$ 0</i>	$(R_3, V_3) \rightarrow (Q, Q)$.

Thus ruling components and embedded vertex components are born in actual one-parameter families of Fitting schemes.

Proof. The determinant of a tensor in each displayed kernel gives the stated Segre incidence. For example

$$\det \left(xI_2 + y \begin{pmatrix} 0 & 1 \\ -t & 0 \end{pmatrix} \right) = x^2 + ty^2,$$

which is secant for $t \neq 0$ and tangent for $t = 0$. Likewise δ restricts to txy on each secant-to-ruling family. The row-scaling families have the stated rank-one limits by inspection.

The residual maximal minors are polynomial in t and in the coefficients of χ_t . Each source and target generic stratum is a nonempty open in its coefficient space. Choosing χ_0 in the target open and a general first-order perturbation $\chi_0 + t\chi_1$ avoids the source bad divisors over $\mathbb{C}(t)$. Hence the two fibres have the primary signatures in Theorem S23.2. Because the ideals come from one polynomial family, the changes displayed in the table are genuine specializations, not comparisons of unrelated normal forms. $\square$

There is an analogous pure-block degeneration. In a fixed target basis let

$$A_t = (f_1, f_2, tf_3) : \text{Sym}^2 H \longrightarrow S_R$$

and choose the remaining mixed and quadratic coefficients generally. For $t \neq 0$, the induced mixed block is generically the rank-three off-Segre type; at $t = 0$, A_H has rank two and the induced mixed block has rank four. Thus the first nontrivial layer specializes from V_1 to V_{22} . This gives an explicit A_H -injective-to-rank-drop family, while the full table of Theorem S23.3 describes the generic pure-block boundary reached after further rank drops.

APPENDIX S24. SHARP GLOBAL DEPTH AND GEOMETRIC BOUNDARY LAWS

The preceding calculations have a global consequence stronger than the uniform exponent five for arbitrary quadratic quotients. The relevant distinction is between the full coefficient space and the open of basepoint-free webs with smooth Jacobian quartic. We first make the Pieri projection quantitative, then use its polynomial nature to pass between all projection ranks. We subsequently identify two global minimal supports and construct an explicit cross-corank incidence family. None of these arguments replaces a primary decomposition by a list of Hilbert functions.

S24.1. A normalized determinant projection.

Lemma S24.1 (Fischer coefficient for the determinant-four line). *Let $X = (x_{ij})$ be a generic 4×4 matrix, put $D = (\det X)^4$, and let $p = [123 \mid 123]^4 x_{44}^4$. The Cauchy-isotypic projection of p onto the line $\mathbb{C}D$ is*

$$\text{pr}_{(4,4,4,4)}(p) = \frac{1}{35}D.$$

In particular the coordinate-ring product of the $(4, 4, 4, 0)$ and (4) Cauchy components has nonzero determinant-four projection.

Proof. On the polynomial ring use the Fischer Hermitian form $\langle x^\alpha, x^\beta \rangle = \alpha! \delta_{\alpha\beta}$. It is invariant under the unitary groups acting on the two matrix factors. Inequivalent Cauchy summands are therefore orthogonal. The monomials of D which use only the northwest 3×3 block and x_{44} are exactly the monomials of p , with the same coefficients. Indeed each of the four determinant factors must then use its x_{44} entry. Hence

$$\langle p, D \rangle = \langle p, p \rangle > 0.$$

Successive application of the classical Cayley identity gives

$$\|\det(X_n)^s\|^2 = \prod_{j=0}^{n-1} \frac{(s+j)!}{j!};$$

see [26]. The northwest variables and x_{44} are disjoint, so

$$\|p\|^2 = 24 \prod_{j=0}^2 \frac{(4+j)!}{j!} = 24883200, \quad \|D\|^2 = \prod_{j=0}^3 \frac{(4+j)!}{j!} = 870912000.$$

Their ratio is $1/35$. This calculates an isotypic projection; it does not assert a coefficient-one identity in an unspecified straightening normalization. $\square$

Theorem S24.2 (Sharp global exponent on the Jacobian open). *For every $R \in G_4^\circ$, the residual ideal on every Schur chart satisfies*

$$d^4 \in J.$$

Consequently

$$\widehat{\mathcal{N}}_R^5 = 0, \quad W_j = \emptyset \quad (j \geq 5).$$

At every projection-corank-three or projection-corank-four point the local nilpotency index is exactly five. Thus the global exponent five for the nilradical is sharp on G_4° .

Proof. The proof of Theorem S18.12, with Lemma S24.1 supplying its explicit nonvanishing, gives a polynomial identity

$$(\det X)^4 \in I_6(\gamma_R \text{Sym}^2 X) \quad \text{in } \mathbb{C}[x_{ij} : 1 \leq i, j \leq 4].$$

It is an identity in the coordinate ring of the entire matrix space, not an identity valid only in the local ring of its origin. On a Schur chart substitute $X = I_H \oplus T$. The maximal minors specialize

to those of $[A_H, B_H(1_H \otimes T), C_H \operatorname{Sym}^2 T]$, and $\det X$ specializes to $d = \det T$. Ideal membership is preserved by this ring map; no flatness hypothesis is involved. Hence $d^4 \in J$ on every chart. Since the failure ideal is dJ , the asserted nilpotence and vanishing of the fifth and subsequent graded layers follow.

The corank-three Schur-symbol exclusion of Lemma S22.4 gives $d^3 \notin J$. The corank-four exclusion in Theorem S18.12 gives the same nonmembership. Together with $d^4 \in J$ these prove exact local index five. The corank-four stratum is nonempty: S_R has dimension six and contains four-planes, whose projections to V are zero. $\square$

Proposition S24.3 (Admissibility of the ambient rank tables). *At projection corank two for $R \in G_4^\circ$, the ranks $a = \operatorname{rank} A_H$ and $b = \operatorname{rank} \bar{B}_H$ necessarily satisfy*

$$a \geq 2, \quad a + b \geq 5.$$

Thus the possible pairs on this open are contained in

$$\{(3, 3), (3, 2), (2, 4), (2, 3)\}.$$

The other rows of the coefficient-space tables remain valid statements about surjective quadratic quotients; they are not strata of the smooth, basepoint-free Jacobian open.

Proof. The dimension of $R \cap \operatorname{Sym}^2 H$ is $3 - a$. If $a \leq 1$, the web contains two independent quadrics in the two H -variables. Along the line $\mathbb{P}(H^\perp)$, their gradients vanish. The Jacobian matrix therefore has rank at most two. Its determinant and all its first derivatives vanish there, contradicting smoothness of Y_R .

The dimension of $R \cap (\operatorname{Sym}^2 H + H \otimes (V/H))$ is $(3 - a) + (4 - b)$. Consequently the restriction of the web to the line $\mathbb{P}((V/H)^*)$ has dimension $a + b - 3$. A vector space of binary quadrics of dimension at most one has a common zero over $\mathbb{C}$, contrary to basepoint freeness. Hence $a + b - 3 \geq 2$. The upper bounds on b are respectively three and four when $a = 3$ and $a = 2$, giving the displayed list. $\square$

S24.2. Two global minimal supports and a resolution. Regard Y_R as a subvariety of $\operatorname{Gr}(3, V) = \mathbb{P}(V^*)$. Let $\mathcal{H}$ be the tautological rank-three bundle on it. Define

$$\tilde{Z}_R = \operatorname{Tot} \operatorname{Hom}(V, \mathcal{H}) = \{(H, T) : H \in Y_R, \operatorname{im} T \subset H\}, \quad \pi : \tilde{Z}_R \longrightarrow \operatorname{End}(V).$$

Let Z_R be its reduced image and let $D_2 \subset \operatorname{End}(V)$ be the determinantal variety of matrices of rank at most two. These constructions refer to the fixed quotient γ_R , not to a choice of presentation of its colons.

Theorem S24.4 (Global minimal supports of the first two layers). *Let $R \in G_4^\circ$, and on the full matrix space put $J = I_6(\gamma_R \operatorname{Sym}^2 T)$, $d = \det T$. Then*

$$\sqrt{(d, J)} = I(Z_R), \quad \sqrt{(d, J : d)} = I(D_2) = I_3(T).$$

The variety Z_R is irreducible of dimension fourteen, and π is a proper birational morphism from a smooth variety. The first layer has generic length one along its unique minimal support Z_R . For $R \in G_4^\dagger$, the second layer has generic length one along its unique minimal support D_2 . These assertions determine minimal supports and their generic lengths; they do not exclude additional embedded primes of either layer.

Proof. For a matrix of rank three with image H , the image of $\operatorname{Sym}^2 T$ is $\operatorname{Sym}^2 H$. The restriction of γ_R to this six-dimensional space has nontrivial kernel exactly when $R \cap \operatorname{Sym}^2 H \neq 0$. If α defines H , contraction $\operatorname{Sym}^2 V \rightarrow V$ by α has kernel $\operatorname{Sym}^2 H$. The determinant of its restriction to R is, up to the fixed nonzero normalization, the Jacobian quartic evaluated at α . Thus the rank-three part of $V(J)$ is exactly the image of the rank-three part of $\tilde{Z}_R$.

Every matrix of rank at most two belongs to $V(J)$, since the symmetric square of its image has dimension at most three. It also belongs to Z_R : the projective space of hyperplanes containing its image has positive dimension and meets the quartic Y_R . Once such a hyperplane is chosen, the matrix is a limit of rank-three maps into that hyperplane. Invertible matrices do not belong to $V(J)$, since γ_R is surjective. This proves the first radical equality.

The total space $\tilde{Z}_R$ is a rank-twelve vector bundle over the smooth irreducible surface Y_R . It is smooth and irreducible of dimension fourteen. Its inclusion in $Y_R \times \text{End}(V)$ is closed, so its projection to matrix space is proper. A rank-three matrix determines its image hyperplane uniquely; on this dense open the inverse is the regular image map to the Grassmannian. Hence π is birational.

We record the local scheme calculation at rank three, since it also controls the second layer. Write a Schur chart as $I_H \oplus t$. Smoothness of the Jacobian implies that A_H has rank five at a point of Y_R ; rank at most four would make every cofactor, and therefore every first derivative of its determinant, vanish. Eliminate an invertible five-minor. The residual ideal is generated by a local equation h of Y_R , the mixed entries tb_i , and a quadratic entry divisible by t^2 . At least one b_i is a unit. Otherwise a nonzero functional on the target cokernel, viewed as a quadratic form on V annihilating R , would vanish on $\text{Sym}^2 H + H \otimes (V/H)$, and would be proportional to α^2 . Then every quadric of R would vanish at $[\alpha]$, contradicting basepoint freeness. It follows that

$$J_{\text{loc}} = (t, h).$$

Thus the first layer is generically reduced on Z_R , whereas $(J : t)$ is the unit ideal at every rank-three point of Z_R . It is also the unit ideal off Z_R in rank three.

At a point of rank at most two, use the homogeneous transverse Schur coordinates. At corank two every residual minor has degree at least three, whereas d has degree two. At corank three the corresponding lower bounds are seven and three; at corank four they are twelve and four. Thus $d \notin J$ even after localization at the transverse origin. If $J : d$ contained a unit there, multiplication by its nonzero constant term would contradict this degree bound. Consequently the support of the second layer is exactly D_2 , proving the second radical equality. On the dense simple-contact, mixed-regular corank-two open supplied by $G_4^\dagger$, the inherited colon calculation gives the reduced transverse vertex for W_2 . Its generic length on the irreducible variety D_2 is therefore one. $\square$

Corollary S24.5 (Fibres of the boundary resolution). *For a matrix T of rank $r \leq 2$, the scheme-theoretic fibre is*

$$\pi^{-1}(T) = Y_R \cap \mathbb{P}((V/\text{im } T)^*).$$

For general matrices of rank two it consists of four reduced points; for general matrices of rank one it is a smooth plane quartic of genus three; over the zero matrix it is the quartic K3 surface Y_R .

Proof. The incidence equation is $\alpha T = 0$, which gives the displayed linear section scheme-theoretically. A general line meets the smooth quartic transversely in four points, and a general hyperplane section is a smooth plane quartic. Adjunction gives genus three in the latter case. At $T = 0$ the incidence equation imposes no condition. $\square$

S24.3. A cross-corank family with explicit components.

Theorem S24.6 (Contact points, plane quartics, and the K3 fibre). *Let $R \in G_4^\dagger$. Choose a general mixed-regular plane $H = \langle e_1, e_2 \rangle$, with a general basis, and extend it to a basis of V . On $B = \mathbb{A}_{u,v}^2$ set*

$$M(u, v) = \text{diag}(u, v, 0, 0).$$

There is a family of four-planes in $V \oplus S_R$ with this projection matrix. Its projection coranks are two on $uv \neq 0$, three on either punctured axis, and four at the origin.

Write $\alpha_1, \dots, \alpha_4$ for the dual homogeneous coordinates. The pullback of the resolution π is

$$\tilde{Z}_B = V(u\alpha_1, v\alpha_2) \subset B \times Y_R.$$

It has seven reduced irreducible components, no embedded components, and the following primary decomposition of ideal sheaves:

$$(u\alpha_1, v\alpha_2) = (u, v) \cap (u, \alpha_2) \cap (\alpha_1, v) \cap (\alpha_1, \alpha_2). \quad (\text{S24.1})$$

The last factor splits as the intersection of the ideals of the four constant contact-point sections. The other components are $\{0\} \times Y_R$, the $u = 0$ axis times $C_2 = Y_R \cap V(\alpha_2)$, and the $v = 0$ axis times $C_1 = Y_R \cap V(\alpha_1)$. The fibre Hilbert polynomials with respect to $\mathcal{O}_{Y_R}(1)$ are respectively

$$4 \quad (uv \neq 0), \quad 4n - 2 \quad (\text{a punctured axis}), \quad 2n^2 + 2 \quad ((u, v) = (0, 0)).$$

The family is flat on each of these strata, but not across them.

Proof. Choose a four-dimensional subspace $E \subset S_R$ and an isomorphism $\iota : V \rightarrow E$. The graphs

$$\mathcal{K}_{u,v} = \{(M(u, v)w, \iota(w)) : w \in V\}$$

are four-planes for every (u, v) , since their second component is injective. They define a morphism to the Grassmannian with the stated ranks. The resolution pulls back by the equations $\alpha M(u, v) = 0$, namely $u\alpha_1 = v\alpha_2 = 0$.

Generality of H makes the four points in $Y_R \cap V(\alpha_1, \alpha_2)$ distinct, and the general basis makes C_1, C_2 smooth plane quartics. They meet transversely at these four points. Locally at each of them, α_1, α_2 are regular parameters on the smooth surface. Equation (S24.1) is then the squarefree monomial identity for (ux, vy) . Away from the four points at least one of α_1, α_2 is a unit, and the same equality reduces to the elementary two-factor intersection. Thus the equality holds as ideal sheaves. The three surface components and four constant sections are distinct reduced irreducible components. The local squarefree description shows that there is no embedded component.

On the torus both linear equations remain; on a punctured axis only one remains; at the origin neither remains. This identifies the fibres. Their Hilbert polynomials are those of four points, a plane quartic of genus three, and a degree-four K3 surface. The restrictions to the displayed strata are products, proving their flatness. The change of Hilbert polynomial proves failure of flatness across them. $\square$

Proposition S24.7 (Intrinsic graded fibre algebra across coranks). *Pull back $\text{gr}_{\hat{\mathcal{N}}_R} \mathcal{O}_{\hat{D}_R}$ along the preceding family of reduced points. After a local trivialization of the conormal line, its algebra on a geometric fibre is*

$$\kappa[\epsilon]/(\epsilon^3) \quad (uv \neq 0), \quad \kappa[\epsilon]/(\epsilon^5) \quad (uv = 0), \quad \deg \epsilon = 1.$$

Thus degrees three and four appear in an actual family passing from projection corank two to three to four.

Proof. Every positive graded piece is a cyclic quotient of the reduced local ring, with the conormal-line twist, and its products are the canonical quotient products of Theorem S18.2. Its residue-field fibre is therefore either zero or one-dimensional. The mixed-regular corank-two calculation gives local nilpotency index three on the torus. Theorem S24.2 gives index five on the punctured axes and at the origin. The nonvanishing pieces are consecutive, and quotient multiplication takes the chosen generator to its powers, giving the stated algebras. $\square$

Remark S24.8 (Two different base-change operations). The preceding proposition pulls back an already formed intrinsic graded algebra. It does not form the nilradical after pulling back the failure scheme to B . In fact $\det M(u, v) = 0$ identically, so the latter pullback loses its transverse nilpotent

information. Likewise, the primary decomposition in (S24.1) is the primary decomposition of the incidence-resolution family, not a claim that all transverse primary components of W_3 and W_4 have been classified. This distinction is essential in a cross-stratum specialization, where kernel formation need not commute with pullback.

S24.4. An explicit infinite family of proper quotient maps. The next result gives an exact primary law and a sharp exponent in all symmetric degrees for quotients induced by a proper quotient of the underlying vector space. Its maximal-minor identity is classical: we use, and explicitly credit, Bruns–Vasconcelos [25, Theorem 1(2)]. The point here is to compute the resulting failure ideal, residual tower, and canonical embedded length in the notation of the present paper.

Theorem S24.9 (Primary law for quotient-induced symmetric powers). *Let $1 \leq k < e$, $r \geq 1$, and let $\rho : V \twoheadrightarrow W$ be a surjection with $\dim V = e$, $\dim W = k$. Set*

$$\gamma = \operatorname{Sym}^r \rho, \quad A = \mathbb{C}[\operatorname{End}(V)], \quad d = \det T, \quad P = I_k(\rho T), \quad a = \binom{k+r-1}{r-1}, \quad h = e - k + 1.$$

If $s = \dim \operatorname{Sym}^r W$ and $J = I_s(\gamma \operatorname{Sym}^r T)$, then

$$J = P^a = P^{(a)}, \quad (J : d^q) = P^{\max(a-q, 0)}, \quad \min\{q : d^q \in J\} = a.$$

Moreover

$$dJ = (d) \cap P^{a+1} \tag{S24.2}$$

is an irredundant primary decomposition. The nilradical of A/dJ has exact index $a + 1$. Its reduced determinant component has generic length one. Its only embedded associated prime is P , and the canonical local torsion length there is

$$\mu_P(A/dJ) = \binom{a+h-1}{h}.$$

For $1 \leq j \leq a$, the j -th positive graded layer is

$$A/(d, P^{a-j+1})(-je),$$

and its length at P is $\binom{a-j+h-1}{h-1}$.

Proof. Choose bases so that ρT consists of the first k rows of a generic $e \times e$ matrix. The maximal-minor identity of Bruns–Vasconcelos gives $J = P^a$ in characteristic zero. We recall why the powers of this maximal-minor ideal are symbolic, so that the primary assertions and colon calculation are explicit.

In the coordinate ring of a generic $k \times e$ matrix, the Cauchy summands are indexed by partitions λ of length at most k . A highest vector is the product of principal minors

$$v_\lambda = \prod_{j=1}^k \Delta_j^{\lambda_j - \lambda_{j+1}}, \quad \lambda_{k+1} = 0.$$

At the generic point of P , the minors Δ_j , $j < k$, are units, whereas Δ_k has order one. This follows also by eliminating an invertible $(k-1)$ -minor: the remaining maximal minors are the parameters of a regular local ring of dimension $h = e - k + 1$. Thus v_λ has order λ_k . The symbolic power is stable under the two general linear groups, so complete reducibility shows that it consists exactly of the summands with $\lambda_k \geq a$. Their highest vectors are divisible by Δ_k^a , and hence belong to P^a . The reverse containment is automatic. This proves $P^{(a)} = P^a$, and the same proof applies to every positive power. Adjoining the remaining rows is a polynomial extension and preserves these identities and primaryness.

In A_P , the full determinant d is a parameter: after the same elimination its linear term in the remaining maximal-minor parameters is nonzero, since the remaining generic rows can complete a rank- $(k-1)$ matrix to rank $e-1$. Hence

$$(P^a A_P : d^q) = P^{\max(a-q, 0)} A_P.$$

This also follows from the polynomial associated graded ring of the regular local ring. Contracting to A proves the asserted colons and exact exponent. In particular $(d) \cap P^{a+1} = d(P^{a+1} : d) = dP^a$. The determinant ideal is prime, P^{a+1} is P -primary, and their distinct radicals satisfy $(d) \subsetneq P$ because $h \geq 2$. Neither component can be removed, so the decomposition is irredundant and determines its associated primes.

Locally at P , write $\mathfrak{m} = PA_P$. Since d is a parameter and $h-1 \geq 1$, the quotient $A_P/(d)$ has no finite-length torsion. Therefore the local torsion of $A_P/(d\mathfrak{m}^a)$ is

$$(d)/(d\mathfrak{m}^a) \simeq A_P/\mathfrak{m}^a,$$

of length $\binom{a+h-1}{h}$. Localizing at (d) makes P the unit ideal, giving generic determinant length one. The residual-colon formula gives the displayed graded layers. Modulo the parameter d , their lengths are the numbers of monomials of degree less than $a-j+1$ in $h-1$ variables, namely $\binom{a-j+h-1}{h-1}$. Their sum is the preceding embedded length. This identifies the canonical embedded torsion, not the noncanonical size of an arbitrarily chosen embedded primary component. $\square$

APPENDIX S25. DEFORMATION TO THE NORMAL CONE: STANDARD MECHANISM AND THE COMPUTED SPECIAL FIBRE

The passage from an ideal to its extended Rees algebra is the standard deformation-to-the-normal-cone construction; see [17, §5.1] and [19, Tag 062Z]. We use it here only as the canonical mechanism that packages the manuscript-specific graded algebra computed in Sections S21 and S22.

Proposition S25.1 (Intrinsic deformation to the normal cone). *Let $\widehat{\mathcal{N}}_R$ be the intrinsic nilradical of $\widehat{D}_R$. On an affine open $\text{Spec } A \subset \widehat{D}_R$, with nilradical $N \subset A$, put*

$$\mathcal{R}_N = A[t, Nt^{-1}] \subset A[t, t^{-1}].$$

These algebras glue and define a flat family

$$\mathfrak{D}_R \longrightarrow \mathbb{A}^1.$$

Over $\mathbb{G}_m$ it is $\widehat{D}_R \times \mathbb{G}_m$, while its special fibre is

$$\mathfrak{D}_{R,0} \simeq \text{Spec}_{\widehat{\Delta}_R}^{\text{gr}_{\widehat{\mathcal{N}}_R}} \mathcal{O}_{\widehat{D}_R} = C_{\widehat{\Delta}_R/\widehat{D}_R}.$$

The construction is functorial for isomorphisms of the abstract failure scheme.

Proof. The extended Rees algebra is the standard construction. Multiplication by t is injective because $\mathcal{R}_N$ is a subring of $A[t, t^{-1}]$. Hence it is torsion-free over the PID $\mathbb{C}[t]$, and therefore flat. Inverting t gives $A[t, t^{-1}]$. Modulo t , the class of nt^{-j} , $n \in N^j$, depends only on $n \bmod N^{j+1}$, giving

$$\widehat{\mathcal{N}}_N/(t) \simeq \bigoplus_{j \geq 0} N^j/N^{j+1}.$$

Localization compatibility glues the construction. Since N is the ideal of the reduction $\widehat{\Delta}_R$ in $\widehat{D}_R$, the special fibre is the normal cone. The nilradical is intrinsic, so the construction is functorial. $\square$

The specific geometric content is not the standard construction in Proposition S25.1, but its bounded and stratified special fibre:

$$\mathrm{gr}_{\widehat{\mathcal{N}}_R} \mathcal{O}_{\widehat{D}_R} = \mathcal{O}_{\widehat{\Delta}_R} \oplus \bigoplus_{j=1}^5 \mathcal{L}^{\otimes j} \otimes \mathcal{O}_{W_j},$$

with the quotient multiplication (S18.4). Projection corank two has the mixed-kernel atlas of Theorem S18.4; all A_H -rank drops have the finite valuation bounds of Proposition S18.10; projection corank three has exact index five on G_4° , by Theorems S18.11 and S24.2; and projection corank four also has exact index five by Theorem S18.12. The five-or-six alternative for general coefficient data outside the smooth-web open is not a dichotomy on G_4° . On this open the summand with $j = 5$ in the displayed universal range is zero.

Flattening is used only to organize coefficient specializations of these explicitly presented finite modules. We use the standard flattening-stratification formalism in [20, Tag 052F]; no novelty is claimed for that formalism either.

Remark S25.2 (Novelty boundary). The standard ingredients in this section are the extended Rees algebra, deformation to the normal cone, and flattening stratification. The manuscript-specific statements are the Fitting factorization, the global determinant-power bound $d^5 \in J$, the nine mixed-kernel normal forms and their graded supports, the corank-three Schur-type exclusion, the corank-four Jacobian-covariant calculation, the collision primary laws, and the polarized reconstruction theorem. These roles are kept separate in the statements and in the referee response.

APPENDIX S26. EXACT SEPARATING PENCILS

The smooth integral witness C_* of (S9.2) provides a direct arithmetic check inside the geometric locus. Its two residual linear forms have determinant 27787/432. We retain below both this exact check and the five-coordinate square-web calculation. Neither is needed for the structural support proof of Theorem S6.13.

Choose a basis x_1, x_2, x_3, x_4 of V , and order the monomial basis of W as

$$(b_1, \dots, b_{10}) = (x_1^2, x_1x_2, x_1x_3, x_1x_4, x_2^2, x_2x_3, x_2x_4, x_3^2, x_3x_4, x_4^2).$$

For increasing indices put $e_{ijkl} = b_i \wedge b_j \wedge b_k \wedge b_l$. Let

$$w_* = e_{1,5,8,10}.$$

The Jacobian map

$$C : E \longrightarrow \det V \otimes \mathrm{Sym}^4 V, \quad q_1 \wedge \dots \wedge q_4 \longmapsto \det(\partial q_i / \partial x_j)$$

is equivariant, with the determinant factor understood in the chosen basis. It is nonzero, since $Cw_* = 16x_1x_2x_3x_4$. Its kernel is E_{175} .

Lemma S26.1 (An exact separating pencil). *The projection of w_* to E_{35} is*

$$Qw_* = \frac{1}{3}e_{1,5,8,10} - \frac{2}{3}e_{1,6,7,9} - \frac{2}{3}e_{2,3,6,10} + \frac{2}{3}e_{2,4,7,8} + \frac{2}{3}e_{3,4,5,9}. \quad (\text{S26.1})$$

On the pencil $aPw_ + bQw_*$, two Pluecker quadrics restrict to*

$$-\frac{2}{9}(b-a)(b+2a), \quad -\frac{4}{9}(b-a)^2. \quad (\text{S26.2})$$

Consequently $\mathcal{L}_ \cap \mathrm{Gr}(4, W)$ is the single reduced point $[w_*]$.*

Proof. Use the Fischer Hermitian forms: a degree- r monomial x^α has squared norm $\alpha!$, and the exterior basis has the induced product norm. The determinant factor has norm one. These forms are unitary invariant. Hence the adjoint C^* is unitary equivariant and, by complexification, is $GL(V)$ -equivariant. Schur's lemma gives $CC^* = c \text{id}$. The only exterior basis vectors contributing to the monomial $x_1 x_2 x_3 x_4$ are the five in (S26.1). Their Jacobian coefficients are respectively 16, -4 , -4 , 4, 4, and their squared norms are 16, 2, 2, 2, 2. The quartic monomial has squared norm one. It follows that

$$c = \frac{16^2}{16} + 4 \frac{4^2}{2} = 48, \quad Q = \frac{C^* C}{48}.$$

Applying this identity to w_* gives (S26.1).

Extend the Pluecker coordinates $z_{i_1 i_2 i_3 i_4}$ alternatingly, with repeated-index coordinates zero. For an ordered triple I and an increasing five-tuple $J = (j_1, \dots, j_5)$, use the quadratic relation

$$F_{I,J}(z) = \sum_{k=1}^5 (-1)^{k-1} z_{I, j_k} z_{J \setminus j_k}.$$

Substitution of (S26.1), together with $Pw_* = w_* - Qw_*$, gives (S26.2) for

$$\begin{aligned} I &= (1, 5, 8), & J &= (1, 6, 7, 9, 10), \\ I &= (1, 6, 7), & J &= (2, 3, 6, 9, 10), \end{aligned}$$

respectively. These two formulas can thus be checked with five nonzero coordinates. Their linear quotients after removing $b - a$ are independent. As ideal sheaves on $\mathbb{P}^1$, they generate $(b - a)$: the two remaining linear forms have no common projective zero. Every other restricted Pluecker quadric vanishes at $[1 : 1]$, so the full restricted ideal sheaf is also $(b - a)$. This proves both uniqueness and reducedness, including at the endpoints and at infinity. $\square$

Remark S26.2 (The same condition at the smooth integral witness). The smooth web C_* of (S9.2) also satisfies the rank-two condition. For its Pluecker vector, in the monomial basis above, the same two quadrics restrict respectively to

$$\frac{(a - b)(339a + 89b)}{24}, \quad \frac{37(a - b)(31a + 17b)}{72}.$$

The determinant of their linear quotients by $b - a$ is $27787/432 \neq 0$. These identities follow by taking the four-by-four minors of the displayed integral coefficient matrix (S9.2), applying $Q = C^* C/48$, and substituting in $F_{I,J}$. They are reproduced by `checks/revision132_exact.py`, together with equivariance for all sixteen matrix-unit generators of $\mathfrak{gl}_4$. The smoothness and basepoint-freeness of this same web are certified independently in Lemma S9.1. The smooth witness checks the geometric family directly. The five-coordinate example gives a transparent normalization check, while the support theorem proves the assertion for every smooth web.

APPENDIX S27. WEIGHTED DETERMINANTAL FLAGS AND THE CLASSICAL ORDER PROFILE

We record explicitly how the retained weighted-matrix calculation fits the classical determinantal theory. This comparison concerns the generic matrix only; the fixed-tensor theorem of §S19 was proved independently.

Let $A = (a_{ij})$ be a generic $q \times p$ matrix over a characteristic-zero field k , with $p \geq q$, and put $R_0 = k[a_{ij}]$, $I_0(A) = R_0$. For a nonincreasing sequence $\rho = (r_1, \dots, r_s)$ write

$$I^\rho = \prod_{\nu=1}^s I_{r_\nu}(A), \quad \gamma_j(\rho) = \sum_{\nu=1}^s \max(r_\nu - j + 1, 0).$$

The classical formula is

$$I^\rho = \bigcap_{j=1}^{r_1} I_j(A)^{(\gamma_j(\rho))}. \quad (\text{S27.1})$$

Here the superscript in parentheses denotes symbolic power. See De Concini–Eisenbud–Procesi [8], and its statement as Theorem 2.4 in Bruns–Conca [5], together with Proposition 2.2 there for the symbolic-order basis. The characteristic hypotheses and attribution are stated on printed pages 13–14 of that exposition.

Proposition S27.1 (Coefficientwise suspension). *In $R_0[t]$, put*

$$F = t \sum_{i=0}^q t^{q-i} I_i(A), \quad P_j = (t, I_j(A)).$$

Then

$$F = (t) \cap \bigcap_{j=1}^q P_j^{(q-j+2)}. \quad (\text{S27.2})$$

This is an irredundant primary decomposition. At a point with $t = 0$ where A has rank a , the nilradical of the corresponding local quotient has index $q - a + 1$.

Proof. In the regular local ring $(R_0)_{I_j}$, the symbolic order of an i -minor is $\max(i - j + 1, 0)$. Adjoining t gives the coefficient rule

$$\sum_{u \geq 0} t^u f_u \in P_j^{(b)} \iff f_u \in I_j^{(\max(b-u, 0))} \text{ for every } u. \quad (\text{S27.3})$$

For precision, the associated graded ring of the maximal ideal $(t, I_j(R_0))_{I_j}$ is a polynomial ring over its residue field; its homogeneous t -degrees are independent. The order of the displayed sum is therefore the minimum of $u + \text{ord}_{I_j}(f_u)$. Contracting the maximal-ideal power proves (S27.3), including the assertion that each contracted symbolic power is primary.

Intersecting with (t) deletes coefficient degree zero. For $1 \leq u \leq q$, put $k = q + 1 - u$. The remaining coefficient conditions on the right of (S27.2) are

$$f_u \in \bigcap_{j=1}^k I_j^{(k-j+1)} = I_k,$$

the one-factor case $\rho = (k)$ of (S27.1). For $u \geq q + 1$ there is no condition. These are exactly the coefficients of F , proving the equality.

For irredundancy, let d_{j-1} be a fixed $(j - 1)$ -minor, with $d_0 = 1$. The element $t^{q-j+1} d_{j-1}^2$ belongs to every component other than $P_j^{(q-j+2)}$. Indeed, its order along P_j is $q - j + 1$, while at P_k for $k < j$ its order is $q - j + 1 + 2(j - k) \geq q - k + 2$, and for $k > j$ its t -order suffices. The element d_q^2 witnesses the necessity of (t) . The associated primes are therefore exactly the displayed radicals.

After inverting an a -minor and Schur reducing, the localized coefficient ideal contains t^{q-a+1} , whereas t^{q-a} has coefficient one in degree $q - a$, where membership would require a one-minor of the residual matrix. That coefficient ideal is proper at the point. Thus the radical (t) has the asserted exact index. $\square$

The exponent $q - j + 2$ is therefore the scalar-variable suspension of the classical minor order $k - j + 1$, taken coefficientwise. This is a direct consequence of the classical profile plus (S27.3); it is not a new straightening theorem. The presentation $I_q[A, tI_q] = \sum_i t^{q-i} I_i(A)$ can also be viewed as substituting a scalar identity block in a larger generic matrix. Such substitution alone does not justify preservation of primary components or flatness. The coefficient proof is what

supplies the exact intersection here, while the ideal $\delta\mathbf{m} + (g_0, \dots, g_4)$ in §S19 requires its separate fixed-tensor analysis.

APPENDIX S28. AN EXPLICIT SYSTEM AND EXACT CERTIFICATE SPECIFICATIONS

The rational system used throughout is the following. It is the same relation system as in the preceding manuscript’s exact smoothness example, now used for a differential and a higher-corank calculation.

$$\begin{aligned} q_0 &= x_0^2 + 2x_0x_1 - 2x_0x_2 + 2x_0x_3 + x_1x_2 + x_1x_3 + x_2x_3 \\ q_1 &= -x_0x_1 - x_0x_3 + x_1^2 - x_1x_2 - x_1x_3 - 2x_2x_3 \\ q_2 &= 2x_0x_1 - x_0x_2 - 2x_0x_3 + 2x_1x_2 + x_2^2 + x_2x_3 \\ q_3 &= x_0x_2 + x_1x_2 - 2x_2x_3 + x_3^2 \end{aligned}$$

For a compact complete specification of its quartic, list the 35 degree-four exponent tuples in descending lexicographic order, starting with $(4, 0, 0, 0)$ and ending with $(0, 0, 0, 4)$. The coefficients of $F = \det(\partial q_i / \partial x_j)$ in that order are the rows of

$$\begin{pmatrix} -8 & -4 & -20 & 16 & 20 & 0 & -28 \\ -2 & -36 & 40 & 20 & -50 & -20 & -26 \\ 72 & -6 & 0 & -66 & -6 & 30 & 4 \\ -14 & -4 & -6 & -12 & 28 & 18 & -56 \\ 28 & 6 & -18 & 8 & -2 & 16 & 2 \end{pmatrix}.$$

Thus this display, together with the fixed monomial order, specifies every coefficient, including zeros. Alternatively, the determinant of the displayed four quadrics is its coordinate-free recipe.

For any list of homogeneous forms $f_1, \dots, f_s$ define the degree- d Macaulay coefficient matrix $\mathcal{M}_d(f)$ as follows. Rows are all degree- d monomials in the same descending lexicographic order. Columns are the forms $x^\beta f_i$, with i outermost and all β of degree $d - \deg f_i$ in descending lexicographic order innermost. The entry in row α is the coefficient of x^α . All entries for this example are integers.

Matrix	Size	Certified rank	Minor modulo 101
$[F, G_{01}, \dots, G_{36}]$	35×25	25	41
$\mathcal{M}_5(q_0, q_1, q_2, q_3)$	56×80	56	44
$\mathcal{M}_9(F_{x_0}, F_{x_1}, F_{x_2}, F_{x_3})$	220×336	220	79

The differential minor consists of the first 25 rows and all columns; its full integer value is (S9.4). The following deterministic rule specifies the other two minors without printing large matrices. Reduce the displayed coefficient matrix modulo 101. Scan columns from left to right. At the next pivot step choose the first remaining row with nonzero entry, exchange it with the next pivot row, normalize that row, and eliminate below. Skip a column if there is no nonzero remaining entry. Record the original row indices in the order selected and the pivot columns. The square submatrix of the *original*, unreduced matrix with exactly these row and column orders has the residue in the final column of the table. Full recorded index lists, along with the coefficient-generating program, are included in `evidence/K3_CERTIFICATES.json` and `checks/exact_k3.py`.

Every step is finite-field arithmetic on integer data. A nonzero residue of a specified square minor proves that integer minor is nonzero, so it proves a rank statement over $\mathbb{Q}$, not merely a rank in one numerical specialization. The Macaulay rank statements imply the homogeneous ideal containments used in Lemma S9.1. The exact integer determinant is also recomputed independently with rational-domain elimination. No floating-point rank, tolerance, or unverified smoothness inference is used.

For the corank-two quotient in (S19.1), one may reproduce its quotient map by multiplying the six-dimensional quadratic quotient by

$$\begin{pmatrix} 0 & -1/2 & -1 & 1 & 0 & 0 \\ 0 & -1/2 & -1 & 0 & 1 & 0 \\ 0 & -3/2 & -2 & 0 & 0 & 1 \end{pmatrix}.$$

This matrix annihilates the three columns of $\text{Sym}^2 H$ and has rank three. It yields the displayed B_H, C_H and the quartic (S19.2) by ordinary multiplication. The factor 2 in the coefficient of uv is essential: it records the square of $ux_2 + vx_3$, not a divided-power convention.

An additional exact regression test uses the normalized matrix (S19.3) with

$$C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1/2 & -3 \end{pmatrix}.$$

Then $h(u, v) = uv(u-v)(u-2v)$, so the four ideals are (a, b) , (c, d) , $(a-c, b-d)$, and $(a-2c, b-2d)$. Elimination computes both sides of (S17.2) over $\mathbb{Q}$, and mutual Gröbner reduction verifies their equality. It also checks that δ^2 is nonzero, that every element of $\mathfrak{m}\delta^2$ vanishes, and that $\delta^3 = 0$. This test checks a concrete instance; the general coefficient-ring proof is Lemma S19.4 and Theorem S19.5, not an extrapolation from that instance.

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REFERENCES

- [1] E. Arrondo and I. Sols, *On congruences of lines in the projective space* (Chapter 6 written in collaboration with M. Pedreira), *Mém. Soc. Math. France (N.S.)* **50** (1992), 1–96. doi:10.24033/msmf.363.
- [2] E. Ballico, On the failure locus of higher order properties of embeddings in projective spaces, *Math. Nachr.* **163** (1993), 5–13. doi:10.1002/mana.19931630102.
- [3] E. Ballico, On the failure cycles for the quadratic normality of a projective variety, *Pacific J. Math.* **172** (1996), 307–313.
- [4] D. Bayer and D. Eisenbud, Ribbons and their canonical embeddings, *Trans. Amer. Math. Soc.* **347** (1995), 719–756. doi:10.1090/S0002-9947-1995-1273472-3.
- [5] W. Bruns and A. Conca, Gröbner bases and determinantal ideals, in *Commutative Algebra, Singularities and Computer Algebra*, NATO Sci. Ser. II Math. Phys. Chem. 115, Kluwer, 2003, 9–66; author preprint arXiv:math/0302058.
- [6] W. Bruns and U. Vetter, *Determinantal Rings*, Lecture Notes in Mathematics 1327, Springer, 1988.
- [7] F. Cossec, Reye congruences, *Trans. Amer. Math. Soc.* **280** (1983), 737–751. doi:10.1090/S0002-9947-1983-0716848-4.
- [8] C. De Concini, D. Eisenbud, and C. Procesi, Young diagrams and determinantal varieties, *Invent. Math.* **56** (1980), 129–165. doi:10.1007/BF01392548.
- [9] I. Dolgachev and S. Kondō, *Enriques Surfaces II*, Springer, 2025, chapter *Reye Congruences*, 67–141. doi:10.1007/978-981-96-1513-1_2.
- [10] I. Dolgachev and I. Reider, On rank 2 vector bundles with $c_1^2 = 10$ and $c_2 = 3$ on Enriques surfaces, in *Algebraic Geometry*, Lecture Notes in Mathematics 1479, Springer, 1991, 39–49. doi:10.1007/BFb0086262.
- [11] J.-M. Drézet, Primitive multiple schemes, *Eur. J. Math.* **7** (2021), 985–1045. doi:10.1007/s40879-020-00447-4.
- [12] D. Huybrechts, *Lectures on K3 Surfaces*, Cambridge Studies in Advanced Mathematics 158, Cambridge University Press, 2016. doi:10.1017/CBO9781316594193.
- [13] C. Ingalls and A. Kuznetsov, On nodal Enriques surfaces and quartic double solids, *Math. Ann.* **361** (2015), 107–133. doi:10.1007/s00208-014-1066-y.
- [14] J. Kollár, *Rational Curves on Algebraic Varieties*, *Ergebnisse der Mathematik und ihrer Grenzgebiete (3)*, vol. 32, Springer, 1996, Chapter IV, §5.

- [15] N. Manolache, Cohen–Macaulay nilpotent schemes, arXiv:math/0312514 (2003).
- [16] G. Martin, G. Mezzedimi, and D. C. Veniani, Nodal Enriques surfaces are Reye congruences, *J. Reine Angew. Math.* **808** (2024), 49–65. doi:10.1515/crelle-2023-0092.
- [17] W. Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 2, Springer, 1998, §5.1.
- [18] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [19] The Stacks Project Authors, *The Stacks Project*, Section 31.20, “The normal cone of an immersion,” Tag 062Z, <https://stacks.math.columbia.edu/tag/062Z>.
- [20] The Stacks Project Authors, *The Stacks Project*, Section 38.21, “Flattening stratifications,” Tag 052F, <https://stacks.math.columbia.edu/tag/052F>.
- [21] The Stacks Project Authors, *The Stacks Project*, Section 31.7, “Relative assassin,” Tag 05AS, <https://stacks.math.columbia.edu/tag/05AS>.
- [22] The Stacks Project Authors, *The Stacks Project*, Section 37.18, “Flat modules and relative assassins,” Tag 0GSF; in particular Lemma 37.18.5, Tag 0GSJ.
- [23] The Stacks Project Authors, *The Stacks Project*, Section 38.18, “A criterion for purity,” Tag 05L2.
- [24] The Stacks Project Authors, *The Stacks Project*, Section 37.25, “Relative assassins,” in particular Lemma 37.25.5, Tag 05KR, <https://stacks.math.columbia.edu/tag/05KR>.
- [25] W. Bruns and W. V. Vasconcelos, Minors of symmetric and exterior powers, *J. Pure Appl. Algebra* **179** (2003), 235–240. doi:10.1016/S0022-4049(02)00115-9; arXiv:math/0111174.
- [26] S. Caracciolo, A. D. Sokal, and A. Sportiello, Algebraic/combinatorial proofs of Cayley-type identities for derivatives of determinants and pfaffians, *Adv. Appl. Math.* **50** (2013), 474–594. doi:10.1016/j.aam.2012.12.001; arXiv:1105.6270.
- [27] The Stacks Project Authors, *The Stacks Project*, Lemma 10.118.2, “Generic freeness,” Tag 051S, <https://stacks.math.columbia.edu/tag/051S>.
- [28] The Stacks Project Authors, *The Stacks Project*, Sections 37.26–37.27, “Reduced fibres” and “Irreducible components of fibres,” <https://stacks.math.columbia.edu/tag/0553>.
- [29] The Stacks Project Authors, *The Stacks Project*, Proposition 35.5.2 (Tag 023T) and Section 35.6 (Tag 0CDQ), fpqc and Galois descent of quasi-coherent modules, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/023T>.
- [30] A. Boralevi, A note on secants of Grassmannians, *Rend. Istit. Mat. Univ. Trieste* **45** (2013), 67–72. <https://rendiconti.dmi.units.it/volumi/45/011.pdf>.
- [31] E. Carlini, Reducing the number of variables of a polynomial, in *Algebraic Geometry and Geometric Modeling*, Springer, 2006, 237–247; author preprint arXiv:math/0507531, Proposition 1.
- [32] J. M. Landsberg and J. Weyman, On the ideals and singularities of secant varieties of Segre varieties, author manuscript (January 2007), Definition 1 and Theorem 3.1. <https://people.tamu.edu/~jml/1-07LWsecseg.pdf>.
- [33] The Stacks Project Authors, *The Stacks Project*, Section 29.36, Unramified morphisms, Tag 02G3, accessed September 23, 2026. <https://stacks.math.columbia.edu/tag/02G3>.
- [34] J. M. Landsberg and G. Ottaviani, *Equations for secant varieties via vector bundles*, arXiv:1010.1825v1 (2010), §6.1; subsequently merged into [35].
- [35] J. M. Landsberg and G. Ottaviani, Equations for secant varieties of Veronese and other varieties, *Ann. Mat. Pura Appl.* (4) **192** (2013), 569–606, doi:10.1007/s10231-011-0238-6; arXiv:1111.4567v1, §10.1.
- [36] J. Sheridan, Divisor varieties of symmetric products, *Int. Math. Res. Not. IMRN* **2022** (2022), no. 13, 9830–9863, doi:10.1093/imrn/rnaa313; arXiv:1906.05465v1, §2.4, Definitions 2.10–2.11.
- [37] S.-J. Cheng and W. Wang, Howe duality for Lie superalgebras, *Compositio Math.* **128** (2001), 55–94; arXiv:math/0008093v1, §2 and Theorem 3.5.
- [38] H. De Bie, D. Eelbode, and M. Roels, The harmonic transvector algebra in two vector variables, *J. Algebra* **473** (2017), 247–282; arXiv:1510.06566v3, §1, equations (1)–(3), and Theorem 2.1.
- [39] R. Kupferman and A. Shachar, A geometric perspective on the Piola identity in Riemannian settings, *J. Geom. Mech.* **11** (2019), 59–76; arXiv:1805.12365v1, equation (1.1) and §3.1.
- [40] C. Fevola, Y. Mandelshtam, and B. Sturmfels, Pencils of quadrics: old and new, *Le Matematiche* **76** (2021), no. 2, 319–335. doi:10.4418/2021.76.2.2; arXiv:2009.04334v2, Theorem 1.1, Corollary 2.1, Remark 2.3, Conjecture 4.5, and Theorem 5.1.
- [41] I. Smith, Floer cohomology and pencils of quadrics, *Invent. Math.* **189** (2012), 149–250. doi:10.1007/s00222-011-0364-1; arXiv:1006.1099, §4.2.
- [42] R. Howe, Remarks on classical invariant theory, *Trans. Amer. Math. Soc.* **313** (1989), 539–570. doi:10.1090/S0002-9947-1989-0986027-X, Theorem 9 and §4(a).

- [43] M. Olive, About Gordan's algorithm for binary forms, arXiv:1403.2283v5 (2015), §2.2, Definition 2.6 and Remark 2.7.
- [44] M. Trevisiol, Normality of closure of orthogonal nilpotent symmetric orbits, arXiv:2105.08680v3 (2022), §1, p. 3, equation (2) and the following dominance criterion.